Don't Try Again: Screening Through Attempt Limits

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Abstract

Licensing and civil-service examinations often cap the number of times a candidate may take them. While economists have long studied how tests should be designed to reveal information, I use mechanism design theory to model the attempt limit as a device for screening candidates' skills through the dynamic allocation of testing opportunities. In a two-period model, agents privately know their skill, and low-skilled agents can acquire skills between attempts. An attempt limit grants a single attempt but lets the agent choose when to use it: high-skilled agents take the test at once, whereas low-skilled agents postpone it and acquire skills in the meantime. The limit sorts agents before testing them, but it is incentive compatible only if the test is not too lenient, and it denies high-skilled agents a second chance. I characterize when the attempt limit is the optimal testing policy, first taking the leniency and accuracy of the test as given, then choosing leniency jointly with the allocation of attempts, and finally choosing costly accuracy as well. I show that, consistent with the model, attempt limits on the U.S. bar exam are tightest at intermediate passing scores and rare where the exam is lenient.

Keywords: attempt limits, test design, screening, mechanism design without transfers, occupational licensing.
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1 Introduction

Standardized examinations decide entry into many regulated professions and public-sector careers. In the United States, aspiring accountants must pass the Uniform CPA Examination, physicians the three-step United States Medical Licensing Examination (USMLE), and lawyers the bar examination. In India, the Civil Services Examination recruits officers for the Indian Administrative Service, the Indian Police Service, and the Indian Foreign Service, and in China the National Civil Service Examination, known as the *Guokao*, recruits officials for central government agencies. These examinations operate at a large scale. In 2024, 74,165 candidates sat for the CPA Examination and 70,436 for the bar examination, and in 2025 more than 55,000 candidates took Step 1 of the USMLE. The Indian Civil Services Examination received 992,599 applications in 2024 for 1,129 vacancies, and 3.4 million applicants qualified to sit for the 2025 *Guokao*, competing for fewer than 40,000 positions.[1]

Many of these examinations limit how many times a candidate may take them. The USMLE allows at most four attempts per Step, down from six before July 2021, and several state medical boards impose tighter limits as a condition for licensure. General-category candidates may sit for the Indian Civil Services Examination at most six times. Among the 40 jurisdictions that administered the Uniform Bar Examination between 2011 and 2024, 18 imposed a limit at some point, ranging from two to six attempts.[2] Why would a licensing authority restrict access to its own test? A limit denies a second chance to candidates who failed by bad luck, and it does not change what the test measures. This paper argues that an attempt limit is a screening device: by making attempts scarce, it changes who takes the test and when, and so sorts candidates before any of them is tested.

Economists have long studied how to design tests that select candidates well. This literature has focused on the grading scheme, that is, how finely to report results, where to set standards, and what to disclose (Lizzeri, 1999; Ostrovsky and Schwarz, 2010; Dubey and Geanakoplos, 2010; Boleslavsky and Cotton, 2015; Harbaugh and Rasmusen, 2018). It has mostly taken a static perspective, in which a candidate takes the test once, and it has

[1]Sources: National Association of State Boards of Accountancy, *The NASBA Report*, 2024 edition; National Conference of Bar Examiners, *2024 Statistics*; USMLE, *Performance Data*, 2025 (29,870 examinees from U.S. schools and 25,660 from non-U.S. schools); Press Information Bureau, Government of India, final results of the Civil Services Examination 2024 (of the 992,599 applicants, 583,213 sat for the preliminary examination and 1,009 were recommended for appointment); State Council of the People's Republic of China, news release of December 1, 2024 (3.416 million qualified applicants for 39,721 positions).

[2]Sources: USMLE, attempt limit policy; Federation of State Medical Boards, *State-Specific Requirements for Initial Medical Licensure* (for example, Texas and Arkansas allow three attempts per Step); Union Public Service Commission, Civil Services Examination notification (candidates from other backward classes are allowed nine attempts, and candidates from scheduled castes and tribes face only the age limit). Bar examination figures are from the data described in Section 2.

largely ignored that candidates decide whether and when to take a test.[3] Both features can be used to screen candidates well before testing them. I use mechanism design to propose a normative framework for a feature of testing that this literature has overlooked: the attempt limit.

In the model, a principal hires an agent who privately knows whether he is high- or low-skilled, and hiring requires passing a pass/fail test that can be offered in two periods. A low-skilled agent who has not passed can acquire skills between the two periods at a private cost. The principal cannot use transfers; instead, she decides which testing opportunities each agent receives. Besides a hiring freeze, in which no one is tested, four regimes allocate testing opportunities: unrestricted testing grants an attempt in each period, a required residency allows testing only in the second period, a stop-loss deadline only in the first, and an attempt limit grants a single attempt that the agent may use in either period. Under the attempt limit, high-skilled agents take the test at once, whereas low-skilled agents postpone it, acquire skills, and take it later. The limit thus sorts agents by the timing of their attempt and induces skill acquisition among those who wait. It is incentive compatible only if the test is not too lenient: when a low-skilled agent is likely to pass anyway, he prefers to gamble on an immediate attempt. It also has a cost, because a high-skilled agent who fails by bad luck cannot try again. The attempt limit is therefore not always optimal, and its value must be assessed against the other regimes.

The value of the attempt limit depends on two further features of the test. Its leniency governs the trade-off between two errors, failing a high-skilled candidate and passing a low-skilled one, and its accuracy determines how much more likely a high-skilled candidate is to pass than a low-skilled one. Both features shape the passing rates of the two types, the welfare of the principal, and the incentive compatibility of the attempt limit itself. Who controls them depends on the institutional context. In medical licensing, the USMLE program designs the examination and sets a single passing standard, while state boards decide how many attempts to accept. In the Uniform Bar Examination, the National Conference of Bar Examiners designs the test, while each jurisdiction sets its own passing score and attempt limit. In civil-service examinations, the government designs the test, chooses the passing threshold, and limits attempts. I therefore study the allocation of testing opportunities in three settings: with leniency and accuracy taken as given, with leniency chosen jointly with the allocation, and with leniency, accuracy, and the allocation chosen together when accuracy is costly.

When the principal chooses only the allocation of testing opportunities, the attempt limit

[3]Exceptions include Rosar (2017), who studies test design when participation is voluntary, and the work on retaking discussed in Section 1.1.

is optimal only if selection needs are high, that is, if the share of high-skilled candidates is small, and the leniency of the test is intermediate. Leniency cannot be too high, for otherwise low-skilled candidates take the test at once and the limit fails to sort them. Nor can it be too low: a test that already rejects most low-skilled candidates leaves little to gain from an additional layer of selection, and the principal prefers to give high-skilled candidates more chances to pass.

When the principal also chooses leniency, the optimal test setting depends on the accuracy of the test. When accuracy is high, the attempt limit is paired with a fully lenient test. An accurate test makes skill acquisition worthwhile, so low-skilled candidates postpone the test and reskill, and the principal can be lenient to pass as many high-skilled candidates as possible without losing the sorting. When accuracy is intermediate, the attempt limit is optimal for intermediate shares of high-skilled candidates, unless the principal is patient enough that delaying every test is cheap. Inducing low-skilled candidates to wait now requires a less lenient test, which is costly when most candidates are high-skilled, and in that case the principal prefers unrestricted testing. When most candidates are low-skilled, she prefers to make everyone wait through a required residency, which induces skill acquisition and allows a fully lenient test. When accuracy is low, a less intuitive setting can be optimal: with a weak pool of candidates, the principal may prefer unrestricted testing with a fully strict test, which gives high-skilled candidates two chances to pass and low-skilled candidates none.

Finally, when testing is costly and the principal chooses accuracy along with leniency and the allocation of attempts, the optimal design depends on the value of a high-skilled hire, relative to the loss from hiring a low-skilled one, and on the share of high-skilled candidates. When this value is small, the principal neither invests in accuracy nor screens candidates: she hires no one when the pool is weak and admits everyone when it is strong. When the value is larger, she invests in accuracy and, since each test is now costly, reduces testing opportunities. The attempt limit is optimal only when the value of a high-skilled hire is high, because the opportunity cost of delaying every test through a required residency then rises. In that case, the principal requires a residency when the pool is weak, admits everyone when it is strong, and adopts the attempt limit in between.

The U.S. bar exam offers evidence consistent with this mechanism. Using sessions of the Uniform Bar Examination between 2011 and 2024, I document three facts in Section 2. First, a higher passing score lowers the passing rate of first-time takers and repeaters alike, as expected from a test whose leniency shifts the passing probability of every candidate. Second, attempt limits leave the passing rate of first-time takers unchanged but raise the passing rate of repeaters, more so the tighter the limit, consistent with a limit that leads candidates to take the test again only when they are prepared. Third, application fees are

unrelated to the passing rate of either group, suggesting that monetary costs do not sort candidates. These facts motivate a model in which the passing score works as the leniency of a test signal and the attempt limit screens candidates without monetary transfers.

1.1 Related Literature

This paper contributes to several strands of the literature. First, it relates to mechanism design without monetary transfers. In the delegation tradition, a principal who cannot pay the agent restricts the set of actions the agent may choose from (Holmström, 1984; Melumad and Shibano, 1991; Martimort and Semenov, 2006; Alonso and Matouschek, 2008; Amador and Bagwell, 2013). A prominent application concerns fiscal rules, in which a government biased toward present spending is bound by a spending limit that trades off commitment against the flexibility to respond to shocks (Amador, Werning, and Angeletos, 2006; Halac and Yared, 2014, 2018), a bias that goes back to the time inconsistency studied by Phelps and Pollak (1968). An attempt limit is a rule of this kind: it restricts the testing opportunities from which the candidate can choose, and it trades off the screening it buys against the flexibility it takes from high-skilled candidates, who lose the chance to retake the test after an unlucky failure. Unlike a spending limit, however, the attempt limit works through self-selection, since it is the timing of the single attempt that reveals the candidate's type. In this respect, the delay that the limit imposes resembles the ordeals that target public programs without prices (Nichols and Zeckhauser, 1982; Alatas et al., 2016): postponing the test is cheap for low-skilled candidates, who use the time to acquire skills, and costly for high-skilled ones, who would only delay their hiring.

Within this literature, the closest papers are those in which the principal can partly verify the agent's private information (Green and Laffont, 1986; Glazer and Rubinstein, 2004; Ben-Porath, Dekel, and Lipman, 2014; Carroll and Egorov, 2019; Dasgupta, 2023). In these models, private information is elicited through evidence that the principal can obtain directly. In my framework, the test is a noisy verification technology, and the key informational instrument is the dynamic allocation of access to it: the principal controls how often and when the agent can be tested, and the agent's choice of when to be tested reveals his type.

Second, the paper is related to the literature on certification, in which a receiver of information designs a test to assess an unobservable quality of a strategic sender (Lizzeri, 1999; Rosar, 2017; Harbaugh and Rasmusen, 2018; Bizzotto, Rüdiger, and Vigier, 2020; Choi and Mukherjee, 2020; Weksler and Zik, 2022; Hancart, 2022; Perez-Richet and Skreta, 2022; Bizzotto and Vigier, 2025). In my setting, the test also serves as a certification device,

but much of this literature focuses on the design of the grading scheme (see also Ostrovsky and Schwarz, 2010; Dubey and Geanakoplos, 2010; Boleslavsky and Cotton, 2015; Zubrickas, 2015). By contrast, I take the grading rule as fixed, pass or fail, and study how other features of the test, namely its leniency, its accuracy, and the number and timing of attempts, can be jointly optimized.

More broadly, the paper is conceptually related to the literature on information design and Bayesian persuasion, in which a designer chooses a signal structure, that is, the distribution of messages conditional on the true state, to influence the actions of receivers (Kamenica and Gentzkow, 2011; Kolotilin et al., 2017; Bergemann and Morris, 2019). In persuasion models, the sender designs signals to move the receiver's posterior beliefs about an exogenous state. In this paper, the principal chooses the distributions of test outcomes for high- and low-skilled candidates through the institutional design of the testing process. These choices determine how informative the test is and how information about skill unfolds over time, but the objective is behavioral rather than informational: the principal does not seek to manipulate beliefs, only to regulate participation and selection through the structure of the testing environment.

Third, the paper is connected to work on standardized testing in college admissions. One branch emphasizes the costs of test preparation, test taking, and applications to rationalize test-optional policies (Garg, Li, and Monachou, 2026; Lee and Suen, 2023; Krishna et al., 2026), and another focuses on affirmative action (Chan and Eyster, 2003; Dessein, Frankel, and Kartik, 2025). I abstract from affirmative action and assume that taking the test is costless; instead, the principal exploits the endogenous cost of testing created by an attempt limit to sort candidates by skill. A related empirical literature shows that retaking a test raises scores and that access to retakes differs across students (Vigdor and Clotfelter, 2003; Frisancho et al., 2016; Goodman, Gurantz, and Smith, 2020). Closest to my question, Krishna, Lychagin, and Frisancho (2018) estimate a structural model of Turkey's university entrance exam and find that limiting retakes would benefit almost all students, because repeat takers push up admission cutoffs, and Bertola (2024) shows that more retake opportunities raise passing probabilities but weaken preparation, so that passing becomes a weaker signal of competence. I take the viewpoint of a principal who uses the number and timing of attempts to screen candidates, and I characterize when limiting attempts is optimal.

Finally, the paper contributes to the literature on occupational licensing (Kleiner, 2000). Pagliero (2011) uses variation in the difficulty of the bar exam across U.S. states to study whether licensing serves the public interest or protects incumbents. My empirical facts complement this evidence by documenting how passing scores, attempt limits, and application fees relate to the passing rates of first-time takers and repeaters, and my model provides a

normative rationale for attempt limits.

The rest of the paper is organized as follows. Section 2 presents the motivating facts from the U.S. bar exam. Section 3 introduces the model. The following sections characterize the optimal test setting when leniency and accuracy are given, when leniency is chosen jointly with the allocation of testing opportunities, and when accuracy is also chosen at a cost. Section 7 shows that the attempt limits of U.S. jurisdictions are consistent with Propositions 3, 5 and 6: limits are rare where the bar exam is lenient and tightest at intermediate passing scores. Section 8 concludes.

2 Motivating Facts

Jurisdictions that license lawyers control three features of the bar exam: how hard it is to pass, how many times a candidate may take it, and how much each attempt costs. This section documents how these features relate to passing rates in the U.S. bar exam and to one another, and four facts emerge. First, a higher passing score lowers the passing rate of first-time takers and repeaters alike. Second, attempt limits leave the passing rate of first-time takers unchanged but raise the passing rate of repeaters, more so the tighter the limit. Third, application fees are unrelated to the passing rate of either group. Fourth, attempt limits are concentrated at intermediate passing scores and are rare where the exam is lenient. These facts motivate the model that follows, in which the passing score governs how likely good and bad candidates are to pass, the attempt limit, rather than a fee, screens candidates, and the limit sorts candidates only when the test is not too lenient.

2.1 Data

The data cover the Uniform Bar Examination sessions held in the 40 jurisdictions, 39 states and the District of Columbia, that administered the UBE between 2011 and 2024. Restricting attention to UBE sessions holds the content and scoring of the exam fixed, so that jurisdictions differ only in the rules they set around it. For each jurisdiction, year, and session (February or July), the data report the number of candidates who sat for the exam and the number who passed, separately for first-time takers and repeaters. An observation is a jurisdiction–year–session cell for one of the two groups, and its passing rate is the ratio of passers to takers. The sample contains 1,336 observations from 668 sessions, covering 343,765 exam sittings, of which 252,123 are by first-time takers and 91,642 by repeaters. The average passing rate is 71.9 percent for first-time takers and 35.5 percent for repeaters.

Each jurisdiction sets three policies, which can change from one session to the next. The

passing score is the minimum UBE score required to pass, on the 400-point scale; it ranges from 256 to 280 and changes 17 times in 14 jurisdictions during the sample. The attempt limit is the total number of times a candidate may sit for the exam in the jurisdiction. Eighteen jurisdictions impose a limit at some point in the sample, ranging from two to six attempts, and seven jurisdictions introduce, remove, or change their limit. I measure the limit in two ways: an indicator equal to one when the jurisdiction has a limit, and its tightness, equal to the inverse of the number of attempts allowed and to zero when there is no limit, so that tightness rises from $1/6$ to $1/2$ as the limit becomes stricter. The passing score and the attempt limit are the same for first-time takers and repeaters. The application fee is what a candidate pays to sit for the exam; it ranges from \$100 to \$1,000, and in 15 jurisdictions repeaters pay a different fee from first-time takers.

2.2 Empirical Specification

To relate passing rates to the three policies, I estimate

$$\begin{aligned} PassRate_{jtsg} = \alpha_j + \tau_t + \nu_s + \beta_0 R_g + \beta_1 Policy_{jts} + \beta_2 Fee_{jtsg} + \beta_3 Score_{jts} \\ + \beta_4 R_g \times Policy_{jts} + \beta_5 R_g \times Score_{jts} + \beta_6 R_g \times Fee_{jtsg} \\ + \beta_7 Covid_{ts} + \varepsilon_{jtsg}, \end{aligned} \tag{1}$$

where j indexes jurisdictions, t years, s sessions, and $g \in \{F, R\}$ the two groups, first-time takers and repeaters. The dependent variable $PassRate_{jtsg}$ is the passing rate of group g. The terms α_j, τ_t, and ν_s are jurisdiction, year, and session fixed effects, and the indicator R_g equals one for repeaters, so that β_0 is the group fixed effect. The variable $Policy_{jts}$ is either the attempt-limit indicator or tightness, $Score_{jts}$ is the passing score in points, Fee_{jtsg} is the application fee paid by group g in hundreds of dollars, and $Covid_{ts}$ flags the July 2020 and February 2021 sessions, which the COVID-19 pandemic disrupted. The error term is ε_{jtsg}.

Because the specification includes jurisdiction fixed effects, the policy coefficients are identified from changes in policy within a jurisdiction over time, net of national movements common to all jurisdictions. The coefficients β_1, β_2, and β_3 measure the effect of each policy on first-time takers, and β_4, β_5, and β_6 measure how the effect differs for repeaters, so that the effects on repeaters are $\beta_1+\beta_4$, $\beta_2+\beta_6$, and $\beta_3+\beta_5$. I estimate each specification without weights, so that every observation counts equally, and with weights equal to the number of takers, so that the coefficients measure the average effect per candidate. Standard errors are clustered by jurisdiction.

2.3 Results

Table 1 reports the estimates. Columns (1) and (2) measure the attempt policy with the limit indicator, and columns (3) and (4) with tightness; columns (2) and (4) weight observations by the number of takers.

Fact 1: a higher passing score lowers the passing rate of both groups. The coefficient on the passing score is negative and significant at the 5 percent level in all four columns: raising the passing score by ten points lowers the passing rate of first-time takers by 6.2 percentage points without weights and by 4.0 percentage points with weights. A higher passing score makes the test less lenient, so fewer candidates clear it. The interaction with the repeater indicator is insignificant in every column and close to zero in the unweighted columns, so the data cannot distinguish the effect of the passing score on repeaters from its effect on first-time takers. The effect on repeaters alone, the sum of the two coefficients, is less precisely estimated, because repeater groups are smaller and their passing rates noisier. Figure 1 shows the relation behind these estimates. For each group, it plots the passing rate against the passing score after removing jurisdiction, year, and session effects and the other policies, so that only variation in the passing score within a jurisdiction over time remains. In both panels the relation is negative and of similar size, with a slope of -0.0040 for first-time takers and -0.0066 for repeaters, in line with the insignificant interaction in Table 1. The passing score therefore shifts the probability of passing for every candidate, whatever the candidate's history.

Fact 2: attempt limits raise the passing rate of repeaters but not of first-time takers. The coefficient on tightness, which measures the effect on first-time takers, is small and insignificant in columns (3) and (4), while its interaction with the repeater indicator is positive and significant at the 5 percent level in both columns. Adding the two, a limit of four attempts, which raises tightness from zero to $1/4$, increases the passing rate of repeaters by 4.1 percentage points without weights and by 6.3 percentage points with weights, against an average repeater passing rate of about one third. The indicator specification tells the same story: an attempt limit raises the passing rate of repeaters by 4.6 percentage points without weights and by 7.5 percentage points with weights.[4] For first-time takers, the indicator coefficient is insignificant without weights and positive with weights, a result driven by the

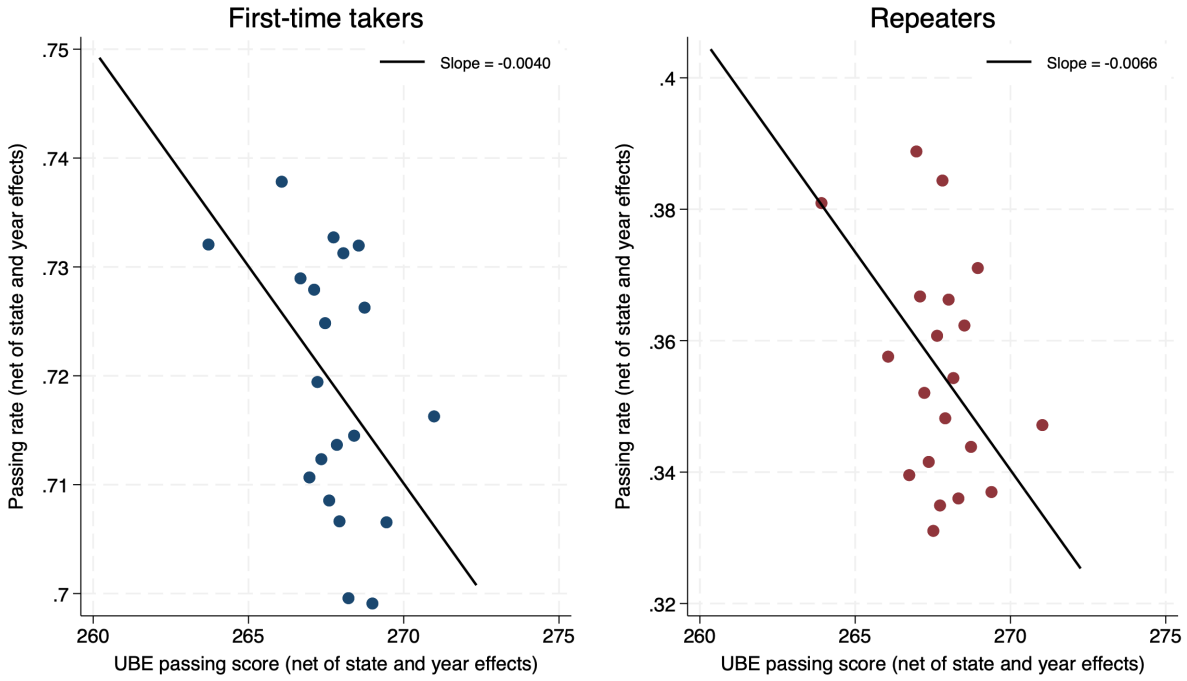
[4]The effect on repeaters, the sum of the policy coefficient and its interaction with the repeater indicator, is significant at the 1 percent level in columns (1) to (3) and at the 5 percent level in column (4).

Table 1: Passing rates and test-setting policies

	Attempt limit		Tightness	
	(1)	(2)	(3)	(4)
Attempt limit	−0.0063	0.0397**		
	(0.0187)	(0.0152)		
Tightness			−0.0019	0.1126
			(0.0628)	(0.0738)
Passing score	−0.0062**	−0.0040**	−0.0063**	−0.0040**
	(0.0026)	(0.0016)	(0.0026)	(0.0016)
Application fee	−0.0058	−0.0012	−0.0052	−0.0012
	(0.0058)	(0.0040)	(0.0057)	(0.0040)
Repeater × Attempt limit	0.0526***	0.0355**		
	(0.0175)	(0.0149)		
Repeater × Tightness			0.1670**	0.1374**
			(0.0755)	(0.0550)
Repeater × Passing score	0.0003	0.0028	0.0008	0.0029
	(0.0023)	(0.0026)	(0.0024)	(0.0025)
Repeater × Application fee	0.0025	0.0023	0.0011	0.0017
	(0.0037)	(0.0042)	(0.0037)	(0.0042)
Repeater	−0.3909***	−0.4173***	−0.3867***	−0.4177***
	(0.0122)	(0.0135)	(0.0126)	(0.0136)
COVID session	0.0647***	0.0531***	0.0647***	0.0534***
	(0.0108)	(0.0081)	(0.0109)	(0.0081)
Weighted by takers	No	Yes	No	Yes
Jurisdiction, year, session FE	Yes	Yes	Yes	Yes
Mean passing rate, first-time takers	0.719	0.761	0.719	0.761
Mean passing rate, repeaters	0.355	0.312	0.355	0.312
Observations	1,336	1,336	1,336	1,336
Jurisdictions	40	40	40	40
R^2	0.743	0.905	0.742	0.904

Note: The dependent variable is the passing rate of first-time takers or repeaters in a jurisdiction, year, and session, for UBE sessions in 39 states and the District of Columbia, 2011–2024. Columns (1) and (2) measure the attempt policy with an indicator for an attempt limit, and columns (3) and (4) with tightness, equal to one over the number of attempts allowed and to zero without a limit. Passing score is the UBE passing score in points out of 400, application fee is the fee paid by the group in hundreds of dollars, and repeater is an indicator for repeaters. Inside the interactions, passing score and application fee are centered at their sample means, so the repeater coefficient is the gap between the two groups at the average passing score and fee and without an attempt limit. All regressions include jurisdiction, year, and session fixed effects and an indicator for the COVID-19 sessions of July 2020 and February 2021. Columns (2) and (4) weight each observation by its number of takers, and the mean passing rates are weighted accordingly. Standard errors clustered by jurisdiction are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 1: Passing score and passing rate, net of fixed effects



Note: Binned scatter plot. For each group separately, the passing rate and the UBE passing score are regressed on jurisdiction, year, and session fixed effects, the COVID indicator, the attempt-limit indicator, and the application fee; the residuals, re-centered at the group means, are averaged within 20 equal-sized bins of the residual passing score. The slope of each line equals the coefficient on the passing score in the regression of the passing rate on the passing score and these controls for that group, with standard errors clustered by jurisdiction of 0.0018 for first-time takers and 0.0059 for repeaters. UBE sessions in 39 states and the District of Columbia, 2011–2024. The left panel reports first-time takers and the right panel repeaters.

District of Columbia, a large jurisdiction that introduced a four-attempt limit in 2019.[5] Figure 2 shows the pattern in the raw data by comparing the mean passing rate under each tightness level with the mean passing rate in jurisdictions without a limit. For first-time takers, the differences change sign across tightness levels, ranging from −5.3 to 5.0 percentage points, and follow no pattern. For repeaters, every tightness level has a higher passing rate than no limit, by 4 to 7 percentage points.[6]

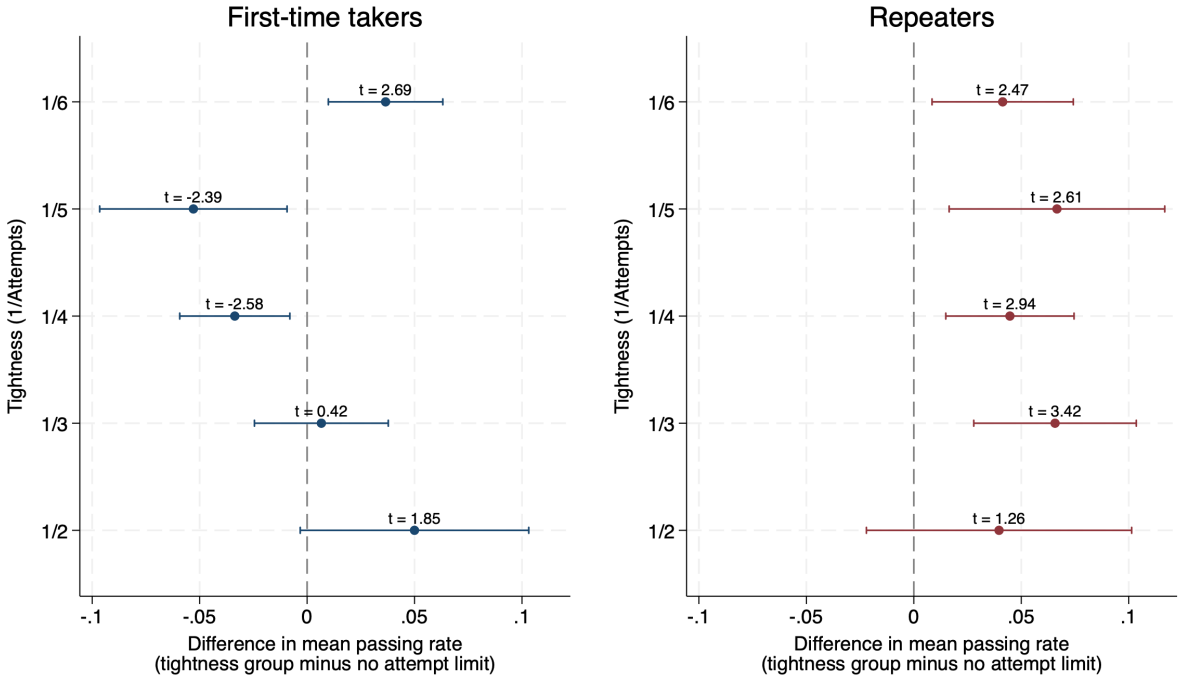
Why would a limit on attempts change the passing rate of repeaters but not that of first-time takers? A first-time taker has every attempt still ahead, so the limit does not bind when he first sits for the exam. If the limit deterred weak candidates from sitting in the first place,

[5]Without the District of Columbia, the coefficient on the attempt-limit indicator in column (2) is 0.007, with a standard error of 0.012.

[6]The confidence intervals in Figure 2 treat sessions as independent. Pooling all tightness levels and clustering standard errors by jurisdiction, the passing rate with a limit is 5.0 percentage points higher than without one for repeaters (standard error 2.2) and 0.3 percentage points lower for first-time takers (standard error 1.8).

the passing rate of first-time takers would rise; the data show no such effect. A repeater, by contrast, has failed at least once, and failing is informative about his chances of passing. When attempts are unlimited, a retake costs little beyond the fee and the time, so candidates keep trying whatever they have learned. When attempts are limited, each retake spends one of a few remaining chances, and the risk of wasting it falls mostly on candidates who expect to fail again. These candidates stop trying, or postpone until they are better prepared, while candidates who expect to pass return at once. The limit therefore tilts the pool of repeaters toward stronger candidates, and their passing rate rises. Part of this rise would occur even if candidates did not respond to the limit: candidates who exhaust their attempts leave the pool of repeaters, and they are mostly those who keep failing. The data cannot separate this mechanical exclusion from the response of candidates' choices; the model makes the latter explicit, by showing when low-skilled candidates postpone their attempt and acquire skills rather than simply running out of attempts. An attempt limit screens candidates through their own choices after a failure, without any monetary payment.

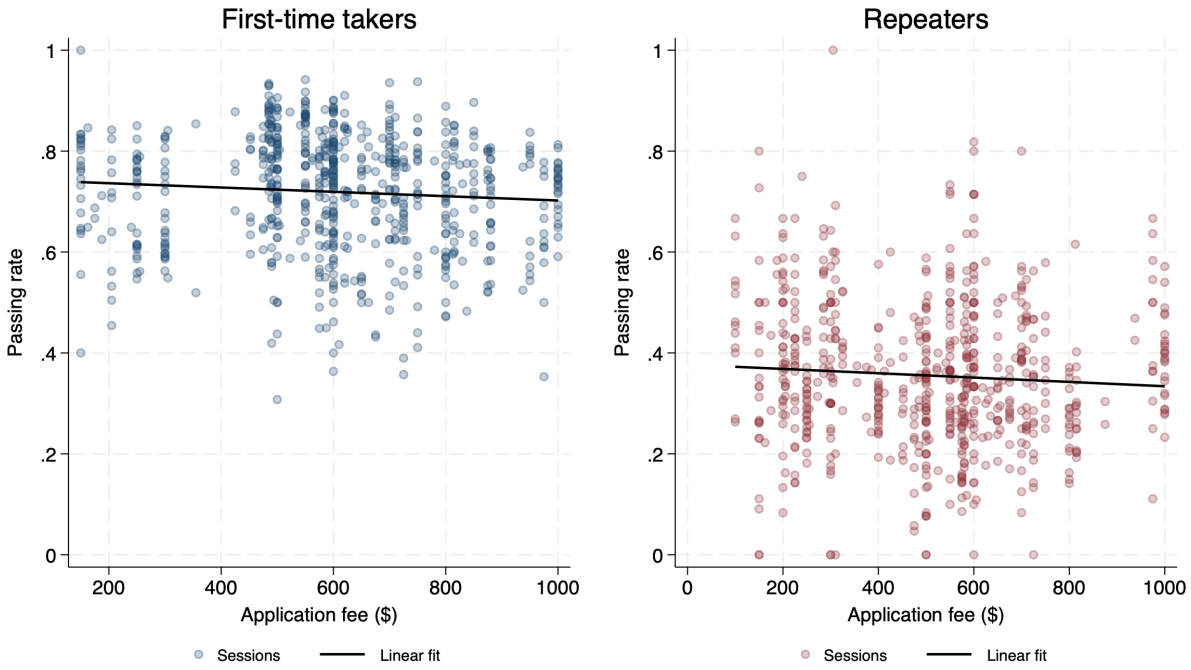
Figure 2: Tightness of the attempt limit and the passing rate



Note: Each point is the mean passing rate of sessions held under the given tightness level minus the mean passing rate of sessions held without an attempt limit, where tightness equals one over the number of attempts allowed. Bars are 95 percent confidence intervals from two-sample t-tests with equal variances, and labels report the t-statistics. The 1/2 level comes from a single jurisdiction. UBE sessions in 39 states and the District of Columbia, 2011–2024. The left panel reports first-time takers and the right panel repeaters.

Fact 3: application fees do not sort candidates. If fees worked as a sorting device, a higher fee would discourage the candidates least likely to pass, whose expected return from sitting for the exam is lowest, and the passing rate would rise with the fee. This should hold especially for repeaters, who have learned from a failure that their chances are lower. The data show no such pattern. The fee coefficient and its interaction with the repeater indicator are small and insignificant in every column: a $100 increase in the fee changes the passing rate of first-time takers by between −0.6 and −0.1 percentage points and that of repeaters by between −0.4 and 0.1 percentage points, and none of these effects is statistically different from zero. Figure 3 shows that the raw relation is flat as well, with a slope of −0.004 per $100 for both groups. Application fees, which never exceed $1,000, are small relative to the cost of law school and to the value of a license, so even weak candidates are willing to pay them. In this range, a fee is a cost that every candidate bears rather than a device that separates candidates.

Figure 3: Application fee and the passing rate



Note: Each dot is the passing rate in one jurisdiction–year–session, plotted against the application fee paid by the group: first-time takers in the left panel and repeaters in the right panel. Lines are least-squares fits on all sessions; the slope is −0.004 per $100 in both panels, with standard errors clustered by jurisdiction of 0.003 and 0.004. UBE sessions in 39 states and the District of Columbia, 2011–2024.

Together, these facts guide the model. The passing score works as the leniency of a test signal: it shifts the probability that good and bad candidates pass, for first-time takers

and repeaters alike. The attempt limit works as a screening mechanism without monetary transfers: by making retakes scarce, it leads candidates who are likely to fail to stay out or prepare before trying again, which raises the quality of those who return. Fees, which could in principle screen candidates through payments, do not play this role in the data, and the model abstracts from them.

3 General Model

Players. Consider a two-period principal–agent model. The agent is hired after passing a test in any of the two periods, and the principal screens out the agent through a test with a stochastic outcome whose distribution depends on the agent's skill. There are two types of agents, $\theta \in \{L, H\}$: low- and high-skilled. Skill level is agent's private info and the initial distribution is $Pr(\theta_1 = H) = \pi$.

Actions. The principal proposes to the agent a menu of contracts $\xi = \{\xi(L), \xi(H)\}$, one intended for each type, and the agent reports type $m \in \{L, H\}$ that might differ from his true θ: each contract $\xi(m)$ allocates testing slots across the two periods and can be characterized as a vector $\xi(m) = \big(\xi_t(m)\big)_{t=1,2}$ specifying whether the agent who has reported type m is tested in period t, where $\xi_t(m) = 1$ means testing and $\xi_t(m) = 0$ means no testing.

The outcome of the test is a binary stochastic event $\omega_t \in \{Pass, Fail\}$, and the probability of passing the test in period t can be modeled as a function of the agent's true type at time t and the contract $\xi(m)$ chosen by reporting m,

$$Pr\big(\omega_t = Pass|\theta_t, \xi(m)\big) = \begin{cases} p_{\theta_t}(\gamma, k), & \xi_t(m) = 1, \\ 0, & \xi_t(m) = 0, \end{cases} \tag{2}$$

where passing rates $p_{\theta_t}(\gamma, k)$ are function of two different parameters, $k \in [0, 1]$ and $\gamma \in [0, 1]$:

$$p_{\theta_t}(\gamma, k) = \begin{cases} (1-k)\gamma, & \theta_t = L, \\ 1 - (1-k)(1-\gamma), & \theta_t = H. \end{cases} \tag{3}$$

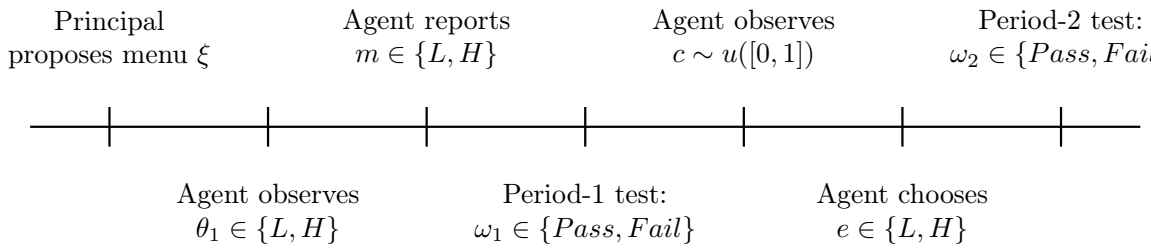
The two parameters play distinct roles. A high-skilled candidate is wrongly failed with probability $(1-k)(1-\gamma)$, and a low-skilled one is wrongly passed with probability $(1-k)\gamma$. The two error rates sum to $1-k$, so k measures the accuracy of the testing technology: raising it reduces both mistakes at once, and at $k = 1$ the test is perfectly revealing. The parameter γ divides the residual error $1-k$ between the two, and so captures a technological

trade-off within the test: a higher γ makes the test better at confirming competence and, by exactly the same amount, worse at detecting its absence. Accordingly $\Delta p_\theta = k$ does not depend on γ, and at $k = 0$ both types pass with the same probability, so the outcome carries no information. Notice that both p_L and p_H are constant over time, meaning that test difficulty does not change over time within the same skill level.

In period 1, the agent is high-skilled with probability π. In the next period, type θ_2 is deterministic, and it depends on the period-1 skill level and on a not verifiable effort level exerted in a skill-acquisition activity, $e \in \{L, H\}$. Formally, I assume that type θ_2 is the outcome of a transition function, $\theta_2 = z_{\theta_1}(e)$: if $\theta_1 = L$, then $z_{\theta_1}(e) = e$; if $\theta_1 = H$, then $z_{\theta_1}(e) = \theta_1$. The interpretation is that the low-type needs to exert high effort to become high-skilled, whereas the high-type retains his skill-level in the next period. High effort carries an individual cost c, drawn from a uniform distribution on $[0, 1]$; let $F(\cdot)$ denote its cumulative distribution function.

Timing. The game unfolds as follows. First, the principal proposes the menu of contracts $\xi = \{\xi(L), \xi(H)\}$. Second, the agent privately learns his type $\theta_1 \in \{L, H\}$ and reports $m \in \{L, H\}$. Then period-1 testing takes place, and test outcome is realized according to probability (2), $\omega_1 \in \{Pass, Fail\}$. If the test is passed, the agent is hired and the game ends; otherwise the game continues to period 2. In the next period, the agent privately observes his skill-acquisition cost type, $c \sim u([0, 1])$, and chooses the effort level in the learning activity, $e \in \{L, H\}$. Finally period-2 testing takes place, and the test outcome is realized according to probability (2), $\omega_2 \in \{Pass, Fail\}$. The game ends after period 2.

Figure 4: Timing of the Game



Payoffs. If the agent passes the test in period $t \in \{1, 2\}$, he is hired and receives a unit wage for infinitely many subsequent periods; otherwise his instant utility is normalized to zero in every period in which he is not hired. In period 2, the agent who failed period-1 test bears skill-acquisition cost $\eta(c, e)$, with $\eta(c, L) = 0$ and $\eta(c, H) = c$.

The principal's payoff depends on the type of the agent who passes the test. If the agent

is hired in period t, the principal receives $u(\theta_t, \omega_t)$ in that period and in every subsequent one, discounted by δ. The flow depends on the outcome and on the agent's competence at the time of passing: $u(\theta_t, \mathit{Pass}) = u(\theta_t)$, with $u(H) = w$ where $0 < w \leq 1$ and $u(L) = -1$, while $u(\theta_t, \mathit{Fail}) = 0$. If the agent never passes, the principal's payoff is zero.

4 Optimal Testing Slot Allocation

This section characterizes the optimal allocation of testing slots. I first introduce the objects on which the definition of the optimal allocation rests: the set of incentive-compatible contract menus, and the principal's welfare over that set. I then turn to the characterization of the set of incentive compatible contract menus, and I derive the optimal allocation. I close with a comparative-statics analysis on how the primitives shift the optimal policy.

Agent's expected utility. To characterize the set of incentive compatible contract menu, I need to define the expected value of a type-θ_1 agent choosing contract $\xi(m)$ from menu ξ. To ease notation, I express passing rates as a function of type θ_t only, $p(\theta_t) = p_{\theta_t}(\gamma, k)$. In case of failure of period-1 test, the expected utility of an agent with skill level θ_1 and cost type c who chooses effort level e is

$$v_{\theta_1}(e; \xi(m), c) = p(z_{\theta_1}(e)) \frac{\xi_2(m)}{1-\beta} - \eta(c, e). \tag{4}$$

Denote by $\varphi_{\theta_1}(\xi(m))$ the probability of passing period-1 test (2) as a function of θ_1 and $\xi(m)$, and let $\hat{e}_{\theta_1}(\xi(m), c)$ be the effort level maximizing (4). The expected utility of a type θ_1 agent choosing contract $\xi(m)$ at the beginning of period 1 is

$$V_{\theta_1}(\xi(m)) = \varphi_{\theta_1}(\xi(m))(1-\beta)^{-1} + \big(1 - \varphi_{\theta_1}(\xi(m))\big)\beta \mathbb{E}\big[v_{\theta_1}\big(\hat{e}_{\theta_1}(\xi(m), c); \xi(m), c\big)\big]. \tag{5}$$

Given $V_{\theta_1}(\xi(m))$, the set of incentive-compatible contract menus is

$$\Xi = \Big\{ \xi = \{\xi(\theta_1)\}_{\theta_1 \in \{L,H\}} \in \{0,1\}^4 \ : \ \forall \theta_1 \in \{L, H\}, \quad \theta_1 \in \arg\max_{m \in \{L,H\}} V_{\theta_1}(\xi(m)) \Big\}. \tag{6}$$

Principal's expected utility. Now that I have defined the set of incentive compatible contract menu, I can focus on the principal welfare function over Ξ. First, let $\mathcal{H}_{\theta_1}(\xi(m))$ be the principal's continuation value when type θ_1 agent chooses contract $\xi(m)$ and fails

period-1 test:

$$\mathcal{H}_{\theta_1}(\xi(m)) = \frac{\xi_2(m)}{1-\delta}\int_0^1 \Big(u(z_{\theta_1}(\hat{e}_{\theta_1}(\xi(m),c)))p(z_{\theta_1}(\hat{e}_{\theta_1}(\xi(m),c)))\Big)dc. \tag{7}$$

The expected welfare of the principal over the incentive-compatible contract menu set Ξ is

$$W(\xi) = \sum_{\theta_1=L,H} Pr(\theta_1)\Big[\varphi_{\theta_1}(\xi(\theta_1))\frac{u(\theta_1)}{1-\delta} + (1-\varphi_{\theta_1}(\xi(\theta_1)))\delta\mathcal{H}_{\theta_1}(\xi(\theta_1))\Big]. \tag{8}$$

Definition 1. *The contract menu $\xi^\star$ is an optimal allocation of testing slots if and only if it maximizes principal's expected welfare among all incentive-compatible menus:*

$$\xi^\star \in \arg\max_{\xi\in\Xi} W(\xi).$$

4.1 Incentive Compatibility

To determine the optimal contract menu, I first characterize the set of incentive-compatible menus. A contract takes one of four forms, according to the periods in which testing is permitted: writing $\chi_n = (\chi_n(1), \chi_n(2))$, these are $\chi_0 = (0,0)$, $\chi_1 = (1,0)$, $\chi_2 = (0,1)$ and $\chi_3 = (1,1)$. In particular, χ_0 permits no test, χ_1 a test in period 1 only, χ_2 a test in period 2 only, and χ_3 a test in both periods. A menu assigns one contract to each type, so there are $2^4 = 16$ menus in total: four offer the same contract, $\xi(L) = \xi(H) = \chi_n$, and twelve menus, in which the two contracts differ, $\xi(L) = \chi_n \neq \chi_{\tilde{n}} = \xi(H)$. The former are trivially incentive compatible, since the two types face identical options. The analysis therefore focuses on the second type of menus. To ease notation, let $\phi(c;k,\beta) = max\{\frac{k}{1-\beta} - c, 0\}$.

Lemma 1. *Consider a contract menu $\xi = \{\xi(L), \xi(H)\}$ such that $\xi(L) \neq \xi(H)$:*

(i) If $\xi \in \Xi$, then $\xi(L) = \chi_2$ and $\xi(H) = \chi_1$;

(ii) Menu $\xi = \{\chi_2, \chi_1\}$ is incentive compatible if and only if

$$\gamma \leq \frac{\beta}{1-k}\int_0^1 \phi(c;k,\beta)\,dc \tag{9}$$

Proof. See Appendix A.1. □

Part (i) says that separation can only work through timing: each type gets exactly one attempt, and the high type gets the earlier one. Part (ii) states that separation is incentive

compatible when the low type's relative net return from test-delaying is high enough. Indeed condition (9) can be rewritten as follows:

$$\frac{\beta}{p_L} \cdot \underbrace{F\left(\frac{\Delta p_\theta}{1-\beta}\right)}_{\textit{Reskilling prob.}} \cdot \underbrace{\left\{\frac{\Delta p_\theta}{1-\beta} - \mathbb{E}\left[c \middle| c \leq \frac{\Delta p_\theta}{1-\beta}\right]\right\}}_{\textit{Expected reskilling net return}} \geq 1.$$

The term $\frac{\Delta p_\theta}{1-\beta} - \mathbb{E}\big[c|c \leq \frac{\Delta p_\theta}{1-\beta}\big]$ is the expected net gain from reskilling measured in period-2 value, multiplied by the probability of reskilling in period 2, $F(\frac{\Delta p_\theta}{1-\beta})$. The factor β discounts this gain to period 1, and dividing by p_L expresses it relative to the pass value the low type could secure at once by testing early. Separation holds precisely when this relative net return clears one: the low type accepts the delayed slot only when waiting and studying returns at least as much as the immediate attempt it forgoes.

4.2 Optimality

At most five contract menus can be incentive compatible: $\xi_\emptyset = \{\chi_0, \chi_0\}$, $\xi_R = \{\chi_2, \chi_2\}$, $\xi_D = \{\chi_1, \chi_1\}$, $\xi_U = \{\chi_3, \chi_3\}$, and $\xi_\Lambda = \{\chi_2, \chi_1\}$, known respectively as *hiring freeze*, *residency requirement*, *stop-loss deadline*, *unrestricted testing*, and *attempt limit*. The principal chooses among them to maximize expected welfare. Say that *testing is efficient* when $\xi^\star \neq \xi_\emptyset$, that is, when the principal finds it worthwhile to test at all. The next two propositions establish when testing is efficient and, when it is, which menu is optimal; both are stated in terms of thresholds that depend on the parameter vector $\varsigma = (w, \delta, \gamma, k)$.

Proposition 1. *Suppose that the attempt limit is not incentive compatible, $\xi_\Lambda \notin \Xi$. Then there exist thresholds $\bar{\pi}_\emptyset(\varsigma) < \underline{\pi}_P(\varsigma) \leq \bar{\pi}_P(\varsigma)$ such that:*

(i) Testing is efficient if and only if $\pi \geq \bar{\pi}_\emptyset(\varsigma)$;

(ii) If testing is efficient, then the optimal allocation of testing slots $\xi^\star$ is given by function

$$\xi_P^\star(\pi, \varsigma) = \begin{cases} \xi_R, & \pi \leq \underline{\pi}_P(\varsigma), \\ \xi_D, & \pi \in (\underline{\pi}_P(\varsigma), \bar{\pi}_P(\varsigma)], \\ \xi_U, & \pi > \bar{\pi}_P(\varsigma). \end{cases}$$

Proof. See appendix A.2 □

When separation is unavailable, the principal can only pool, and the three testing menus differ in how many attempts they grant and when. As good candidates become more abundant she loosens both margins in turn. At low priors she uses ξ_R: delaying everyone to

period 2 converts a fraction ρ of the low types, and the cost of making the high types wait is small when there are few of them. As π rises that cost grows, and ξ_D takes over: testing once in period 1 forgoes conversion but hires the high types immediately. Finally ξ_U prevails, granting both attempts to both types once the pool is good enough that the extra low-type leakage is outweighed by the high type's retake. The ξ_D region may be empty: when reskilling is sufficiently likely, low types are an asset in period 2 and ξ_U dominates ξ_D at every prior.

Proposition 2. *Suppose that attempt limit is incentive compatible, $\xi_\Lambda \in \Xi$. Then there exist thresholds $\underline{\pi}_\emptyset(\varsigma) < \pi_S(\varsigma)$, with $\underline{\pi}_\emptyset(\varsigma) \leq \bar{\pi}_\emptyset(\varsigma)$, such that:*

(i) Testing is efficient if and only if $\pi \geq \underline{\pi}_\emptyset(\varsigma)$;

(ii) If testing is efficient, then the optimal allocation of testing slots is given by function

$$\xi_S^\star(\pi;\varsigma) = \begin{cases} \xi_\Lambda, & \pi \leq \pi_S(\varsigma), \\ \xi_U, & \pi > \pi_S(\varsigma). \end{cases}$$

Proof. See appendix A.3 □

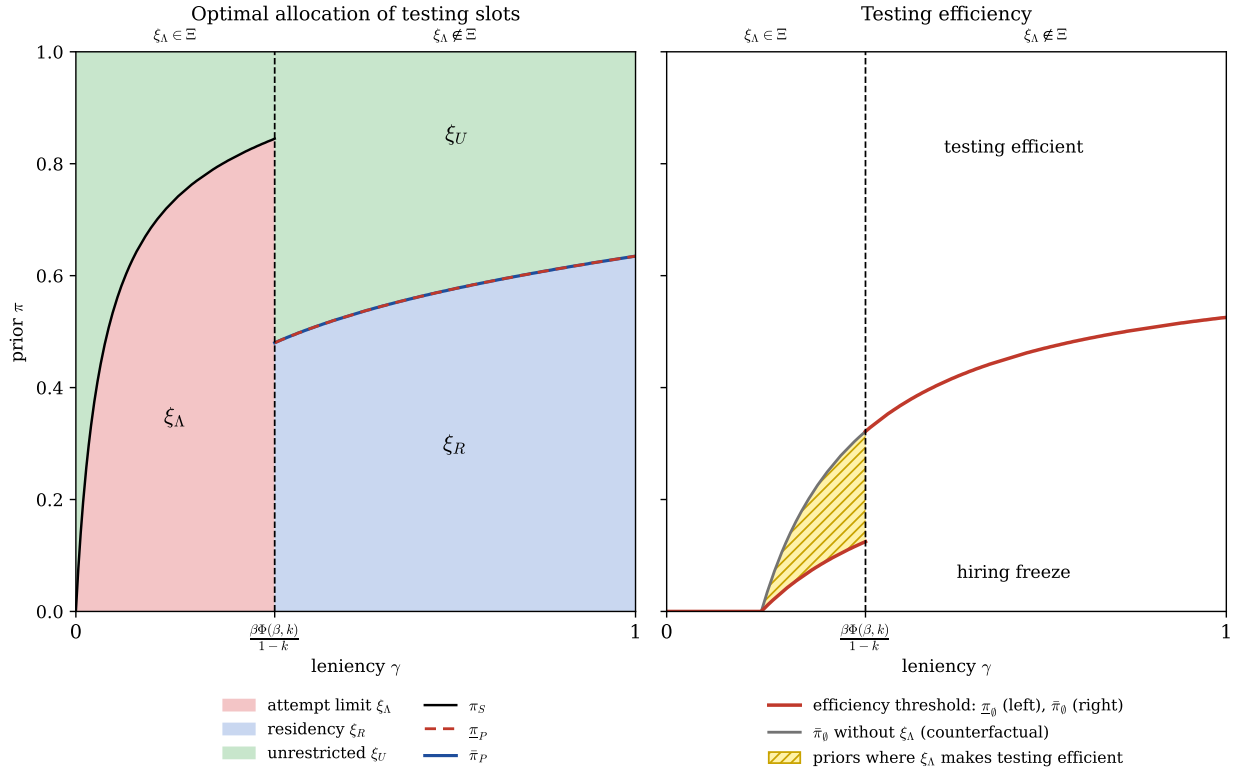
When separation is available the picture simplifies, because ξ_Λ dominates both restrictive pooling menus outright. All three grant each candidate a single attempt and differ only in its timing: ξ_D places it in period 1 for everyone, so the low type is tested while still unskilled and never reaches the stage at which he might reskill; ξ_R places it in period 2 for everyone, which suits the low type but delays the high type for no reason. The attempt limit assigns each type the timing that suits it, and so dominates both. Only the hiring freeze and unrestricted testing remain, and the intermediate region belongs to ξ_Λ throughout. Propositions 1 and 2 take the leniency of the test as given. Before asking for which values of γ the attempt limit is optimal, it is useful to record how the thresholds respond to it. Define $\Phi(k,\beta) \equiv \int_0^1 \phi(c;k,\beta)\,dc$ and denote by $\tilde{\gamma} = \frac{\beta\Phi(k,\beta)}{1-k}$ the incentive compatibility threshold of the attempt limit.

Lemma 2. *Every prior-threshold in Proposition 1 and 2 is non-decreasing in γ.*

Proof. See appendix A.4. □

A more lenient test adds leakage without adding information: both passing rates rise by the same amount, so the gap $\Delta p_\theta = k$, and with it the reskilling probability, are unchanged. At any given prior the case for restricting attempts therefore strengthens. Every threshold compares a more restrictive menu against a less restrictive one, and each shifts upward: the

Figure 5: Optimal Allocation of Testing Slots and Testing Efficiency



Note: The figure plots the optimal allocation of testing slots (left panel) and the testing efficiency thresholds (right panel) as functions of the test's leniency γ, for the test technology $p_L = (1-k)\gamma$, $p_H = 1-(1-k)(1-\gamma)$. The dashed line marks the highest leniency at which the attempt limit is incentive compatible, $\gamma = \beta\Phi(\beta,k)/(1-k)$: to its left $\xi_\Lambda \in \Xi$ (Proposition 2), to its right $\xi_\Lambda \notin \Xi$ (Proposition 1). In the left panel, the thresholds π_S, $\underline{\pi}_P$ and $\bar{\pi}_P$ separate the optimal testing menus. In the right panel, red curves are the efficiency thresholds; to the left of the dashed line, the grey curve is the threshold the principal would face without the attempt limit, $\bar{\pi}_\emptyset$, and the hatched area shows the priors at which the attempt limit makes testing efficient. Parameters: $\beta = 0.7$, $\delta = 0.18$, $w = 0.15$, $k = 0.2$.

whole policy map translates in the same direction as the test becomes more lenient, and the ordering of menus in π never rearranges.

Propositions 1 and 2 take the incentive-compatibility status of the attempt limit as given and characterize the optimal policy on either side of it. But that status is itself determined by the parameters: whether ξ_Λ belongs to Ξ depends on the same passing rates that govern which menu the principal would choose. The next result therefore treats the two together, asking for which false-positive rates the attempt limit is both feasible and optimal. Before stating the next result, denote $\bar{\varsigma} = (w, \delta, \beta, k)$, and $\bar{\gamma} \equiv min\{1, \tilde{\gamma}\}$.

Proposition 3. *Assume $k \in (0,1)$, and let $\pi_\Lambda \equiv \pi_S(\bar{\gamma}, \bar{\varsigma})$:*

(i) If $\pi \leq \pi_\Lambda$, there exists a non-empty closed interval $\mathcal{I}(\pi) \subset (0, \bar{\gamma}]$ such that the attempt limit is the optimal allocation of testing slots if and only if $\gamma \in \mathcal{I}(\pi)$.

(ii) If $\pi > \pi_\Lambda$, the attempt limit is never the optimal allocation.

Proof. See appendix A.5. □

The two parts say different things. Part (i) states that the false-positive rate must take intermediate values: $\mathcal{I}(\pi)$ is bounded above because a test that admits unqualified candidates too often destroys the low type's willingness to wait, and bounded below because a test that almost never admits them leaves nothing for the attempt limit to prevent. Part (ii) establishes a necessary condition on the pool itself: the proportion of high-skilled candidates cannot be too high. Denying a retake costs $\delta p_H w(1 - p_H)$ per high type, so when good candidates are abundant that cost outweighs the leakage the attempt limit blocks, whatever the test. The screening motive must be strong enough – that is, competent candidates scarce enough – for restricting attempts to be worth its cost.

5 Endogenous Leniency

So far the leniency of the test has been taken as given. In many settings, however, it is chosen by the tester herself, who designs the test and sets the passing threshold. This section therefore lets the principal choose the leniency γ together with the allocation of testing slots: at the beginning of the game she announces both the menu of contracts and the leniency of the test, and the game then unfolds as before, with the accuracy k held fixed. Hence let $V_{\theta_1}(\xi(m);\gamma)$ denote the expected utility of type θ_1 agent defined in (5) and expressed as a function of both ξ and γ. Likewise, let $W(\xi,\gamma)$ be the expected welfare of the principal over the set of incentive compatible test settings, defined as

$$\Pi = \Big\{ (\xi,\gamma) \in \{0,1\}^4 \times [0,1] \;:\; \forall \theta_1 \in \{L,H\}, \quad \theta_1 \in \arg\max_{m\in\{L,H\}} V_{\theta_1}(\xi(m);\gamma) \Big\}. \tag{10}$$

Definition 2. *Pair $(\xi^\star,\gamma^*)$ is an optimal test setting if and only if it maximizes principal's expected welfare among all incentive-compatible test settings:*

$$(\xi^\star,\gamma^*) \in \arg\max_{(\xi,\gamma)\in\Pi} W(\xi,\gamma).$$

Because a change in leniency shifts both passing rates by the same amount, the principal's welfare is linear in γ under every menu except unrestricted testing, whose retakes make it quadratic. Denote by $\gamma_U^\star$ the optimal leniency under unrestricted testing. The attempt limit differs from the other menus in one respect: by Lemma 1, it is incentive compatible only if

the test is not too lenient, namely only if $\gamma \leq \tilde{\gamma}(k, \beta)$. Whether the principal can sort the two types therefore depends, in general, on the leniency she chooses.

Definition 3. *Sorting is leniency-independent if and only if* $(\xi_\Lambda, \gamma) \in \Pi$ *for every* $\gamma \in [0, 1]$.

Since the incentive ceiling is increasing in accuracy, sorting is leniency-independent if and only if $\tilde{\gamma}(k, \beta) \geq 1$, that is, if and only if $k \geq \bar{k} \equiv 1 - \frac{\beta(1+\beta)}{2}$.

Proposition 4. *Suppose the principal chooses* (ξ, γ) : *(i) Testing is efficient*; *(ii) If* $k \geq \bar{k}$, *then* $(\xi^*, \gamma^*) = (\xi_\Lambda, 1)$; *(iii) If* $k < \bar{k}$, *then* $(\xi^*, \gamma^*) \in \big\{(\xi_\Lambda, \tilde{\gamma}),\ (\xi_R, 1),\ (\xi_D, 1),\ (\xi_U, \gamma_U^\star)\big\}$.

Proof. See appendix A.6. □

Testing is always efficient because, however small its accuracy, the test can be made strict enough that no unqualified candidate passes while some competent ones do. When the test is accurate, the attempt limit and the most lenient pass mark are complements: every competent candidate passes at once, so denying a retake costs nothing, and every deferred low type reskills. When it is less accurate, the incentive ceiling binds, and the attempt limit, if chosen, comes with the most lenient test the low type will tolerate: the pass mark is set to sustain incentives rather than to sort. Part (iii) leaves open which of the four settings prevails when the test is not accurate.

When sorting is not leniency-independent, the principal can no longer raise the leniency of the test freely under the attempt limit, and the value of the low type becomes central to her choice. A low type who fails or skips the period-1 test reskills with probability $\rho \equiv F(\frac{k}{1-\beta}) = \min\{\frac{k}{1-\beta}, 1\}$, and is tested again in period 2. Let

$$X(\gamma, k) \equiv \rho\, p_H(\gamma, k)\, w - (1 - \rho)\, p_L(\gamma, k)$$

denote the principal's expected flow value from such a deferred low type when the test has leniency γ: with probability ρ he has reskilled, passes at rate $p_H(\gamma, k)$ and is worth w; otherwise he passes at rate $p_L(\gamma, k)$ and costs 1. Raising the leniency lets more deferred low types through, some of whom have reskilled and some of whom have not, so whether leniency helps or hurts on this margin depends on how many reskill.

Definition 4. *A test is* leniency-rewarding *if and only if* $\partial X(\gamma, k)/\partial\gamma > 0$.

Substituting the passing rates gives $\partial X(\gamma, k)/\partial\gamma = (1 - k)\kappa$, where $\kappa \equiv \rho(1 + w) - 1$ is the expected value of one additional pass of a deferred low type. The slope is therefore constant in γ, and its sign depends only on accuracy: the test is leniency-rewarding if and only if $k > \frac{1-\beta}{1+w}$ Under a leniency-rewarding test, enough deferred low types reskill that the principal gains from every extra pass she grants them.

Proposition 5. *Assume a leniency-rewarding test with accuracy $k < \bar{k}$. There exists a threshold $\bar{\delta}(k) < k + \beta\Phi(\beta, k)$, such that*

(i) If $\delta < \bar{\delta}(k)$, there exist priors $0 < \pi_1(\bar{\varsigma}) < \pi_2(\bar{\varsigma}) < 1$ so that the optimal setting is

$$(\xi^*, \gamma^*) = \begin{cases} (\xi_R, 1), & \pi < \pi_1(\bar{\varsigma}), \\ (\xi_\Lambda, \tilde{\gamma}), & \pi_1(\bar{\varsigma}) \leq \pi < \pi_2(\bar{\varsigma}), \\ (\xi_U, \gamma_U^\star), & \pi \geq \pi_2(\bar{\varsigma}). \end{cases}$$

(ii) If $\delta \geq \bar{\delta}(k)$, there exists a prior $\pi_0(\bar{\varsigma}) \in (0, 1)$ such that the optimal setting is $(\xi_R, 1)$ if $\pi < \pi_0(\bar{\varsigma})$, and $(\xi_U, \gamma_U^\star)$ otherwise.

Proof. See Appendix A.7 □

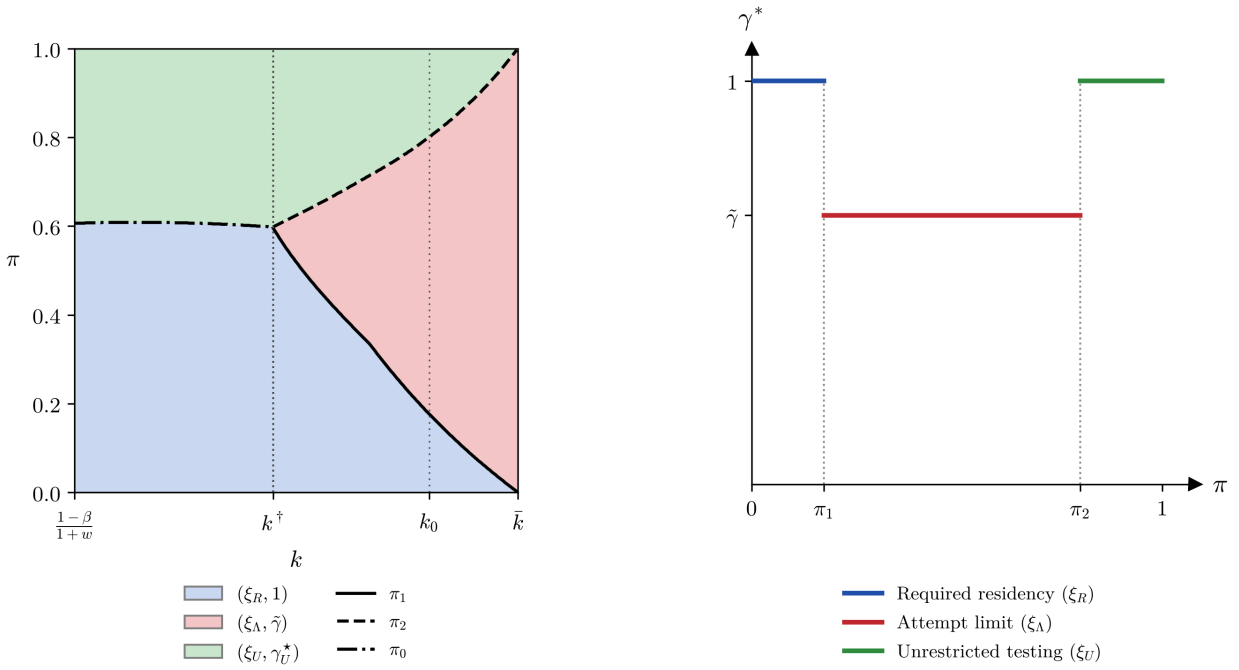
Proposition 5 shows how the optimal test setting changes with the strength of the pool when the test is leniency-rewarding. When competent candidates are scarce, residency is optimal. Delaying everyone costs little because most of the value comes from low types. A leniency-rewarding test also means that a lenient test is worth more on deferred low types: those who reskill pass for sure, and that outweighs the few unqualified candidates who get through. When competent candidates are abundant, unrestricted testing is optimal. With retakes, no competent candidate is ever turned away for good, and that matters most when such candidates make up most of the pool.

The attempt limit can only fill the space between these two menus, and whether it does depends on the principal's patience. Compared with residency, it gains only on competent candidates: a share $k + \beta\Phi(k, \beta)$ of them is hired today rather than all of them tomorrow, which pays off only if $\delta < k + \beta\Phi(k, \beta)$. It loses on low types, who face a stricter test than under residency. Compared with unrestricted testing, it gains by keeping low types out of early attempts, but it permanently fails the competent candidates who miss their single attempt. The threshold $\bar{\delta}(k)$ weighs both comparisons at once. It therefore lies strictly below $k + \beta\Phi(k, \beta)$: beating residency is necessary, but it is not enough. When $\delta \geq \bar{\delta}(k)$, residency gives way directly to unrestricted testing, and no prior makes the attempt limit optimal. Finally, the stop-loss deadline is never optimal in this regime. Since $X(1, k) > 0$, deferring low types to a lenient test is better than screening them out early.

Definition 5. *A test is* leniency-tolerant *if $X(1, k) > 0$*; leniency-intolerant *if $X(1, k) \leq 0$.*

The value of a deferred low type, $X(\gamma, k)$, is always positive under a strict test: when $\gamma = 0$ only those who have reskilled pass, so that $X(0, k) = \rho k w > 0$. The definition

Figure 6: Optimal test setting when the test rewards leniency



Note: The left panel shows the optimal test setting in the plane of accuracy k and prior π; the dotted line at k_0 marks the accuracy of the right panel, and $k^\dagger$ is the lowest accuracy at which the attempt limit is optimal at some prior. The right panel shows the optimal leniency γ^* as a function of the share π of high-skilled candidates, at accuracy k_0, with each segment coloured by the optimal attempt policy. Parameters: $\beta = 0.5$, $w = 1$, $\delta = 0.5$, $k_0 = 0.55$.

asks whether this value survives once the test is fully lenient. How it changes on the way there depends on whether the test is leniency-rewarding. If it is, each additional pass of a deferred low type is a gain, so his value rises with leniency, and the test is always leniency-tolerant. If the test is not leniency-rewarding, each additional pass is a loss, since too few deferred low types reskill, so his value declines as the test becomes more lenient. It may remain non-negative at full leniency, in which case the test is leniency-tolerant and a lenient retake still pays on average, or it may turn negative, in which case giving a deferred low type another chance at a lenient test does more harm than good. A test can therefore be leniency-intolerant only if it is not leniency-rewarding. In terms of accuracy, a test is leniency-intolerant if and only if its accuracy is low enough, $k < k_X$, where

$$k_X \equiv \frac{(2 + w - \beta) - \sqrt{(2 + w - \beta)^2 - 4(1 - \beta)}}{2},$$

with $0 < k_X < \frac{1-\beta}{1+w}$. Leniency-intolerant tests are thus the least accurate, leniency-rewarding tests the most accurate, and tests that are not leniency-rewarding but leniency-tolerant have intermediate accuracy. The distinction matters at high priors, where the test is lenient

because nearly every candidate is competent, and the only remaining question is whether to grant a retake.

Under test that is not leniency-rewarding, whether the attempt limit or residency can be optimal depends on how patient the principal is, through two thresholds:

$$\delta_\Lambda(k,\beta) \equiv \min\left\{\frac{\tilde{\gamma}}{k},\ \frac{\tilde{\gamma}}{\tilde{\gamma}+k(1-k)(1-\tilde{\gamma})}\right\}, \tag{11}$$

$$\delta_R(k,\beta) \equiv \max\left\{k+\beta\Phi(k,\beta),\ \frac{k}{1-k(1-k)}\right\}. \tag{12}$$

Both come from comparisons with unrestricted testing under the strictest test, $\gamma = 0$. Such a test passes no low type in period 1, so it defers low types just as the attempt limit does, and it then retests them strictly, which is what the principal prefers when leniency is costly. The attempt limit therefore cannot do better on low types and must win on competent candidates. Consider those who would fail a strict test. The attempt limit, at its ceiling, hires a share $\tilde{\gamma}$ of them at once; strict unrestricted testing hires a share k of them through the retake, one period later. The attempt limit wins only if $\tilde{\gamma} > \delta k$, which gives the first term of δ_Λ. When the principal is patient enough that residency also beats the attempt limit on competent candidates, the attempt limit must beat both alternatives at the same prior, which gives the second term. Residency, for its part, tests deferred low types with the most lenient test and is therefore worse than both alternatives on low types. It hires every competent candidate, but only in period 2, so it can be optimal only if the principal values this more than the attempt limit's immediate pass rate, $k+\beta\Phi(k,\beta)$, and more than the immediate pass and retake of a strict unrestricted test, $k+\delta k(1-k)$. These two requirements give the two terms of δ_R.

Proposition 6. *Suppose that the test is not leniency-rewarding. If $k < \frac{1-\beta}{1+w}$, there exist thresholds*

$$0 \le \pi_1(\bar{\varsigma}) \le \pi_2(\bar{\varsigma}) \le \pi_3(\bar{\varsigma}) \le \pi_4(\bar{\varsigma}) \le \pi_D(\bar{\varsigma}) \le 1$$

such that:

(i) Test setting $(\xi_\Lambda, \tilde{\gamma})$ is optimal at some prior only if $\delta \le \delta_\Lambda(k,\beta)$, and whenever it is, it is optimal if and only if $\pi_1(\bar{\varsigma}) \le \pi \le \pi_2(\bar{\varsigma})$;

(ii) Test setting $(\xi_R, 1)$ is optimal at some prior only if $\delta \ge \delta_R(k,\beta)$, and whenever it is, it is optimal if and only if $\pi_3(\bar{\varsigma}) \le \pi \le \pi_4(\bar{\varsigma})$;

(iii) Test setting $(\xi_D, 1)$ is optimal if and only if $\pi \ge \pi_D(\bar{\varsigma})$, with $\pi_D(\bar{\varsigma}) < 1$ if and only if the test is leniency-intolerant;

(iv) At any other prior, test setting $(\xi_U, \gamma_U^\star)$ *is optimal.*

If $k = \frac{1-\beta}{1+w}$*, there exists a threshold* $\bar{\pi}(\bar{\varsigma}) \in [0,1]$ *such that test setting* $(\xi_U, \gamma_U^\star)$ *is optimal if and only if* $\pi \geq \bar{\pi}(\bar{\varsigma})$*. For* $\pi \leq \bar{\pi}(\bar{\varsigma})$*, test setting* $(\xi_\Lambda, \tilde{\gamma})$ *is optimal if* $\delta \leq \tilde{p}_H$*, and test setting* $(\xi_R, 1)$ *is optimal if* $\delta \geq \tilde{p}_H$*.*

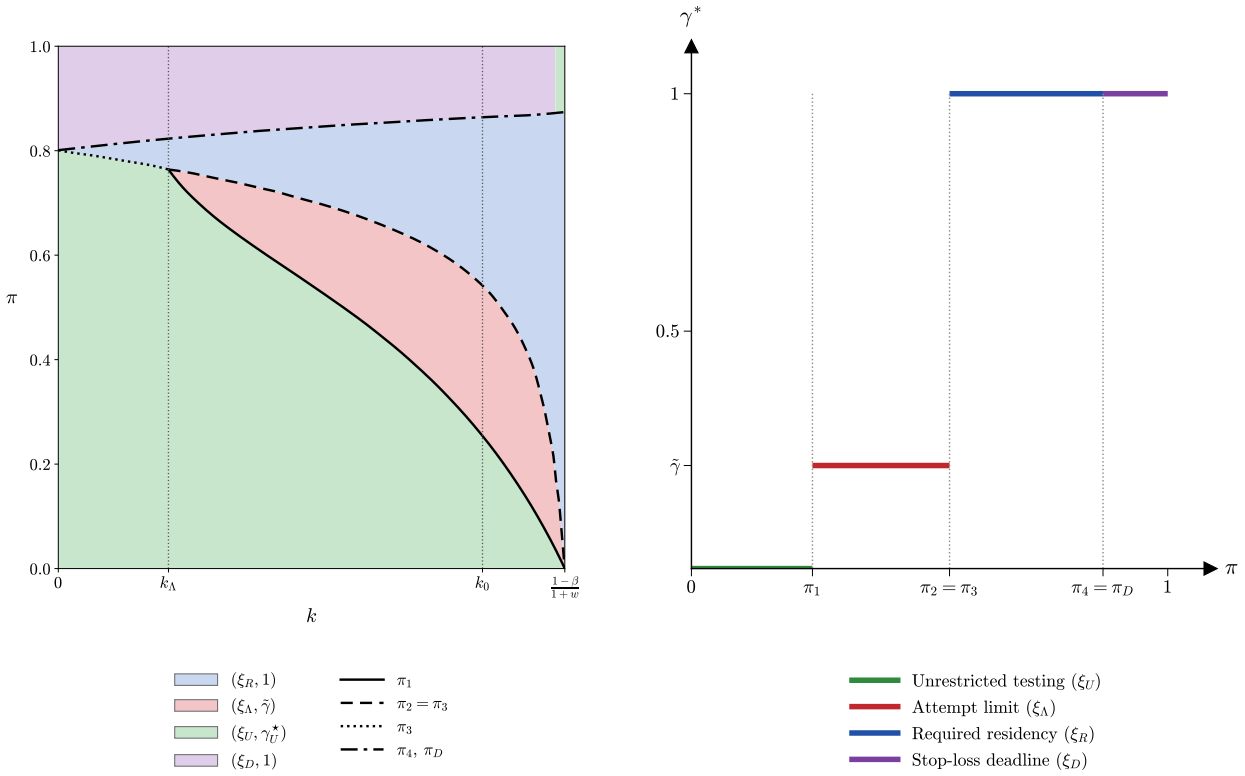
Proof. See Appendix A.8. □

Proposition 6 departs from the previous results in two respects. First, the order of the attempt limit and residency is reversed relative to Proposition 5: the attempt limit now serves the lower priors, and residency, when it is used at all, serves higher ones. The reason lies in how each menu treats a deferred low type. Both defer him and retest him after he has had the chance to reskill, but residency retests him at full leniency, whereas the attempt limit uses the stricter test $\tilde{\gamma}$. When leniency is rewarded, a lenient retest is valuable, because it lets more of the reskilled candidates through; residency therefore dominates when low types are common. When leniency is penalized, the same lenient retest mostly lets through those who did not reskill, so the stricter retest of the attempt limit is the better way to handle a pool dominated by low types. Residency can only prevail where high types matter more, and only if waiting one period to pass them for sure is worth more than passing them now with probability $\tilde{p}_H$, which requires $\delta > \delta_R(k, \beta)$. Second, unlike in Proposition 1, the stop-loss deadline is no longer an intermediate option but the optimal one when competent candidates are abundant, provided the test is leniency-intolerant. With a fixed test, unrestricted testing eventually prevails because it grants failed high types a second attempt. Once leniency is a choice, however, a principal facing mostly high types sets the test at full leniency, so that high types pass at the first attempt and gain nothing from a retake. A second attempt then matters only for the low types who failed, and retesting them leniently is harmful precisely when the test is leniency-intolerant. Cutting off the retake through a stop-loss deadline is then the best allocation: when the screening instrument can be made lenient, limiting attempts is valuable not when competent candidates are scarce, but when they are the norm.

6 Endogenous Accuracy

So far the accuracy of the test has been taken as given. In many settings, however, the tester can make the test more informative, by adding questions, assessors or practical tasks, at a cost. This section therefore lets the principal choose the accuracy k together with the leniency γ and the allocation of testing slots: at the beginning of the game she announces the menu of contracts, the leniency and the accuracy of the test, and the game then unfolds

Figure 7: Optimal test setting when the test does not reward leniency



Note: The left panel shows the optimal test setting in the plane of accuracy k and prior π; the dotted line at k_0 marks the accuracy of the right panel, and k_Λ is the lowest accuracy at which the attempt limit is optimal at some prior. The right panel shows the optimal leniency γ^* as a function of the share π of high-skilled candidates, at accuracy k_0, with each segment coloured by the optimal attempt policy. Parameters: $\beta = 0.9$, $w = 0.25$, $\delta = 0.45$, $k_0 = 0.067$.

as before. Accuracy is costly: every time a candidate sits the test, the principal pays k. The cost is therefore paid per sitting, so that a setting that lets candidates sit the test more often, or earlier, also spends more on testing.

Define a testing policy by contract menu ξ and information design (γ, k), the combination of test leniency and accuracy, and denote by $V_{\theta_1}(\xi(m); \gamma, k)$ the expected utility of a type θ_1 agent defined in (5) as a function of the type m reported by the agent and the testing policy (ξ, γ, k) set by the principal. The set of incentive compatible testing policies is given by

$$\mathcal{T} = \Big\{ (\xi, \gamma, k) \in \{0,1\}^4 \times [0,1]^2 \ : \ \forall \theta_1 \in \{L, H\}, \quad \theta_1 \in \underset{m \in \{L,H\}}{\arg\max}\, V_{\theta_1}(\xi(m); \gamma, k) \Big\}.$$

Let $\mathcal{H}_{\theta_1}(\xi(m), \gamma, k)$ be the principal's continuation value when type θ_1 agent chooses contract

$\xi(m)$ and does not pass the test in period 1, after setting leniency γ and accuracy k:

$$\mathcal{H}_{\theta_1}(\xi(m),\gamma,k) = \xi_2(m)\left[\frac{1}{1-\delta}\int_0^1 \Big(u(z_{\theta_1}(\hat{e}_{\theta_1}(\xi(m),c)))p(z_{\theta_1}(\hat{e}_{\theta_1}(\xi(m),c)))\Big)dc - k\right].$$

The expected welfare of the principal over the set $\mathcal{T}$ of incentive-compatible testing policies is given by

$$W\big(\xi,\gamma,k\big) = \sum_{\theta_1=L,H} Pr(\theta_1)\left[\varphi_{\theta_1}(\xi(\theta_1))\frac{u(\theta_1)}{1-\delta} + [1-\varphi_{\theta_1}(\xi(\theta_1))]\delta\mathcal{H}_{\theta_1}(\xi(\theta_1),\gamma,k) - \xi_1(\theta_1)k\right].$$

Definition 6. *Triple $(\xi^\star,\gamma^\star,k^\star)$ is an optimal testing policy if and only if it maximizes the principal's expected welfare over the est of incentive-compatible testing policies:*

$$(\xi^\star,\gamma^\star,k^\star) \in \underset{(\xi,\gamma,k)\in\mathcal{T}}{\arg\max}\, W(\xi,\gamma,k).$$

The optimal testing policy depends heavily on how the principal treats candidates who are incompetent at the outset. Residency is the menu built for them: no one is tested in the first period, an incompetent candidate decides whether to reskill, and every candidate sits the test once in the second period. Under an information design (γ,k), a deferred incompetent candidate reskills with probability $\rho(k) = \min\{\frac{k}{1-\beta},1\}$, and he is then admitted with probability $p_H(\gamma,k)$ and is worth w. Otherwise he stays incompetent, is admitted with probability $p_L(\gamma,k)$, and costs 1. His expected value to the principal is therefore $X(\gamma,k) = \rho(k)\,p_H(\gamma,k)\,w - \big(1-\rho(k)\big)\,p_L(\gamma,k)$, and the test he sits costs $(1-\delta)k$ in the normalized units of W. The difference between these two terms is his net contribution under residency, and it is what decides whether deferring him can pay.

Definition 7. *Residency is* worthwhile *if and only if*

$$\max_{(\gamma,k)\in[0,1]^2}\Big[X(\gamma,k) - (1-\delta)k\Big] > 0.$$

Residency is thus worthwhile when some information design makes a deferred incompetent candidate a net gain to the principal. The maximum in Definition 7 is never negative, because the design $(\gamma,k)=(0,0)$ passes no one and costs nothing, and so yields exactly zero. It turns out that the maximum can take only two values: zero, or a single strictly positive value that is attained by one design. Which of the two arises depends on the parameters through one inequality.

Lemma 3. *Residency is worthwhile if and only if $w > (1-\delta)(1-\beta)$.*

Proof. See appendix A.11. □

The design $(1, 1-\beta)$ is the best possible treatment of a deferred incompetent candidate. An accuracy of $1-\beta$ is the lowest one that makes him reskill for sure, and full leniency then admits him for sure once he is competent, so he is worth exactly w. Raising accuracy above $1-\beta$ only adds cost. Lowering it saves cost, but some candidates then do not reskill, and a lenient test admits them while they are still incompetent. Lowering leniency fails candidates who did reskill. Lemma 3 therefore says that residency fails to be worthwhile exactly when this best case does not pay: even under the test that makes a deferred incompetent candidate reskill for sure and then passes him, his value w does not cover the cost of the test, $(1-\delta)(1-\beta)$. In that case every informative design loses money on him, and the best the principal can do with him in isolation is not to test at all.

Whether residency is worthwhile concerns incompetent candidates alone, and it is not the same as whether residency is optimal. When residency is not worthwhile, it can still be chosen at some priors, because deferring incompetent candidates may lose less than admitting them.

The rest of the analysis distinguishes the two cases. With accuracy chosen by the principal, the optimal testing policy depends on the primitives $\zeta \equiv (w, \delta, \beta)$: the value of a competent hire, the discount factor, and the agent's patience. Thresholds on the prior depend on ζ, while thresholds on w depend only on (δ, β). Under residency, the test that the principal chooses is fully lenient whenever residency is optimal, and its accuracy balances two forces. A more accurate test induces more deferred incompetent candidates to reskill and filters out more of those who do not, but every candidate sits it, so its cost is borne by the whole pool. The accuracy that balances them is

$$k_R(\pi) = \tfrac{1}{2}\Big[2 + w - \beta - \frac{(1-\delta)(1-\beta)}{1-\pi}\Big],$$

which falls with the prior: the fewer incompetent candidates there are, the less accuracy is worth buying.

Proposition 7. *Suppose that residency is not worthwhile. There exist thresholds* $0 < \underline{\pi}_R(\zeta) \leq \frac{1}{1+w} \leq \bar{\pi}_R(\zeta) < 1$ *such that:*

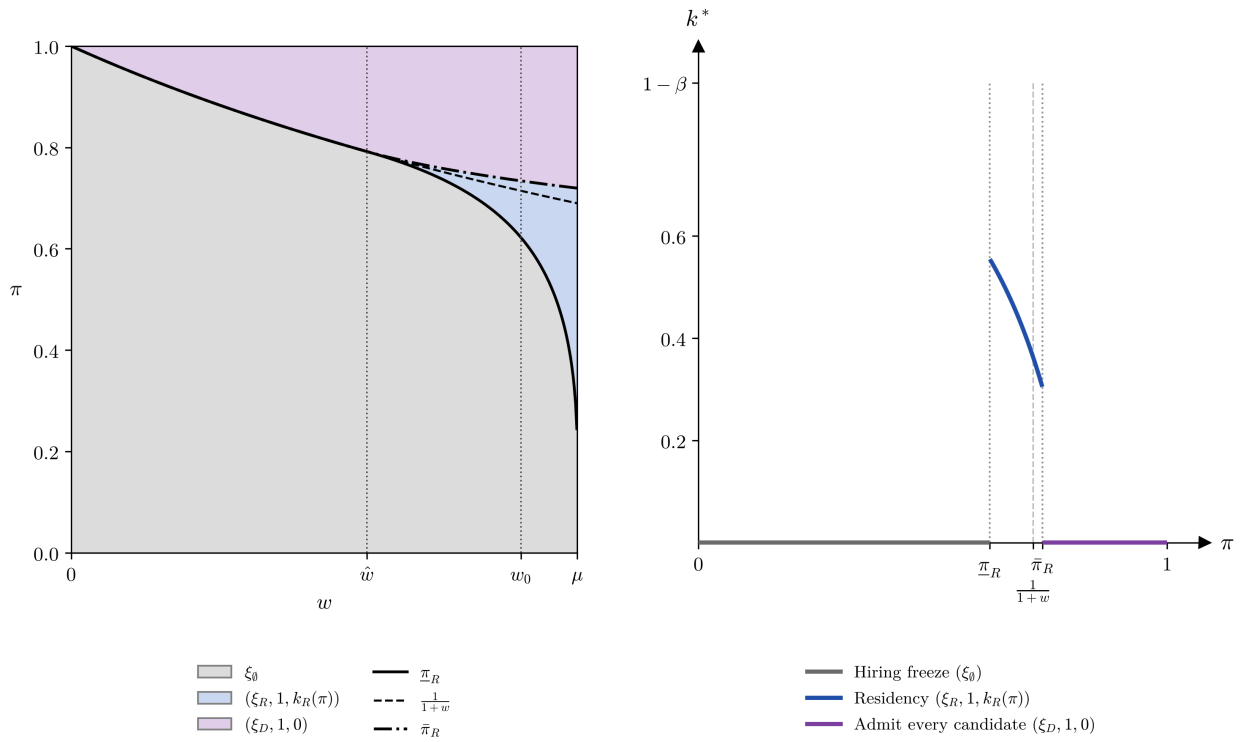
(i) Hiring freeze $\xi_\emptyset$ *is optimal for* $\pi \leq \underline{\pi}_R(\zeta)$*, admitting every candidate,* $(\xi_D, 1, 0)$*, is optimal for* $\pi \geq \bar{\pi}_R(\zeta)$*, and residency* $\big(\xi_R, 1, k_R(\pi)\big)$ *is optimal otherwise;*

(ii) There exists an interior point $\hat{w}$ *such that* $\underline{\pi}_R(\zeta) < \bar{\pi}_R(\zeta)$ *if and only if* $w > \hat{w}$*.*

Proof. See Appendix A.10. □

Proposition 7 shows that, when the value of a competent hire is small relative to the cost of an incompetent one, the principal has little reason to screen. Screening is costly, and what it buys, the chance to tell competent candidates from incompetent ones, is worth little when w is low. Below $\hat{w}$, the principal therefore gives up on screening altogether and treats all candidates alike: she hires no one when the pool is weak, $\pi \leq \frac{1}{1+w}$, and admits everyone without testing when it is strong. The switch happens at the prior at which a candidate admitted blindly is worth exactly zero. Above $\hat{w}$, screening pays only for intermediate pools, where hiring no one and admitting everyone both give up the most. There the principal turns to residency, with a fully lenient test and an accuracy strictly below $1-\beta$. This test does not screen out incompetent candidates directly. Instead it induces some of them to reskill before they are admitted, and so improves the pool she ends up hiring from. When residency is worthwhile, its optimal accuracy may hit a ceiling. Beyond $1-\beta$, every deferred incompetent candidate already reskills, so further accuracy only adds cost. The optimal accuracy under residency is therefore $k_R^*(\pi) \equiv \min\{k_R(\pi), 1-\beta\}$.

Figure 8: Optimal test design when residency is not worthwhile



Note: The left panel shows the optimal test design in the plane of the value w of a high-skilled hire and prior π, for w up to $\mu = (1-\delta)(1-\beta)$; the dotted lines mark $\hat{w}$ and the value w_0 of the right panel. The right panel shows the optimal accuracy k^* as a function of the share π of high-skilled candidates, at $w = w_0$, with each segment coloured by the optimal test design. Parameters: $\beta = 0.1$, $\delta = 0.5$, $w_0 = 0.40$.

Proposition 8. *Suppose that residency is worthwhile. There exist thresholds* $w^* > (1-\delta)\bar{k}$ *and* $\pi_D(\zeta) \in (0,1)$ *such that* $(\xi_D, 1, 0)$ *is optimal if and only if* $\pi \geq \pi_D(\zeta)$*, and for* $\pi \leq \pi_D(\zeta)$ *:*

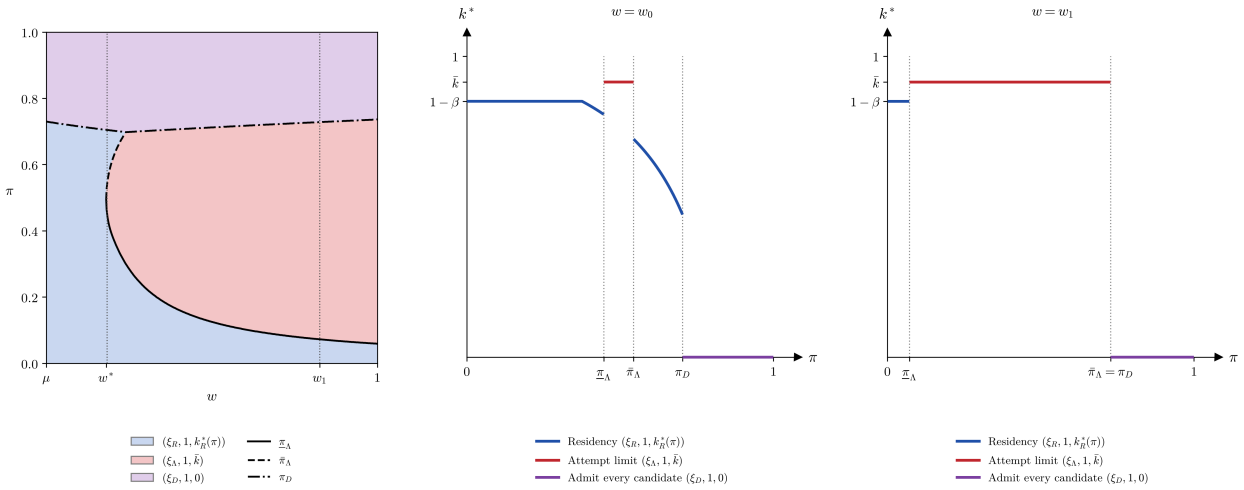
(i) If $w \leq w^*$*, residency* $\left(\xi_R, 1, k_R^*(\pi)\right)$ *is optimal;*

(ii) If $w > w^*$*, there exist thresholds* $0 < \underline{\pi}_\Lambda(\zeta) < \bar{\pi}_\Lambda(\zeta) \leq \pi_D(\zeta)$ *such that* $(\xi_\Lambda, 1, \bar{k})$ *is optimal for* $\underline{\pi}_\Lambda(\zeta) \leq \pi \leq \bar{\pi}_\Lambda(\zeta)$*, and* $\left(\xi_R, 1, k_R^*(\pi)\right)$ *is optimal otherwise.*

Proof. See Appendix A.11 □

Three forces shape Proposition 8. First, once the principal bears the cost of testing, unrestricted testing is never optimal, because restricting testing opportunities and designing the test are complements. When opportunities are scarce, a competent candidate has a single chance to pass, and failing him by mistake loses him. The principal therefore makes the test fully lenient, so that competent candidates pass for sure. Leniency, however, also lets incompetent candidates through, and the principal offsets it with accuracy, which lowers their passing probability without failing anyone competent. Under unrestricted testing this pairing breaks down: a competent candidate who fails can try again, so leniency is less valuable, while accuracy must be paid for at every sitting. Whenever the principal screens, she therefore restricts testing opportunities, through either residency or the attempt limit. Second, the value of a competent hire determines whether the attempt limit is ever used. Its

Figure 9: Optimal test design when residency is worthwhile



Note: The left panel shows the optimal test design in the plane of the value w of a high-skilled hire and prior π, for w above $\mu = (1-\delta)(1-\beta)$; w^* is the lowest value at which the attempt limit is optimal at some prior, and the dotted line marks w_1. The centre and right panels show the optimal accuracy k^* as a function of the share π of high-skilled candidates, at $w = w_0$ and $w = w_1$, with each segment coloured by the optimal test design. Parameters: $\beta = 0.15$, $\delta = 0.5$, $w_0 = 0.531$, $w_1 = 0.9$.

only advantage over residency is speed: competent candidates are hired in the first period

rather than after a period of waiting. Its drawback is cost: incentive compatibility forces an accuracy of $\bar{k}$, above the highest accuracy residency ever uses, $1-\beta$, and every candidate pays for it. The gain from speed grows with w, while the extra cost does not, so the attempt limit is optimal at some prior only when w exceeds w^*.

Third, the prior determines where each policy is used. Under residency, accuracy benefits only incompetent candidates, by inducing them to reskill and by screening out those who do not, while its cost is paid on every candidate. Residency's optimal accuracy therefore falls as the prior rises. When π is low, many candidates are incompetent, and residency buys enough accuracy to make all of them reskill; the attempt limit has few competent candidates to hire early, so residency is optimal. At intermediate priors, the attempt limit hires a larger share of competent candidates without delay, while residency still pays for high accuracy, so the attempt limit can take over. As π rises further, residency lowers its accuracy and its testing costs, whereas the attempt limit keeps paying for accuracy $\bar{k}$ on a pool that hardly needs screening. When w is only slightly above w^*, the attempt limit's lead is narrow, and these savings let residency become optimal again before admitting every candidate takes over. For larger values of w, the attempt limit's advantage is stronger, and residency's accuracy is also higher, so it saves less; the attempt limit then remains optimal until admitting every candidate becomes optimal.

7 An Empirical Prediction

The model predicts where attempt limits should appear. A limit screens candidates only if low-skilled candidates prefer to postpone their attempt and acquire skills. When the test is lenient, they pass often enough by taking it at once, and the limit fails to sort them; when the test is strict, few of them pass, and the limit adds little to the selection that the test already performs. With leniency given, a limit is therefore optimal only at intermediate leniency (Proposition 3). The same holds when the principal chooses leniency together with the attempt policy, as long as the test is not too accurate: whenever a limit is optimal, it comes with leniency $\tilde{\gamma}$, which lies between the leniency $\gamma_U^\star$ chosen under unrestricted testing and the leniency $\gamma = 1$ chosen with a residency requirement or a stop-loss deadline (Propositions 5 and 6). With an accurate test, by contrast, the limit comes with a fully lenient test, so the prediction concerns tests of limited accuracy. A higher passing score makes the test stricter, since it lowers the passing rate of first-time takers and repeaters alike (Fact 1). Across jurisdictions, the model therefore predicts that attempt limits are rare or loose where the passing score is low or high, and tightest in between. This section takes the prediction to the U.S. bar exam.

Because the prediction holds whether leniency is given or chosen, it does not depend on how jurisdictions set the two instruments. U.S. jurisdictions set both, and mostly one at a time: of the 25 policy changes in the sample described below, 17 concern the passing score and 8 the attempt limit; no jurisdiction changed both in the same session, and only Arizona changed both, loosening its limit in 2021 and lowering its passing score in 2023. To compare passing scores across jurisdictions, I use sessions of the Uniform Bar Examination (UBE), which all participating jurisdictions grade on the same 400-point scale while each sets its own passing score. The sample contains 668 sessions in 39 states and the District of Columbia between 2011 and 2024, with one observation per jurisdiction and session, since the passing score and the attempt limit are the same for first-time takers and repeaters.

To relate the attempt policy to the passing score, I estimate

$$Policy_{jts} = \tau_t + \nu_s + \beta_L \, Lenient_{jts} + \beta_S \, Strict_{jts} + \varepsilon_{jts}, \tag{13}$$

where j, t and s index jurisdictions, years and sessions (February or July), τ_t and ν_s are year and session fixed effects, $Lenient_{jts}$ and $Strict_{jts}$ are indicators for passing scores of at most 264 and at least 272, and ε_{jts} is an error term. The outcome $Policy_{jts}$ is either the indicator $Limit_{jts}$ of an attempt limit, in which case equation (13) is a linear probability model, or the tightness of the limit, $Tightness_{jts}$, equal to one over the number of attempts allowed and to zero without a limit. Coefficients β_L and β_S measure the difference from sessions in the same year and season with a passing score between 266 and 270. To test for a non-monotonic relationship without grouping passing scores, I also estimate

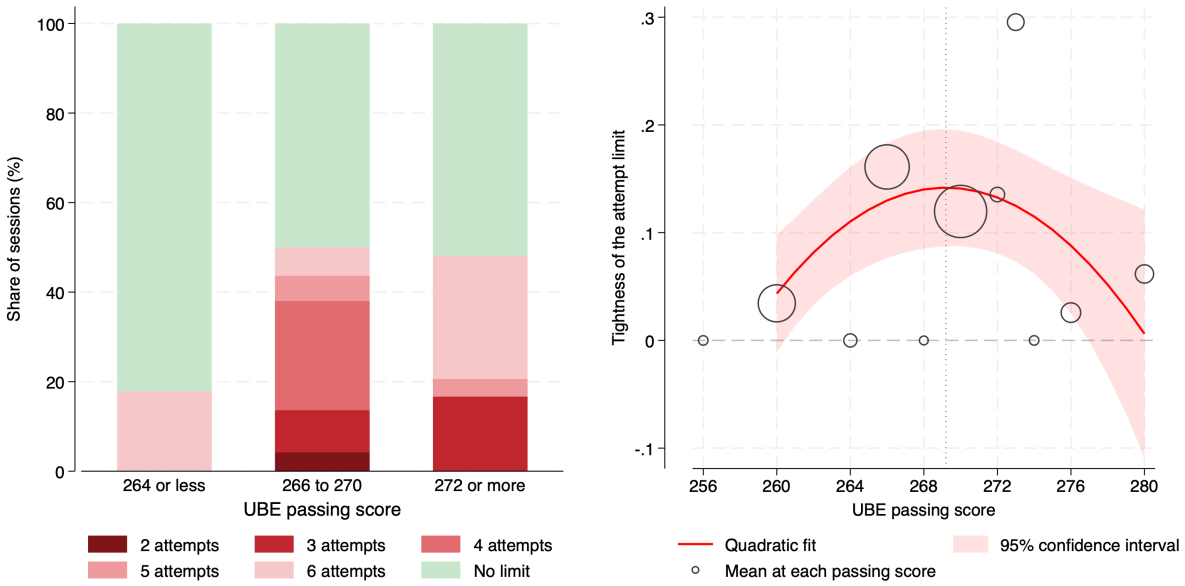
$$Tightness_{jts} = \tau_t + \nu_s + \beta_P \, (Score_{jts} - 268) + \beta_Q \, (Score_{jts} - 268)^2 + \varepsilon_{jts}, \tag{14}$$

where $Score_{jts}$ is the passing score, centred at the middle of the intermediate group, so that the fitted curve peaks at a passing score of $268 - \beta_P/(2\beta_Q)$. Since the passing score and the attempt limit rarely change within a jurisdiction, these regressions omit jurisdiction fixed effects, unlike equation (1), and compare jurisdictions with one another. I estimate both equations by ordinary least squares, with standard errors clustered by jurisdiction. Table 2 reports the estimates, and Figure 10 shows how attempt policies are distributed across passing scores.

Limits are rare where the exam is lenient: sessions with a passing score of 264 or less are 32 percentage points less likely to be held under a limit than sessions with a passing score between 266 and 270, and their tightness is lower by 0.11, as much as the sample mean (columns 1 and 2). Every limit in force at a lenient passing score allows six attempts, whereas limits of four attempts or fewer appear only at passing scores between 266 and 273.

At strict passing scores, of 272 or more, limits are about as common as at intermediate ones but looser, and the decline in tightness is imprecisely estimated. The relationship between tightness and the passing score is therefore not monotonic. A linear term in the passing score is small and insignificant (column 3), whereas the square term in equation (14) is negative and significant, and the fitted curve peaks at a passing score of 269, with a 95 percent confidence interval from 267 to 272 that lies inside the observed range (column 4). The test of Lind and Mehlum (2010), which requires the fitted slope to be positive at the lowest passing score and negative at the highest, rejects a monotonic relationship ($p = 0.010$).[7]

Figure 10: Attempt policy by passing score



Note: Share of sessions held under each attempt policy, for three ranges of the UBE passing score, with limits stacked from the tightest, at the bottom, to no limit, at the top. The number of sessions in each range is reported above its bar; the lenient and strict ranges each comprise eight jurisdictions, and the intermediate range 33. UBE sessions in 39 states and the District of Columbia, 2011–2024.

The evidence is consistent with the prediction, with three qualifications. First, it is correlational: jurisdictions with lenient or strict passing scores may differ in other ways that also shape their attempt policy, including, when the two instruments are chosen together,

[7]The two-lines test of Simonsohn (2018), which fits separate lines below and above a passing score of 268 without imposing a quadratic shape, confirms the increase (one-sided $p = 0.018$) but finds the decrease only at the margin of significance (one-sided $p = 0.052$). Extending the sample to all states and the District of Columbia between 2007 and 2024, with passing scores converted to the 400-point scale, gives the same pattern with a sharper decline: sessions with a lenient passing score are 29 percentage points less likely to be held under a limit ($p = 0.010$), sessions with a strict one 17 percentage points less likely ($p = 0.25$), and the test of Lind and Mehlum (2010) again rejects a monotonic relationship ($p = 0.008$). Probit and logit average marginal effects for column (1) are nearly identical to the linear probability estimates.

the primitives that determine the range of passing scores at which a limit sorts candidates. Second, each tail of the distribution of passing scores comprises only eight jurisdictions, so a few of them weigh heavily in the comparison; at strict passing scores, for instance, the only limit of fewer than five attempts is Arizona's, which it loosened from three attempts to six in 2021. Third, the prediction finds firmer support at lenient passing scores, where limits are rare, than at strict ones, where limits are as common as at intermediate passing scores and only looser.

Table 2: Attempt limits and the passing score

	Attempt limit	Tightness		
	(1)	(2)	(3)	(4)
Lenient passing score	−0.322*	−0.107***		
	(0.172)	(0.038)		
Strict passing score	−0.037	−0.034		
	(0.230)	(0.060)		
Passing score			0.0029	
			(0.0037)	
Passing score − 268				0.0028
				(0.0028)
(Passing score − 268)^2				−0.00116***
				(0.00040)
Peak of the fitted curve				269.2
95% confidence interval of the peak				[266.6, 271.8]
Lind–Mehlum test, p-value				0.010
Year and session FE	Yes	Yes	Yes	Yes
Mean of the dependent variable	0.430	0.109	0.109	0.109
Observations	668	668	668	668
Jurisdictions	40	40	40	40
R^2	0.075	0.101	0.023	0.102

Note: An observation is a jurisdiction and session, for UBE sessions in 39 states and the District of Columbia, 2011–2024. The dependent variable is an indicator for an attempt limit in column (1), a linear probability model, and tightness, equal to one over the number of attempts allowed and to zero without a limit, in columns (2) to (4). Lenient and strict passing score are indicators for passing scores of 264 or less and of 272 or more, so the omitted group is sessions with a passing score between 266 and 270. Passing score is the UBE passing score in points out of 400. The confidence interval of the peak uses the delta method. The test of Lind and Mehlum (2010) rejects a monotonic relationship when the fitted slope is significantly positive at the lowest passing score in the sample, 256, and significantly negative at the highest, 280. All regressions are estimated by ordinary least squares and include year and session fixed effects. Standard errors clustered by jurisdiction are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

8 Conclusion

This paper provides a normative framework for the use of attempt limits in standardized testing. An attempt limit grants each candidate a single attempt but lets him choose when to

use it. High-skilled candidates take the test at once, whereas low-skilled candidates postpone it and acquire skills in the meantime, so that the limit sorts candidates before any of them is tested and without any monetary transfer. The limit is incentive compatible only if the test is not too lenient, and it is costly because high-skilled candidates lose a second chance after an unlucky failure. Whether it is optimal depends on the share of high-skilled candidates, on the leniency and accuracy of the test, and on the value of a high-skilled hire. When leniency and accuracy are given, the attempt limit is optimal only if selection needs are high and leniency is intermediate. When leniency is chosen as well, an accurate test calls for an attempt limit with a fully lenient test, a test of intermediate accuracy calls for an attempt limit, with intermediate leniency, only for intermediate shares of high-skilled candidates, and an inaccurate test may call for unrestricted testing with a fully strict test. When accuracy is costly, the attempt limit is optimal only if a high-skilled hire is valuable enough.

Two features of the U.S. bar exam fit this framework. Attempt limits are associated with higher passing rates among repeaters but not among first-time takers, whereas application fees are unrelated to passing rates, which motivates modelling the limit, rather than a fee, as the screening device (Section 2). Moreover, attempt limits are tightest where the passing score is intermediate and rare where it is low, as the model predicts for a test of limited accuracy (Section 7). This evidence is correlational: it shows that the attempt policies of U.S. jurisdictions fit the model, not that the limit sorts candidates in the way the model describes. Measuring how much of the effect of a limit on repeaters comes from candidates' choices, rather than from the exclusion of those who exhaust their attempts, requires a dynamic model of repeated attempts, which I calibrate to bar exam data in a companion project.

Reality may depart from this framework in an important respect. The tester is often a jurisdiction-specific gatekeeper, such as a state board of bar examiners, while human capital is mobile. A candidate who exhausts his attempts in one jurisdiction may sit for the exam in another, so that an attempt limit set by a single jurisdiction may fail to deter weak candidates from trying again. A jurisdiction that loses the attempt limit as a screening device must then rely more heavily on the other features of the test, such as its passing score. Portable test scores, such as those of the Uniform Bar Examination, may help jurisdictions keep track of attempts and enforce their limits. At the same time, jurisdictions that compete for mobile professionals may have incentives to loosen their test settings, lowering passing scores or relaxing attempt limits to attract candidates. How competition among gatekeepers shapes attempt limits and the other features of the test is the subject of ongoing work.

A Appendix

A.1 Proof of Lemma 1

I split the proof into two parts: first, I show that only (χ_2, χ_1) can be a separating incentive-compatible menu; secondly, I illustrate its incentive compatibility condition. Before starting, notice that the continuation value of a period-1 low type agent choosing any contract χ_n with $\chi_n(2) = 1$ is equal to

$$\mathcal{V} \equiv \int_0^1 \max\Big\{ \frac{p_L}{1-\beta},\ \frac{p_H}{1-\beta} - c \Big\}\, dc = \frac{p_L}{1-\beta} + \int_0^1 \max\Big\{ \frac{\Delta p_\theta}{1-\beta} - c,\, 0 \Big\}\, dc. \tag{A.1}$$

Step 1. *Only (χ_2, χ_1) can be a separating incentive-compatible menu.*

Proof. Two observations eliminate most menus. First, any menu that assigns the no-test contract χ_0 to the high type violates his constraint, since $V_H(\chi_0) = 0$ while every other contract yields $V_H(\chi_n) > 0$, for $n > 0$. Second, any menu that assigns χ_0 to the low type violates his constraint for the same reason, as $V_L(\chi_0) = 0$ while the contract offered to the high type yields the low type a strictly positive payoff. Together these rule out the six menus involving contract χ_0.

Among the remaining six separating menus, which assign χ_2, χ_1, and χ_3 across the two types, the high type's constraint requires that his contract deliver him at least as much as the low type's. Since $V_H(\chi_3) \geq V_H(\chi_1) \geq V_H(\chi_2)$, any menu in which the high type is offered χ_2 while the low type is offered χ_1 or χ_3, or is offered χ_1 while the low type is offered χ_3, violates his constraint. This eliminates (χ_1, χ_2), (χ_3, χ_2), and (χ_3, χ_1).

Menus (χ_2, χ_3) and (χ_1, χ_3) instead fail the low type's constraint, because χ_3 permits testing in both periods, so it weakly dominates every other contract. In either case mimicking the high type yields him $V_L(\chi_3) = \frac{p_L}{1-\beta} + (1-p_L)\beta\mathcal{V}$, whereas truthful behaviour yields him respectively $V_L(\chi_2) = \beta\mathcal{V}$ and $V_L(\chi_1) = \frac{p_L}{1-\beta}$. In the case of menu (χ_2, χ_3), the low type pretends to be the high type because

$$V_L(\chi_3) - V_L(\chi_2) = \frac{p_L}{1-\beta} + (1-p_L)\beta\mathcal{V} - \beta\mathcal{V} = p_L\Big(\frac{1}{1-\beta} - \beta\mathcal{V}\Big) \geq 0.$$

Indeed, for every $c \in [0,1]$ it holds that $\max\big\{\frac{p_L}{1-\beta},\, \frac{p_H}{1-\beta} - c\big\} \leq \frac{p_H}{1-\beta}$, implying $\mathcal{V} \leq \frac{p_H}{1-\beta}$. Hence $\beta\mathcal{V} \leq \frac{\beta p_H}{1-\beta} < \frac{1}{1-\beta}$ since $\beta < 1$ and $p_H \leq 1$. Intuitively, χ_3 gives the low type everything χ_2 gives him plus a period-1 attempt, and that attempt is worth having because an immediate pass beats a discounted one. In the case of menu (χ_1, χ_3), the low type pretends to be the

high type because

$$V_L(\chi_3) - V_L(\chi_1) = \frac{p_L}{1-\beta} + (1-p_L)\beta\mathcal{V} - \frac{p_L}{1-\beta} = (1-p_L)\beta\mathcal{V} \geq 0,$$

since $\mathcal{V} \geq \frac{p_L}{1-\beta} > 0$ and $p_L < 1$. Here χ_3 gives the low type the same period-1 attempt as χ_1 plus a period-2 continuation he would otherwise forgo. This leaves (χ_2, χ_1) as the only candidate. □

Step 2. *The menu $\xi = (\chi_2, \chi_1)$ is incentive compatible if and only if*

$$\gamma \leq \frac{\beta}{1-k} \int_0^1 \phi(c; k, \beta)\, dc \tag{A.2}$$

Proof. The high-type incentive compatibility constraint, $V_H(\chi_1) \geq V_H(\chi_2)$, holds because

$$V_H(\chi_1) - V_H(\chi_2) = \frac{p_H}{1-\beta} - \frac{\beta p_H}{1-\beta} = \frac{p_H(1-\beta)}{1-\beta} = p_H \;\geq\; 0.$$

The low-type constraint is $V_L(\chi_2) \geq V_L(\chi_1)$: by equality (A.1), this is equivalent to

$$\frac{\beta p_L}{1-\beta} + \beta \int_0^1 \max\left\{\frac{\Delta p_\theta}{1-\beta} - c, 0\right\} dc \;\geq\; \frac{p_L}{1-\beta} \iff \beta \int_0^1 \phi(c; k, \beta)\, dc \;\geq\; p_L.$$

Dividing by $1-k$ yields (9). □

Together, Steps 1 and 2 establish Lemma 1. □

A.2 Proof of Proposition 1

Let $\rho \equiv F\left(\frac{\Delta p_\theta}{1-\beta}\right)$ be the probability that a low-skilled agent reskills, and denote by $X \equiv \rho\, p_H w - (1-\rho)\, p_L$ the expected value to the principal of a low-skilled agent who holds a period-2 testing slot: with probability ρ he reskills, passes at rate p_H and is worth w; otherwise he stays low-skilled, passes at rate p_L and costs 1. Note that $-p_L \leq X \leq p_H w$.

Step 1: welfare under each menu. Since $\xi_\Lambda \notin \Xi$, the candidates are $\xi_\emptyset$, ξ_R, ξ_D and ξ_U. Under $\xi_\emptyset$ nobody is tested, so $W(\xi_\emptyset) = 0$. The expected welfare of the principal under ξ_D,

ξ_R, and ξ_U is respectively

$$W(\xi_D) = \frac{\pi\, p_H w - (1-\pi)p_L}{1-\delta},$$
$$W(\xi_R) = \frac{\delta\big[\pi\, p_H w + (1-\pi)X\big]}{1-\delta},$$
$$W(\xi_U) = \frac{\big[\pi\, p_H w\big(1+(1-p_H)\delta\big) + (1-\pi)\big(-p_L + (1-p_L)\delta X\big)\big]}{1-\delta}.$$

Each of these is an affine function of π. Denote by ψ_ξ the slope of $W(\xi)$ in π: $\psi_\emptyset = 0$,

$$\psi_D = \frac{p_H w + p_L}{1-\delta},$$
$$\psi_R = \frac{\delta(1-\rho)(p_H w + p_L)}{1-\delta},$$
$$\psi_U = \frac{[p_H w\big(1+(1-p_H)\delta\big) + p_L - (1-p_L)\delta X]}{1-\delta}.$$

Lemma 4. *If menu ξ is optimal at some π_1 and menu ξ' is optimal at some $\pi_2 > \pi_1$, then $\psi_{\xi'} \geq \psi_\xi$. Consequently each menu is optimal on at most one interval, and the optimal menus appear in increasing order of slope as π rises.*

Proof. Each $W(\xi)$ is a straight line in π, and at each π the principal selects the highest one. Optimality at π_1 and at π_2 gives $W(\xi)(\pi_1) \geq W(\xi')(\pi_1)$ and $W(\xi')(\pi_2) \geq W(\xi)(\pi_2)$, so the affine function $W(\xi') - W(\xi)$ is non-positive at π_1 and non-negative at π_2, and therefore has non-negative slope: $\psi_{\xi'} \geq \psi_\xi$. Hence, as π rises, the optimal menu can only be replaced by one with a weakly larger slope, and each menu is optimal on at most one interval. □

Step 2: ordering the menus. First, $\psi_D - \psi_R = (p_H w + p_L)(1-\delta+\delta\rho)(1-\delta)^{-1} > 0$; hence $0 \leq \psi_R < \psi_D$ and, by Lemma 4, ξ_R precedes ξ_D.

For the last comparison, let $\Delta(\pi) \equiv W(\xi_U) - W(\xi_D)$ be an affine function with slope

$$\psi_U - \psi_D = \frac{\delta[p_H w(1-p_H) - (1-p_L)X]}{1-\delta},$$

whose sign is ambiguous. There are two cases. If $\psi_U > \psi_D$, then by Lemma 4 the menus appear in the order $\xi_\emptyset, \xi_R, \xi_D, \xi_U$. If instead $\psi_U \leq \psi_D$, note that $\Delta(1) = \delta\, p_H w(1-p_H)(1-\delta)^{-1} > 0$, since at $\pi = 1$ the two menus differ only in the high type's retake, which ξ_U grants; as Δ is affine with non-positive slope, $\Delta(\pi) = \Delta(1) - (1-\pi)\big(\psi_U - \psi_D\big) \geq \Delta(1) > 0$ for every $\pi \in [0,1]$, so ξ_U strictly dominates ξ_D and the ξ_D region is empty. In either case the order $\xi_\emptyset, \xi_R, \xi_D, \xi_U$ is respected.

Step 3: the efficiency threshold. $W(\xi_R)$ is affine with $\psi_R \geq 0$ and $W(\xi_R) = \delta\, p_H w(1-\delta)^{-1} > 0$ at $\pi = 1$, so it crosses zero at most once, at $\bar{\pi}_\emptyset = -X(p_H w - X)^{-1}$, which lies in $(0,1)$ when $X < 0$ and is non-positive when $X \geq 0$. Testing is therefore efficient if and only if $\pi \geq \bar{\pi}_\emptyset$. Letting π_R and π_D denote the two remaining crossing points, we obtain $\bar{\pi}_\emptyset < \pi_R \leq \pi_D$ with the stated partition. □

A.3 Proof of Proposition 2

Step 1: welfare under the attempt limit. Since $\xi_\Lambda \in \Xi$, the high type tests in period 1 and the low type in period 2, so

$$W(\xi_\Lambda) = \frac{\pi\, p_H w + (1-\pi)\,\delta X}{1-\delta}, \qquad \psi_\Lambda = \frac{p_H w - \delta X}{1-\delta},$$

with ρ and X as in Appendix A.2. The welfare of the remaining menus is unchanged.

Step 2: ξ_R and ξ_D are dominated. Two direct computations give $W(\xi_\Lambda) - W(\xi_R) = \pi p_H w > 0$ and $W(\xi_\Lambda) - W(\xi_D) = (1-\pi)[\delta\rho\, p_H w + p_L\big(1-\delta(1-\rho)\big)](1-\delta)^{-1} > 0$. Intuitively, ξ_Λ gives each type the single attempt that suits it; compared to ξ_R it tests the high type a period earlier, and relative to ξ_D it defers the low type to the period in which he may reskill.

Hence, only $\xi_\emptyset$, ξ_Λ and ξ_U remain, and their slopes are strictly ordered: first of all, $\psi_\Lambda = (p_H w - \delta X)(1-\delta)^{-1} \geq p_H w(1-\delta)(1-\delta)^{-1} = p_H w > 0$ because $p_H w \geq X$; secondly, $\psi_U - \psi_\Lambda = [\delta\, p_H w(1-p_H) + p_L(1+\delta X)](1-\delta)^{-1} > 0$, since $X \geq -p_L$ and $\delta p_L < 1$ imply $1 + \delta X > 0$. Hence $0 < \psi_\Lambda < \psi_U$, and by Lemma 4 the optimal menu is $\xi_\emptyset$, then ξ_Λ, then ξ_U as π rises.

Step 3: the middle region is non-degenerate. Evaluating at the endpoints $\pi = 0$ and $\pi = 1$, the welfare difference $W(\xi_\Lambda) - W(\xi_U)$ is respectively $p_L(1+\delta X)(1-\delta)^{-1} > 0$ and $-\delta\, p_H w(1-p_H)(1-\delta)^{-1} < 0$. So ξ_Λ beats ξ_U at $\pi = 0$ and loses at $\pi = 1$; since the difference is affine, they cross exactly once, at some $\pi_\Lambda \in (0,1)$. Furthermore $W(\xi_\Lambda)$ is strictly increasing with $W(\xi_\Lambda) = p_H w(1-\delta)^{-1} > 0$ at $\pi = 1$, so it crosses zero at most once, at $\underline{\pi}_\emptyset = -\delta X(p_H w - \delta X)^{-1}$, which lies in $(0,1)$ when $X < 0$ and is non-positive when $X \geq 0$. Testing is therefore efficient if and only if $\pi \geq \underline{\pi}_\emptyset$. Since ξ_Λ lies above both $\xi_\emptyset$ and ξ_U immediately to the left of π_Λ, we have $\underline{\pi}_\emptyset < \pi_\Lambda$, which gives the stated partition.

Step 4: comparison with the pooling threshold. Let $\bar{\pi}_\emptyset \in [0,1]$ be the lowest prior at which some pooling menu, ξ_R, ξ_D or ξ_U, yields non-negative welfare; it exists because $W(\xi_D) = p_H w(1-\delta)^{-1} > 0$ at $\pi = 1$ and each $W(\xi)$ is continuous in π. At $\pi = \bar{\pi}_\emptyset$ that

pooling menu is still available to the principal, so her maximal welfare is non-negative there and testing is efficient. By Step 3, testing is efficient if and only if $\pi \geq \underline{\pi}_\emptyset$; hence $\underline{\pi}_\emptyset \leq \bar{\pi}_\emptyset$. $\square$

A.4 Proof of Lemma 2

Throughout, $p_H = 1-(1-k)(1-\gamma)$ and $p_L = (1-k)\gamma$, so $\partial p_L/\partial\gamma = \partial p_H/\partial\gamma = 1-k \geq 0$, and $\partial X/\partial\gamma = (1-k)[\rho w - (1-\rho)]$ both constant in γ. Recall that ρ is free of γ.

Step 1: the two efficiency thresholds and $\underline{\pi}_P$. Differentiating the closed forms of Appendices A.2 and A.3, each derivative is a fraction with a squared denominator, so only the numerators matter. These are

$$\begin{aligned} \underline{\pi}_\emptyset &: \ \delta k w (1-k)(1-\rho), \\ \bar{\pi}_\emptyset &: \ k w (1-k)(1-\rho), \\ \underline{\pi}_P &: \ k w (1-k)(1-\delta)\big[1-\delta(1-\rho)\big], \end{aligned}$$

each a product of non-negative factors, since $0 < k < 1$, $w > 0$, $\rho \leq 1$, $\delta < 1$ and $1-\delta(1-\rho) > 0$. Hence all three derivatives are non-negative. They vanish when $\rho = 1$, in which case $X = p_H w$ is constant in γ and the thresholds do not move.

Step 2: a common form for π_S and $\bar{\pi}_P$. Write $A \equiv \delta p_H w (1-p_H)$. Collecting terms,

$$W(\xi_\Lambda) - W(\xi_U) = (1-\pi)N - \pi A, \qquad W(\xi_D) - W(\xi_U) = (1-\pi)M - \pi A,$$

with $N = p_L(1+\delta X)$ and $M = -\delta(1-p_L)X$, so that

$$\pi_S = \frac{N}{A+N}, \qquad \bar{\pi}_P = \frac{M}{A+M}, \qquad \frac{\partial}{\partial\gamma}\left(\frac{Z}{A+Z}\right) = \frac{AZ' - ZA'}{(A+Z)^2}.$$

It therefore suffices to sign $AZ' - ZA'$ for $Z \in \{N, M\}$.

Step 3: the threshold π_S. Dividing $AN' - NA'$ by $\delta w (1-k)^2 > 0$ gives the quadratic $\alpha_0 + \alpha_1\gamma + \alpha_2\gamma^2$ with

$$\alpha_0 = k\big(1+\delta k\rho w\big), \qquad \alpha_1 = 2\delta k(1-k)\big[\rho(1+w)-1\big], \qquad \alpha_2 = (1-k)B,$$

where $B \equiv 1 + \delta\rho w(1-k) + \delta(2k-1)(1-\rho)$. Clearly $\alpha_0 > 0$. For α_2, if $k \geq \frac{1}{2}$ every term of B is non-negative, while if $k < \frac{1}{2}$ then $(1-2k)(1-\rho) \leq 1$ gives $B \geq 1-\delta > 0$; hence $\alpha_2 > 0$.

If $\alpha_1 \geq 0$ the quadratic is positive on $\gamma \geq 0$ and we are done. If $\alpha_1 < 0$, then $\rho(1+w) < 1$, so $\big[1-\rho(1+w)\big]^2 < 1$ and hence

$$\alpha_1^2 = 4\delta^2 k^2 (1-k)^2 \big[1-\rho(1+w)\big]^2 < 4\delta^2 k^2 (1-k)^2.$$

Moreover $\alpha_0 \geq k$ always, since $\delta k \rho w \geq 0$, so $4\alpha_0\alpha_2 \geq 4k(1-k)B$. It therefore suffices that $\delta^2 k(1-k) < B$. For $k \geq \frac{1}{2}$ this holds since $B \geq 1 > \frac{1}{4} \geq \delta^2 k(1-k)$; for $k < \frac{1}{2}$ the factor $2k-1$ is negative, so, dropping the non-negative term $\delta\rho w(1-k)$ and using $1-\rho \leq 1$,

$$B \;\geq\; 1-\delta(1-2k)(1-\rho) \;\geq\; 1-\delta(1-2k) \;=\; 1-\delta+2\delta k.$$

On the other side, $\delta < 1$ and $1-k < 1$ give $\delta^2 k(1-k) < \delta k$, which is smaller than $2\delta k$, which in turn is at most $1-\delta+2\delta k$ because $\delta < 1$. Hence $\delta^2 k(1-k) < B$, the discriminant $\alpha_1^2 - 4\alpha_0\alpha_2$ is negative and the quadratic is strictly positive.

Step 4: the threshold $\bar{\pi}_P$. This threshold is relevant only where $X < 0$, which is where $M > 0$. Put $s \equiv 1-p_H = (1-k)(1-\gamma)$, which ranges over $[0, 1-k]$ as γ runs over $[0,1]$. Dividing $AM' - MA'$ by $\delta^2 k w(1-k) > 0$ gives

$$D(1-2s) + E\,s^2, \qquad D \equiv (1-k)(1-\rho) - \rho w, \qquad E \equiv 2(1-\rho) - \rho w,$$

and $E - D = (1-\rho)(1+k) \geq 0$, so $E \geq D$.

I first show $D \geq 0$. From $X < 0$ I have $\rho p_H w < (1-\rho)p_L$, hence $\rho w < (1-\rho)\frac{p_L}{p_H}$ and

$$D > (1-\rho)\left[(1-k) - \frac{p_L}{p_H}\right] = (1-\rho)\,\frac{k\,(1-p_H)}{p_H} \;\geq\; 0,$$

using $p_L = p_H - k$ and $p_H \leq 1$. If $s \leq \frac{1}{2}$ both terms of $D(1-2s) + Es^2$ are non-negative. If $s > \frac{1}{2}$, then $D(1-2s) \leq 0$ and $E\,s^2 \geq D\,s^2 \geq D(2s-1)$: the first inequality by $E \geq D \geq 0$ and the second because $s^2 - (2s-1) = (s-1)^2 \geq 0$. In both cases the expression is non-negative.

Hence $AZ' - ZA' \geq 0$ for $Z \in \{N, M\}$, and all five thresholds are non-decreasing in γ. □

A.5 Proof of Proposition 3

Throughout, $\bar{\varsigma}$ is fixed with $k \in (0,1)$, the thresholds of Proposition 2 are written as functions of γ, and ρ and X are as in Appendix A.2.

Step 1: incentive compatibility and the bound π_Λ. By Lemma 1, $\xi_\Lambda \in \Xi$ if and only if $\beta\Phi(k,\beta) \geq p_L$, where Φ does not depend on γ because $\Delta p_\theta = k$. Substituting $p_L = (1-k)\gamma$, this holds if and only if $\gamma \leq \tilde{\gamma}(k,\beta)$; since $\gamma \in [0,1]$, the attempt limit is incentive compatible exactly on $[0,\bar{\gamma}]$. On this range $\pi_S(\gamma,\bar{\varsigma}) = N/(A+N)$, with $A \equiv \delta p_H w(1-p_H) \geq 0$ and $N \equiv p_L(1+\delta X) \geq 0$ not both zero, since that would require $p_H = 1$ and $p_L = 0$, that is $k = 1$. Hence π_S is well defined on $[0,\bar{\gamma}]$, and $\pi_\Lambda = \pi_S(\bar{\gamma},\bar{\varsigma})$ is well defined.

Step 2: the set $\mathcal{I}(\pi)$. Fix $\pi \leq \pi_\Lambda$, recalling that $\pi \in (0,1)$, and let

$$\mathcal{I}(\pi) \;\equiv\; \big\{\, \gamma \in [0,\bar{\gamma}] \;:\; \underline{\pi}_\emptyset(\gamma,\bar{\varsigma}) \leq \pi \leq \pi_S(\gamma,\bar{\varsigma}) \,\big\},$$

the set of leniencies at which the attempt limit is preferred both to the hiring freeze and to unrestricted testing. Every $\gamma \in [0,\bar{\gamma}]$ satisfies $\xi_\Lambda \in \Xi$, so Proposition 2 applies on this range and implies that $\xi^\star = \xi_\Lambda$ if and only if $\gamma \in \mathcal{I}(\pi)$; for $\gamma \in (\bar{\gamma},1]$ the attempt limit is not incentive compatible and cannot be optimal.

Step 3: $\mathcal{I}(\pi)$ is a non-empty closed interval in $(0,\bar{\gamma}]$. At $\gamma = 0$ we have $p_L = 0$, so $N = 0$, while $A = \delta k w(1-k) > 0$; hence $\pi_S(0,\bar{\varsigma}) = 0$. By Lemma 2, π_S is continuous and strictly increasing on $[0,\bar{\gamma}]$, so it maps $[0,\bar{\gamma}]$ one-to-one onto $[0,\pi_\Lambda]$. Since $\pi > 0 = \pi_S(0,\bar{\varsigma})$ and $\pi \leq \pi_\Lambda$, there is a unique $\gamma_S(\pi) \in (0,\bar{\gamma}]$ with $\pi_S(\gamma_S(\pi),\bar{\varsigma}) = \pi$, and

$$\big\{\gamma \in [0,\bar{\gamma}] :\; \pi \leq \pi_S(\gamma,\bar{\varsigma})\big\} = \big[\gamma_S(\pi),\, \bar{\gamma}\big].$$

Proposition 2 orders the two thresholds, so $\underline{\pi}_\emptyset(\gamma_S(\pi),\bar{\varsigma}) < \pi_S(\gamma_S(\pi),\bar{\varsigma}) = \pi$. By Lemma 2, $\underline{\pi}_\emptyset$ is continuous and non-decreasing, so the set of $\gamma \in [0,\bar{\gamma}]$ with $\underline{\pi}_\emptyset(\gamma,\bar{\varsigma}) \leq \pi$ is a closed interval $[0,\gamma_\emptyset(\pi)]$, non-empty since it contains $\gamma_S(\pi)$, with $\gamma_\emptyset(\pi) \geq \gamma_S(\pi)$. Hence $\mathcal{I}(\pi) = \big[\gamma_S(\pi),\ \gamma_\emptyset(\pi)\big]$, a non-empty closed interval contained in $(0,\bar{\gamma}]$.

Step 4: necessity of $\pi \leq \pi_\Lambda$. Suppose $\pi > \pi_\Lambda$. If $\gamma > \bar{\gamma}$ then $\xi_\Lambda \notin \Xi$ by Step 1, so ξ_Λ cannot be optimal. If instead $\gamma \leq \bar{\gamma}$, then $\pi > \pi_\Lambda = \pi_S(\bar{\gamma},\bar{\varsigma}) \geq \pi_S(\gamma,\bar{\varsigma})$ by Lemma 2, and Proposition 2 then gives $\xi^\star = \xi_U \neq \xi_\Lambda$. Hence the attempt limit is optimal for no leniency. □

A.6 Proof of Proposition 4

Throughout, $k \in (0,1)$, $p_H = 1-(1-k)(1-\gamma)$ and $p_L = (1-k)\gamma$, and ρ and X are as in Appendix A.2. Since $\Delta p_\theta = k$, the reskilling probability $\rho = \min\{\frac{k}{1-\beta}, 1\}$ does not depend

on γ, so p_H, p_L and X are affine in γ. Let $\bar{W}(\xi,\gamma) \equiv (1-\delta)\,W(\xi,\gamma)$; since $1-\delta>0$, $\bar{W}$ ranks test settings exactly as W does. The five menus yield $\bar{W}(\xi_\emptyset,\gamma)=0$,

$$\begin{aligned}
\bar{W}(\xi_R,\gamma) &= \delta\big[\pi p_H w + (1-\pi)X\big],\\
\bar{W}(\xi_D,\gamma) &= \pi p_H w - (1-\pi)p_L,\\
\bar{W}(\xi_\Lambda,\gamma) &= \pi p_H w + (1-\pi)\delta X,\\
\bar{W}(\xi_U,\gamma) &= \pi p_H w\big[1+(1-p_H)\delta\big] + (1-\pi)\big[-p_L+(1-p_L)\delta X\big].
\end{aligned}$$

The first four are affine in γ, while the last is quadratic. The pooling menus are incentive compatible at every leniency, and the attempt limit if and only if $\gamma \le \tilde{\gamma}$.

Step 1: testing is efficient. At $\gamma=0$ we have $p_L=0$ and $p_H=k$, so $\bar{W}(\xi_D,0)=\pi k w > 0=\bar{W}(\xi_\emptyset,\gamma)$. Since $(\xi_D,0)\in\Pi$, the hiring freeze is strictly dominated at every leniency, and testing is efficient.

Step 2: the strictest test favours unrestricted testing. At $\gamma=0$ we also have $X=\rho k w$, so

$$\begin{aligned}
\bar{W}(\xi_U,0)-\bar{W}(\xi_R,0) &= \pi k w(1-\delta k) > 0,\\
\bar{W}(\xi_U,0)-\bar{W}(\xi_D,0) &= \delta k w\big[\pi(1-k)+(1-\pi)\rho\big] > 0,\\
\bar{W}(\xi_U,0)-\bar{W}(\xi_\Lambda,0) &= \pi\delta k w(1-k) > 0.
\end{aligned}$$

Hence none of ξ_R, ξ_D and ξ_Λ is optimal at $\gamma=0$.

Step 3: the incentive ceiling. Ceiling $\frac{\beta\Phi(k,\beta)}{1-k}$ is continuous and strictly increasing in k on $(0,1)$, since $\Phi(k,\beta)$ is positive and non-decreasing in k and $\frac{1}{1-k}$ is strictly increasing. At $k=1-\beta$ we have $\frac{k}{1-\beta}=1$, so $\Phi=\frac{1}{2}$ and $\tilde{\gamma}=\frac{\beta/2}{\beta}=\frac{1}{2}$; hence $\tilde{\gamma}<1$ for every $k\le 1-\beta$. For $k>1-\beta$, $\Phi(k,\beta)=\frac{k}{1-\beta}-\frac{1}{2}$, and

$$\tilde{\gamma}\ge 1 \iff \beta\Big(\frac{k}{1-\beta}-\frac{1}{2}\Big)\ge 1-k \iff \frac{k}{1-\beta}\ge 1+\frac{\beta}{2} \iff k\ge(1-\beta)\Big(1+\frac{\beta}{2}\Big).$$

Define

$$\bar{k} \;\equiv\; (1-\beta)\Big(1+\frac{\beta}{2}\Big) \;=\; 1-\frac{\beta(1+\beta)}{2},$$

the accuracy at which the incentive ceiling reaches one. Since $\bar{k}-(1-\beta)=\frac{\beta(1-\beta)}{2}>0$ and $\bar{k}<1$, the threshold lies in $(1-\beta,1)$, and $\tilde{\gamma}\ge 1$ if and only if $k\ge\bar{k}$. In that case $k>1-\beta$, so $\rho=1$ and $X=p_H w\ge 0$.

Step 4: accurate tests, $k \geq \bar{k}$. By Step 3, $\gamma = 1$ is feasible for the attempt limit, and $\rho = 1$. At $\gamma = 1$ we have $p_H = 1$, $p_L = 1 - k$ and $X = w$, so $\bar{W}(\xi_\Lambda, 1) = \pi w + (1 - \pi)\delta w$. I show that this strictly exceeds the welfare of every other test setting.

Residency and deadline. Since their welfare is affine in γ, it suffices to compare at $\gamma \in \{0, 1\}$. With $\rho = 1$,

$$\begin{aligned}
\bar{W}(\xi_\Lambda, 1) - \bar{W}(\xi_R, 1) &= (1 - \delta)\pi w > 0, \\
\bar{W}(\xi_\Lambda, 1) - \bar{W}(\xi_R, 0) &= \pi w(1 - \delta k) + (1 - \pi)\delta w(1 - k) > 0, \\
\bar{W}(\xi_\Lambda, 1) - \bar{W}(\xi_D, 1) &= (1 - \pi)(\delta w + 1 - k) > 0, \\
\bar{W}(\xi_\Lambda, 1) - \bar{W}(\xi_D, 0) &= \pi w(1 - k) + (1 - \pi)\delta w > 0.
\end{aligned}$$

Attempt limit at other leniencies. With $X = p_H w$, $\bar{W}(\xi_\Lambda, \gamma) = p_H w\big[\pi + (1 - \pi)\delta\big]$, which is strictly increasing in γ. Hence $\bar{W}(\xi_\Lambda, \gamma) < \bar{W}(\xi_\Lambda, 1)$ for every $\gamma < 1$.

Unrestricted testing. At $\gamma = 1$, $\bar{W}(\xi_\Lambda, 1) - \bar{W}(\xi_U, 1) = (1 - \pi)(1 - k)(1 + \delta w) > 0$. For $\gamma < 1$, write $s \equiv 1 - p_H > 0$, $A \equiv \delta p_H w\, s$ and $N \equiv p_L(1 + \delta X) \geq 0$. Collecting terms, $\bar{W}(\xi_U, \gamma) - \bar{W}(\xi_\Lambda, \gamma) = \pi A - (1 - \pi)N$, while $\bar{W}(\xi_\Lambda, 1) - \bar{W}(\xi_\Lambda, \gamma) = s\, w\big[\pi + (1 - \pi)\delta\big]$. Since $\delta p_H \leq 1$, $\bar{W}(\xi_\Lambda, 1) - \bar{W}(\xi_U, \gamma) = s\, w\big[\pi + (1 - \pi)\delta\big] - \pi A + (1 - \pi)N \geq s\, w\, \pi\, (1 - \delta p_H) + s\, w\, (1 - \pi)\delta > 0$.

Together with Step 1, $(\xi_\Lambda, 1)$ strictly dominates every other test setting, so $(\xi^\star, \gamma^\star) = (\xi_\Lambda, 1)$.

Step 5: less accurate tests, $k < \bar{k}$. By Step 3, $\tilde{\gamma} < 1$, so the attempt limit is feasible exactly on $[0, \tilde{\gamma}]$. Consider any optimal test setting $(\xi^\star, \gamma^\star)$; by Step 1, $\xi^\star \neq \xi_\emptyset$. Suppose $\xi^\star \in \{\xi_R, \xi_D\}$. Its welfare is affine in γ on $[0, 1]$: if its slope is zero, it equals its value at $\gamma = 0$, which is strictly dominated by Step 2; otherwise its maximum is attained only at an endpoint, and the endpoint $\gamma = 0$ is ruled out by Step 2, so $\gamma^\star = 1$. The same argument applied on $[0, \tilde{\gamma}]$ shows that $\xi^\star = \xi_\Lambda$ requires $\gamma^\star = \tilde{\gamma}$. Finally, if $\xi^\star = \xi_U$, then $\gamma^\star = \gamma_U^\star$ by definition. Hence $(\xi^\star, \gamma^\star) \in \big\{(\xi_\Lambda, \tilde{\gamma}),\ (\xi_R, 1),\ (\xi_D, 1),\ (\xi_U, \gamma_U^\star)\big\}$. □

A.7 Proof of Proposition 5

Fix β, w and an accuracy k inducing strong reskilling, so that $\frac{1-\beta}{1+w} < k < \bar{k}$. As in the proof of Proposition 4, work with $\bar{W} \equiv (1 - \delta)W$, which ranks test settings exactly as W does. For sake of simplicity, write $p_H(\gamma)$ and $p_L(\gamma)$ for the passing rates at leniency γ, and let

$$\tilde{p}_H \equiv p_H(\tilde{\gamma}) = k + \beta\Phi(k, \beta), \qquad \kappa \equiv \rho(1 + w) - 1, \qquad X(\gamma) \equiv \rho\, p_H(\gamma) w - (1 - \rho)\, p_L(\gamma),$$

with $X_\Gamma \equiv X(\tilde{\gamma})$ and $X_1 \equiv X(1)$.

Lemma 5. *Fix β, w and $k \in (0, \bar{k})$, and let $\kappa \equiv \rho(1+w) - 1$.*

(i) The term κ is positive if and only if the test induces strong reskilling, that is, $k > \frac{1-\beta}{1+w}$.

(ii) The value of a deferred low type is $X(\gamma) = \rho k w + (1-k)\kappa\gamma$. Hence, under strong reskilling, X is strictly increasing in γ and $X(\gamma) \geq \rho k w > 0$; in particular $X_1 > 0$.

(iii) Ceiling $\tilde{\gamma} \in (0,1)$, so that $\tilde{p}_H = k + \beta\Phi(k,\beta) \in (k,1)$ and $X_1 - X_\Gamma = (1-\tilde{p}_H)\kappa$.

(iv) The slope of the value of a deferred low type is $\partial X(\gamma,k)/\partial\gamma = (1-k)\kappa$. Hence $\partial X(\gamma,k)/\partial\gamma > 0$ if and only if $\kappa > 0$, that is, if and only if $k > \frac{1-\beta}{1+w}$.

Proof. (i) If $k \geq 1-\beta$, then $\rho = 1$ and $\kappa = w > 0$, while $k \geq 1-\beta > \frac{1-\beta}{1+w}$. If $k < 1-\beta$, then $\rho = \frac{k}{1-\beta}$, and $\kappa > 0$ if and only if $\frac{k}{1-\beta} > \frac{1}{1+w}$, that is, $k > \frac{1-\beta}{1+w}$. (ii) Substituting $p_H = k + (1-k)\gamma$ and $p_L = (1-k)\gamma$ gives $X(\gamma) = \rho k w + (1-k)\gamma\big[\rho w - (1-\rho)\big] = \rho k w + (1-k)\kappa\gamma$. The remaining claims follow from $\kappa > 0$ and $k < 1$. (iii) The ceiling $\tilde{\gamma} = \beta\Phi(k,\beta)/(1-k)$ is positive because $k > 0$, and it is below one because $k < \bar{k}$. Therefore $\tilde{p}_H = k + (1-k)\tilde{\gamma} \in (k,1)$, and by (ii), $X_1 - X_\Gamma = (1-k)(1-\tilde{\gamma})\kappa = (1-\tilde{p}_H)\kappa$. (iv) Differentiating the expression in (ii) gives $\partial X(\gamma)/\partial\gamma = (1-k)\kappa$, which has the sign of κ because $k < 1$; the equivalence with $k > \frac{1-\beta}{1+w}$ then follows from (i). □

Step 1: three candidates. By Proposition 4(iii), the optimal test setting belongs to $\mathcal{C} = \{(\xi_\Lambda, \tilde{\gamma}), (\xi_R, 1), (\xi_D, 1), (\xi_U, \gamma_U^\star)\}$. The four test settings are identified by the allocation of testing slots: thus, let $W_\xi(\pi)$ denote the welfare of the principal adopting test setting $(\xi,\gamma) \in \mathcal{C}$, and represent it as a function of parameter π; furthermore, let $\bar{W}_\xi(\pi) = (1-\delta)W_\xi(\pi)$. The stop-loss deadline is weakly dominated:

$$\bar{W}_U(\pi) - \bar{W}_D(\pi) \;\geq\; \bar{W}(\xi_U, 1) - \bar{W}_D(\pi) = (1-\pi)\,\delta k X_1 \;\geq\; 0,$$

with strict inequality for $\pi < 1$ by Lemma 5(ii). The choice is therefore among $(\xi_R, 1)$, $(\xi_\Lambda, \tilde{\gamma})$, and (ξ_U, γ_U^*). Note that $\bar{W}_R$ and $\bar{W}_\Lambda$ are affine in π, whereas $\bar{W}_U$ is the upper envelope of the affine functions $\bar{W}(\xi_U, \gamma)$.

Step 2: residency versus unrestricted testing. For every γ,

$$\bar{W}_R(\pi) - \bar{W}_U(\gamma;\pi) = \pi A_\gamma + (1-\pi)B_\gamma,$$

where $B_\gamma \equiv \delta\big[X_1 - X(\gamma)\big] + p_L(\gamma)\big[1 + \delta X(\gamma)\big] > 0$ holds for $\gamma < 1$ because $X_1 > X(\gamma)$ and $X(\gamma) > 0$, and it holds for $\gamma = 1$ because $p_L(1) = 1 - k > 0$. Hence $\{\pi : \bar{W}_R(\pi) > \bar{W}(\xi_U, \gamma)\}$

is an initial segment of $[0,1]$ that contains 0. Since the maximum defining $\bar{W}_U$ is attained, $\bar{W}_R(\pi) > \bar{W}_U(\pi)$ if and only if $\bar{W}_R(\pi) > \bar{W}(\xi_U, \gamma)$ for all γ. The set $\{\pi : \bar{W}_R(\pi) > \bar{W}_U(\pi)\}$ is therefore an intersection of initial segments containing 0, and so is itself an initial segment containing 0. It is relatively open by continuity, and it excludes $\pi = 1$, where $\bar{W}_R(1) = \delta w < w = \bar{W}_U(1)$. Hence there is $\pi_B \in (0,1)$ such that $\bar{W}_R(\pi) > \bar{W}_U(\pi)$ if and only if $\pi < \pi_B$.

Lemma 6. *Assume a test inducing strong reskilling. For every $\gamma \in [0,1]$, let*

$$a_\gamma \equiv p_H(\gamma) - \tilde{p}_H, \qquad b_\gamma \equiv p_H(\gamma)\big(1 - p_H(\gamma)\big), \qquad n_\gamma \equiv X_\Gamma - \big(1 - p_L(\gamma)\big)X(\gamma),$$
$$M_\gamma \equiv w\big(a_\gamma + \delta b_\gamma\big), \qquad N_\gamma \equiv p_L(\gamma) + \delta\, n_\gamma.$$

Then the welfare gap between the attempt limit and unrestricted testing at leniency γ is $\bar{W}_\Lambda(\pi) - \bar{W}(\xi_U, \gamma) = (1-\pi)\, N_\gamma - \pi\, M_\gamma$, and $M_\gamma \le 0$ implies $N_\gamma > 0$.

Proof. On a high type, unrestricted testing yields $p_H(\gamma)w\big[1+\delta(1-p_H(\gamma))\big]$ and the attempt limit yields $\tilde{p}_H w$; the difference is M_γ. On a low type, the attempt limit yields δX_Γ and unrestricted testing yields $-p_L(\gamma) + \delta(1 - p_L(\gamma))X(\gamma)$; the difference is N_γ. Weighting each type by its prior gives the decomposition.

Suppose now that $M_\gamma \le 0$. Since $b_\gamma \ge 0$, this requires $a_\gamma \le -\delta b_\gamma \le 0$. Equality $a_\gamma = 0$ would force $b_\gamma = 0$, which is impossible, since then $p_H(\gamma) = \tilde{p}_H \in (0,1)$ by Lemma 5(iii). Hence $a_\gamma < 0$, that is, $p_H(\gamma) < \tilde{p}_H$, and so $\gamma < \tilde{\gamma}$. By Lemma 5(ii), the value X is strictly increasing and positive, so $X(\gamma) < X_\Gamma$. Therefore $n_\gamma = \big[X_\Gamma - X(\gamma)\big] + p_L(\gamma)X(\gamma) > 0$, and $N_\gamma = p_L(\gamma) + \delta n_\gamma > 0$. □

Step 3: attempt limit versus unrestricted testing. Fix a leniency γ. By Lemma 6, the welfare gap $\bar{W}_\Lambda(\pi) - \bar{W}(\xi_U, \gamma) = (1-\pi)N_\gamma - \pi M_\gamma$ is affine in π. If $N_\gamma > 0$, the gap is positive at $\pi = 0$, so the set of priors at which the attempt limit beats unrestricted testing at leniency γ is an initial segment of $[0,1]$ containing 0. If instead $N_\gamma \le 0$, Lemma 6 gives $M_\gamma > 0$, so the gap is negative on $(0,1]$ and non-positive at 0, and the set is empty.

Since the maximum defining $\bar{W}_U$ is attained, the attempt limit beats unrestricted testing at prior π, that is, $\bar{W}_\Lambda(\pi) > \bar{W}_U(\pi)$, if and only if it beats unrestricted testing at every leniency. The set $\{\pi : \bar{W}_\Lambda(\pi) > \bar{W}_U(\pi)\}$ is therefore an intersection of initial segments, and hence itself an initial segment or empty. It is relatively open, because $\bar{W}_U$ is continuous in π. It excludes $\pi = 1$, because $\bar{W}_\Lambda(1) = \tilde{p}_H w < w = \bar{W}_U(1)$. Consequently, either the attempt limit never beats unrestricted testing, or there exists $\pi_2 \in (0,1)$ such that $\bar{W}_\Lambda(\pi) > \bar{W}_U(\pi)$ if and only if $\pi < \pi_2$. In the latter case, consider the gap

$$G(\pi) \equiv \bar{W}_\Lambda(\pi) - \bar{W}_U(\pi) = \min_\gamma \big[\bar{W}_\Lambda(\pi) - \bar{W}(\xi_U, \gamma)\big] = \min_\gamma \big[(1-\pi)N_\gamma - \pi M_\gamma\big],$$

which is concave in π. Indeed, take any $\pi', \pi'' \in [0,1]$ and $\lambda \in [0,1]$, and let $\pi_\lambda \equiv \lambda\pi' + (1-\lambda)\pi''$. For every γ, the bracket is affine in π and is at least G at π' and at π'', so its value at π_λ is at least $\lambda G(\pi') + (1-\lambda)G(\pi'')$. This bound does not depend on γ, hence it also holds for the minimum: $G(\pi_\lambda) \geq \lambda G(\pi') + (1-\lambda)G(\pi'')$. The gap G is positive at 0 and zero at π_2. For any $\pi \in (\pi_2, 1]$, write $\pi_2 = \lambda \cdot 0 + (1-\lambda)\pi$ with $\lambda \in (0,1)$. Concavity gives $0 = G(\pi_2) \geq \lambda G(0) + (1-\lambda)G(\pi)$, and since $G(0) > 0$, we obtain $G(\pi) < 0$, that is, $\bar{W}_\Lambda(\pi) < \bar{W}_U(\pi)$. Unrestricted testing is thus strictly preferred at every prior above π_2.

Step 4: attempt limit versus residency. We have $\bar{W}_\Lambda(\pi) - \bar{W}_R(\pi) = \pi\, w(\tilde{p}_H - \delta) - (1-\pi)\,\delta\Delta_X$. If $\delta \geq \tilde{p}_H$, this is non-positive on $[0,1]$, so the attempt limit never beats residency. If $\delta < \tilde{p}_H$, it is strictly increasing in π, negative at 0 and positive at 1. Hence $\bar{W}_\Lambda > \bar{W}_R$ if and only if $\pi > \pi_A$, where $\pi_A \equiv \frac{\delta\Delta_X}{\delta\Delta_X + w(\tilde{p}_H - \delta)}$.

Step 5: the optimal test setting. Let

$$\mathcal{D} \equiv \left\{\delta \in (0,1) :\ \bar{W}_\Lambda(\pi) > \max\left\{\bar{W}_R(\pi), \bar{W}_U(\pi)\right\} \text{ for some } \pi \in [0,1]\right\}$$

be the set of discount factors at which the attempt limit is strictly optimal at some prior. By Steps 3 and 4, a discount factor δ belongs to $\mathcal{D}$ if and only if $\delta < \tilde{p}_H$,the set $\{\pi : \bar{W}_\Lambda(\pi) > \bar{W}_U(\pi)\} = [0, \pi_2)$ is non-empty, and $\pi_A < \pi_2$.

Case $\delta \in \mathcal{D}$. Set $\pi_1 \equiv \pi_A$, so that $0 < \pi_1 < \pi_2 < 1$. At π_1 the attempt limit and residency yield the same welfare, and both exceed unrestricted testing, since $\bar{W}_R(\pi_1) = \bar{W}_\Lambda(\pi_1) > \bar{W}_U(\pi_1)$; by Step 2, this implies $\pi_1 < \pi_B$. Three regions follow. On $[0, \pi_1)$, Steps 2 and 4 give $\bar{W}_R(\pi) > \bar{W}_\Lambda(\pi)$ and $\bar{W}_R(\pi) > \bar{W}_U(\pi)$, so $(\xi_R, 1)$ is optimal. On $[\pi_1, \pi_2)$, the attempt limit yields $\bar{W}_\Lambda(\pi) \geq \bar{W}_R(\pi)$, with equality only at π_1, and $\bar{W}_\Lambda(\pi) > \bar{W}_U(\pi)$, so $(\xi_\Lambda, \tilde{\gamma})$ is optimal. On $[\pi_2, 1]$, unrestricted testing yields $\bar{W}_U(\pi) \geq \bar{W}_\Lambda(\pi) > \bar{W}_R(\pi)$, the last inequality because $\pi \geq \pi_2 > \pi_A$, so $(\xi_U, \gamma_U^\star)$ is optimal.

Case $\delta \notin \mathcal{D}$. Set $\pi_0 \equiv \pi_B \in (0,1)$. On $[0, \pi_0)$, Step 2 gives $\bar{W}_R(\pi) > \bar{W}_U(\pi)$. Moreover, $\bar{W}_R(\pi) \geq \bar{W}_\Lambda(\pi)$ there: otherwise $\bar{W}_\Lambda(\pi) > \bar{W}_R(\pi) > \bar{W}_U(\pi)$ at some prior, contradicting $\delta \notin \mathcal{D}$. Hence $(\xi_R, 1)$ is optimal. On $[\pi_0, 1]$, Step 2 gives $\bar{W}_U(\pi) \geq \bar{W}_R(\pi)$. Moreover, $\bar{W}_U(\pi) \geq \bar{W}_\Lambda(\pi)$ there: otherwise $\bar{W}_\Lambda(\pi) > \bar{W}_U(\pi) \geq \bar{W}_R(\pi)$ at some prior, again a contradiction. Hence $(\xi_U, \gamma_U^\star)$ is optimal.

It remains to show that $\mathcal{D} = (0, \bar{\delta}(k))$ for some $\bar{\delta}(k) \in \big(0,\ k + \beta\Phi(k,\beta)\big)$.

Step 6: the set $\mathcal{D}$ is an initial interval below $\tilde{p}_H$. *(a) Reformulation.* By Step 4, the attempt limit never beats residency when $\delta \geq \tilde{p}_H$, so $\mathcal{D} \subset (0, \tilde{p}_H)$. Fix $\delta < \tilde{p}_H$. By Step 3, the attempt limit is never optimal at $\pi = 1$, so it suffices to look at priors $\pi \in [0,1)$. For

these, write the odds as $r \equiv \pi/(1-\pi) \in [0,\infty)$, and let $r_A \equiv \frac{\delta\Delta_X}{w(\tilde{p}_H-\delta)} > 0$ be the odds at the threshold π_A of Step 4. Dividing the welfare gaps by $1-\pi > 0$, Steps 3 and 4 yield

$$\bar{W}_\Lambda(\pi) > \bar{W}_R(\pi) \iff r > r_A, \qquad \bar{W}_\Lambda(\pi) > \bar{W}(\xi_U,\gamma) \iff N_\gamma > r\,M_\gamma.$$

The attempt limit beats unrestricted testing at prior π if and only if it beats it at every leniency, since the maximum defining $\bar{W}_U(\pi)$ is attained. Hence $\delta \in \mathcal{D}$ if and only if some odds $r > r_A$ satisfy $N_\gamma > rM_\gamma$ for every $\gamma \in [0,1]$.

I now show that this holds if and only if $N_\gamma - r_A M_\gamma > 0$ for every $\gamma \in [0,1]$ that is, if and only if the inequality holds at the threshold odds r_A themselves. For necessity, take odds $r > r_A$ that work for every γ. If $M_\gamma \geq 0$, then $r_A M_\gamma \leq rM_\gamma < N_\gamma$. If $M_\gamma < 0$, then $N_\gamma > 0$ by Lemma 6, while $r_A M_\gamma < 0$, so again $N_\gamma > r_A M_\gamma$. For sufficiency, the odds r_A themselves cannot be used, because at r_A the attempt limit only ties with residency. I must move slightly above r_A, and the step must be small enough for every leniency at once. Let $\phi(\gamma) \equiv N_\gamma - r_A M_\gamma$. This function is continuous and positive on the compact set $[0,1]$, so it attains a minimum $\eta > 0$, and $\phi(\gamma) \geq \eta$ for every γ: the margin is uniform in γ. Moreover, $|M_\gamma| \leq w\big(|a_\gamma| + b_\gamma\big) \leq 1 + \frac{1}{4} < 2$. Take $r = r_A + \eta/4 > r_A$. Then, for every γ, $N_\gamma - rM_\gamma = \phi(\gamma) - \frac{\eta}{4}M_\gamma \;\geq\; \eta - \frac{\eta}{4}\cdot 2 = \frac{\eta}{2} > 0$, so these odds satisfy both requirements, and $\delta \in \mathcal{D}$.

Finally, multiplying $N_\gamma - r_A M_\gamma > 0$ by $w(\tilde{p}_H - \delta) > 0$ and dividing by w removes the fraction in r_A. Since $N_\gamma = p_L(\gamma) + \delta n_\gamma$ and $M_\gamma = w(a_\gamma + \delta b_\gamma)$, I obtain

$$\delta \in \mathcal{D} \iff \delta < \tilde{p}_H \text{ and } g_\gamma(\delta) > 0 \;\text{ for every } \gamma \in [0,1],$$

where $g_\gamma(\delta) \equiv (\tilde{p}_H - \delta)\big(p_L(\gamma) + \delta n_\gamma\big) - \delta\Delta_X\big(a_\gamma + \delta b_\gamma\big) = \tilde{p}_H p_L(\gamma) + \delta\big(\tilde{p}_H n_\gamma - p_L(\gamma) - \Delta_X a_\gamma\big) - \delta^2\big(n_\gamma + \Delta_X b_\gamma\big)$. The quantities $p_L(\gamma)$, n_γ, a_γ, b_γ, $\tilde{p}_H$ and Δ_X depend on (k,β,w,γ) but not on δ. Each g_γ is therefore a polynomial of degree at most two in δ.

(b) Single crossing. We show that, for every γ, positivity of g_γ at some $\delta' \in (0,\tilde{p}_H)$ implies positivity at every $\delta \in (0,\delta')$.

At $\gamma = 0$, we have $p_L(0) = 0$, $a_0 = k - \tilde{p}_H = -\beta\Phi(k,\beta) < 0$ and $n_0 = X_\Gamma - X(0) > 0$. The polynomial thus factors as $g_0(\delta) = \delta\,\ell(\delta)$, where $\ell(\delta) \equiv \tilde{p}_H n_0 + \Delta_X\beta\Phi(k,\beta) - \delta\big(n_0 + \Delta_X b_0\big)$ is strictly decreasing. If $g_0(\delta') > 0$, then $\ell(\delta') > 0$, hence $\ell(\delta) > \ell(\delta') > 0$ and $g_0(\delta) > 0$ for every $\delta < \delta'$.

For $\gamma > 0$, setting $\delta = 0$ gives $g_\gamma(0) = \tilde{p}_H p_L(\gamma) > 0$. Let $g_\gamma(\delta') > 0$ and $0 < \delta < \delta'$. Two cases arise, according to the sign of the δ^2-coefficient. If $n_\gamma + \Delta_X b_\gamma \geq 0$, the function g_γ is concave. Writing $\delta = \lambda\cdot 0 + (1-\lambda)\delta'$ with $\lambda \in (0,1)$, concavity gives $g_\gamma(\delta) \geq \lambda g_\gamma(0) + (1-\lambda)g_\gamma(\delta') > 0$. If instead $n_\gamma + \Delta_X b_\gamma < 0$, the function g_γ is convex, and I

first show that it is negative at $\tilde{p}_H$. Since $\Delta_X b_\gamma \geq 0$, the case requires $n_\gamma < 0$, that is, $(1 - p_L(\gamma))X(\gamma) > X_\Gamma > 0$. As $1 - p_L(\gamma) \leq 1$, this implies $X(\gamma) > X_\Gamma = X(\tilde{\gamma})$, and since X is strictly increasing, $\gamma > \tilde{\gamma}$. Hence $p_H(\gamma) > \tilde{p}_H$ and $a_\gamma > 0$. At $\delta = \tilde{p}_H$ the first term of g_γ vanishes, so $g_\gamma(\tilde{p}_H) = -\tilde{p}_H \Delta_X \big(a_\gamma + \tilde{p}_H b_\gamma\big) < 0$. Now suppose, by contradiction, that $g_\gamma(\delta) \leq 0$. Since $\delta < \delta' < \tilde{p}_H$, write $\delta' = \lambda\delta + (1-\lambda)\tilde{p}_H$ with $\lambda \in (0,1)$. Convexity gives $g_\gamma(\delta') \leq \lambda g_\gamma(\delta) + (1-\lambda) g_\gamma(\tilde{p}_H) < 0$, contradicting $g_\gamma(\delta') > 0$. Hence $g_\gamma(\delta) > 0$.

In all cases, positivity at δ' carries over to every smaller discount factor. By (a), membership in $\mathcal{D}$ requires $g_\gamma > 0$ for every γ simultaneously. Therefore, if $\delta' \in \mathcal{D}$ and $0 < \delta < \delta'$, then $g_\gamma(\delta) > 0$ for every γ, and $\delta \in \mathcal{D}$.

(c) Openness. The function $\underline{g}(\delta) \equiv \min_{\gamma \in [0,1]} g_\gamma(\delta)$ is continuous, as the minimum over a compact set of a function jointly continuous in (γ, δ). By (a), the set $\mathcal{D} = \{\delta \in (0, \tilde{p}_H) : \underline{g}(\delta) > 0\}$ is therefore open.

(d) Non-emptiness. Since $g_\gamma(0) = \tilde{p}_H p_L(\gamma)$ vanishes at $\gamma = 0$, positivity at $\delta = 0$ is not uniform in γ, and small leniencies require a separate argument. Expanding the first product in g_γ gives

$$g_\gamma(\delta) = (\tilde{p}_H - \delta)\, p_L(\gamma) + \delta\, h(\gamma, \delta), \qquad h(\gamma, \delta) \equiv (\tilde{p}_H - \delta) n_\gamma - \Delta_X \big(a_\gamma + \delta b_\gamma\big).$$

By (b), $a_0 = -\beta\Phi(k, \beta)$ and $n_0 > 0$, so that $h(0,0) = \tilde{p}_H n_0 + \Delta_X \beta \Phi(k, \beta) > 0$. The function h is a polynomial in (γ, δ), hence continuous, and there exist $\gamma_0 > 0$ and $\delta_0 \in (0, \tilde{p}_H)$ such that $h > 0$ on $[0, \gamma_0] \times [0, \delta_0]$. For $\gamma \in [0, \gamma_0]$ and $\delta \in (0, \delta_0]$, the first term of $g_\gamma(\delta)$ is non-negative and the second is positive, so $g_\gamma(\delta) > 0$. For $\gamma \in [\gamma_0, 1]$, the rewriting above gives $g_\gamma(\delta) = g_\gamma(0) + \delta\big[h(\gamma, \delta) - p_L(\gamma)\big]$, where $g_\gamma(0) = \tilde{p}_H(1-k)\gamma \geq \tilde{p}_H(1-k)\gamma_0 \equiv \mu > 0$. The bracket is continuous on the compact set $[\gamma_0, 1] \times [0, \delta_0]$, and hence bounded in absolute value by some $K > 0$. Therefore $g_\gamma(\delta) \geq \mu - \delta K > 0$ for every $\delta < \mu/K$. Choosing $\delta_1 \in (0, \delta_0]$ with $\delta_1 < \mu/K$, we obtain $g_\gamma(\delta) > 0$ for all $\gamma \in [0,1]$ and all $\delta \in (0, \delta_1]$. By (a), $(0, \delta_1] \subset \mathcal{D}$, and so $\bar{\delta}(k) > 0$.

(e) Upper bound. At $\gamma = 1$, we have $p_H(1) = 1$, so $a_1 = 1 - \tilde{p}_H$ and $b_1 = 0$. Evaluating at $\delta = \tilde{p}_H$, $g_1(\tilde{p}_H) = -\tilde{p}_H \Delta_X (1 - \tilde{p}_H) < 0$ by Lemma 5(iii). By continuity, there is $\varepsilon > 0$ such that $g_1 < 0$ on $[\tilde{p}_H - \varepsilon, \tilde{p}_H]$. By (a), these discount factors lie outside $\mathcal{D}$.

Conclusion. Let $\bar{\delta}(k) \equiv \sup \mathcal{D}$. By (d), $\bar{\delta}(k) \geq \delta_1 > 0$, and by (e), $\bar{\delta}(k) \leq \tilde{p}_H - \varepsilon < k + \beta\Phi(k, \beta)$. By (b), every discount factor below an element of $\mathcal{D}$ belongs to $\mathcal{D}$, so $(0, \bar{\delta}(k)) \subset \mathcal{D}$. By (c), the set $\mathcal{D}$ is open, so it does not contain its supremum. Hence $\mathcal{D} = (0, \bar{\delta}(k))$ with $\bar{\delta}(k) \in \big(0,\, k + \beta\Phi(k, \beta)\big)$. □

A.8 Proof of Proposition 6

Fix β, w, δ and an accuracy k inducing a test that is not leniency-rewarding, and work with $\bar{W} \equiv (1-\delta)W$, which ranks test settings as W does. Since k is fixed, write $p_H(\gamma)$, $p_L(\gamma)$ and $X(\gamma)$ for $p_H(\gamma, k)$, $p_L(\gamma, k)$ and $X(\gamma, k)$, and let $\tilde{p}_H \equiv p_H(\tilde{\gamma})$, $X_\Gamma \equiv X(\tilde{\gamma})$ and $X_1 \equiv X(1)$. All welfare functions below are continuous in π, so we define them on $[0,1]$; the statement concerns their restriction to $(0,1)$.

Step 0: preliminaries. *(a) Parameters.* The test is not leniency-rewarding, so $\kappa = \rho(1+w) - 1 \leq 0$. This rules out $k \geq 1-\beta$, for which $\rho = 1$ and $\kappa = w > 0$. Hence $k < 1-\beta < \bar{k}$, so that $\rho = \frac{k}{1-\beta} \in (0,1)$ and, by Lemma 5(iii), $\tilde{\gamma} \in (0,1)$ and $\tilde{p}_H = k + (1-k)\tilde{\gamma} = k + \beta\Phi(k,\beta) \in (k,1)$. In particular, $\beta\Phi(k,\beta) = (1-k)\tilde{\gamma}$. Moreover, $\kappa = \frac{k(1+w)}{1-\beta} - 1$, so that $\kappa < 0$ if $k < \frac{1-\beta}{1+w}$, which is part (a) of the proposition, and $\kappa = 0$ if $k = \frac{1-\beta}{1+w}$, which is part (b).

(b) Value of a deferred low type. By Lemma 5(ii), $X(\gamma) = \rho k w + (1-k)\kappa\gamma$, which is non-increasing in γ, with $X(0) = \rho k w > 0$, $X(0) - X_\Gamma = (1-k)|\kappa|\tilde{\gamma} \geq 0$, and $X_\Gamma - X_1 = (1-k)|\kappa|(1-\tilde{\gamma}) \geq 0$. Moreover, $X_1 = \rho w - (1-\rho)(1-k) > -(1-k)$, because $\rho > 0$.

(c) Candidate settings. Because $k < \bar{k}$, Proposition 4(iii) implies that the maximal welfare at prior π is $\mathcal{W}(\pi) \equiv \max\{\bar{W}_\Lambda(\pi), \bar{W}_R(\pi), \bar{W}_D(\pi), \bar{W}_U(\pi)\}$, where $\bar{W}_\xi(\pi)$ denotes the welfare of the principal adopting test setting $(\xi,\gamma) \in \{(\xi_\Lambda,\tilde{\gamma}), (\xi_R,1), (\xi_D,1), (\xi_U,\gamma_U^\star)\}$; in particular, $\bar{W}_D(\pi) = \pi w - (1-\pi)(1-k)$. Moreover, $\bar{W}_U(\pi) \equiv \max_{\gamma\in[0,1]} \bar{W}(\xi_U,\gamma)$, with $\bar{W}(\xi_U,\gamma) = \pi H(\gamma) + (1-\pi)L(\gamma)$ denoting the welfare of unrestricted testing at leniency γ:

$$H(\gamma) \equiv p_H(\gamma) w\big[1 + \delta\big(1 - p_H(\gamma)\big)\big], \qquad L(\gamma) \equiv -p_L(\gamma) + \delta\big(1 - p_L(\gamma)\big)X(\gamma).$$

The functions H and L are polynomials in γ, so the maximum defining $\bar{W}_U$ is attained. Each $\bar{W}(\xi_U,\gamma)$ is affine in π with slope bounded by $K \equiv \max_\gamma |H(\gamma) - L(\gamma)|$, so $\bar{W}_U$ is Lipschitz, and hence continuous, on $[0,1]$.

(d) Two facts at the top. The function $h(p) \equiv wp\big(1 + \delta(1-p)\big)$ satisfies $h'(p) = w(1 + \delta - 2\delta p) \geq w(1-\delta) > 0$ on $[0,1]$, and $H(\gamma) = h(p_H(\gamma))$ with $p_H(1) - p_H(\gamma) = (1-k)(1-\gamma)$. By the mean value theorem, $H(1) - H(\gamma) \geq w(1-\delta)(1-k)(1-\gamma)$ for every $\gamma \in [0,1]$, and $H(1) = w$. Hence $\bar{W}_U(1) = w$, while $\bar{W}_\Lambda(1) = \tilde{p}_H w < w$, $\bar{W}_R(1) = \delta w < w$ and $\bar{W}_D(1) = w$; thus $\mathcal{W}(1) = \bar{W}_U(1) = \bar{W}_D(1)$. Finally, a direct computation gives $\bar{W}(\xi_U,1) - \bar{W}_D(\pi) = (1-\pi)\,\delta k X_1$.

Step 1: optimality sets. For each menu $\xi \in \{\xi_\Lambda, \xi_R, \xi_D, \xi_U\}$, let $\Delta_\xi \equiv \{\pi \in [0,1] : \bar{W}_\xi(\pi) = \mathcal{W}(\pi)\}$ be the set of priors at which ξ is optimal; these sets cover $[0,1]$.

(a) Concavity. For $\xi \in \{\xi_\Lambda, \xi_R, \xi_D\}$, the function $f_\xi(\pi) \equiv \bar{W}_\xi(\pi) - \bar{W}_U(\pi) = \min_\gamma \big[\bar{W}_\xi(\pi) - \bar{W}(\xi_U, \gamma)\big]$ is continuous and concave in π. To see concavity, take $\pi', \pi'' \in [0,1]$, $\lambda \in [0,1]$ and $\pi_\lambda \equiv \lambda\pi' + (1-\lambda)\pi''$. For every γ, the bracket is affine in π and is at least f_ξ at π' and at π'', so its value at π_λ is at least $\lambda f_\xi(\pi') + (1-\lambda) f_\xi(\pi'')$. Taking the minimum over γ gives $f_\xi(\pi_\lambda) \geq \lambda f_\xi(\pi') + (1-\lambda) f_\xi(\pi'')$.

(b) Shape. For $\xi \in \{\xi_\Lambda, \xi_R, \xi_D\}$, the set Δ_ξ is the intersection of $\{\pi \in [0,1] : f_\xi(\pi) \geq 0\}$, a closed interval because f_ξ is concave and continuous, with the sets $\{\pi \in [0,1] : \bar{W}_\xi(\pi) - \bar{W}_{\xi'}(\pi) \geq 0\}$, $\xi' \in \{\xi_\Lambda, \xi_R, \xi_D\} \setminus \{\xi\}$, which are closed intervals because $\bar{W}_\xi(\pi) - \bar{W}_{\xi'}(\pi)$ is affine. Hence Δ_ξ is a closed interval $[a_\xi, b_\xi]$, possibly empty or a single point.

Step 2: ordering. *(a) Attempt limit against residency.* By Step 0(b), the difference

$$B(\pi) \equiv \bar{W}_\Lambda(\pi) - \bar{W}_R(\pi) = -\pi w(\delta - \tilde{p}_H) + (1-\pi)\,\delta(1-k)|\kappa|(1-\tilde{\gamma})$$

is affine in π, with $B(0) = \delta(1-k)|\kappa|(1-\tilde{\gamma}) \geq 0$. If $\kappa < 0$, then $B(0) > 0$, since $\tilde{\gamma} < 1$. The set of priors at which B is non-positive is therefore either empty or an interval $[\pi^\dagger, 1]$ with $\pi^\dagger > 0$, and in the latter case B is non-negative exactly on $[0, \pi^\dagger]$. Since $B \geq 0$ on Δ_{ξ_Λ} and $B \leq 0$ on Δ_{ξ_R}, every prior in Δ_{ξ_Λ} is no larger than every prior in Δ_{ξ_R}; in particular, $b_{\xi_\Lambda} \leq a_{\xi_R}$ whenever both sets are non-empty. If $\kappa = 0$, then $B(\pi) = \pi w(\tilde{p}_H - \delta)$, which is non-negative on $[0,1]$ if $\delta \leq \tilde{p}_H$ and non-positive on $[0,1]$ if $\delta \geq \tilde{p}_H$.

(b) stop-loss deadline last. By Step 0(d), $1 \in \Delta_{\xi_D}$, so that $\Delta_{\xi_D} = [a_{\xi_D}, 1]$. For $\xi \in \{\xi_\Lambda, \xi_R\}$, the affine function $\bar{W}_\xi - \bar{W}_D$ is positive at $\pi = 0$, because $\delta X_\Gamma \geq \delta X_1 > -\delta(1-k) \geq -(1-k)$ by Step 0(b), and negative at $\pi = 1$ by Step 0(d). Its root $\pi^\ddagger_\xi \in (0,1)$ satisfies $\Delta_\xi \subset [0, \pi^\ddagger_\xi]$ and $\Delta_{\xi_D} \subset [\pi^\ddagger_\xi, 1]$, so that $b_\xi \leq a_{\xi_D}$ whenever $\Delta_\xi \neq \emptyset$.

Step 3: the stop-loss deadline. *Leniency-tolerant tests.* If $X_1 > 0$, Step 0(d) gives $\bar{W}_U(\pi) \geq \bar{W}(\xi_U, 1) > \bar{W}_D(\pi)$ for every $\pi \in [0,1)$. The stop-loss deadline is then optimal only at $\pi = 1$, and $a_{\xi_D} = 1$.

Leniency-intolerant tests. Let $X_1 \leq 0$, and let $M_L \geq 0$ be a Lipschitz constant of the polynomial L on $[0,1]$. For every $\gamma \in [0,1]$ and $\pi \in [0,1]$, Step 0(d) yields

$$\begin{aligned}\bar{W}(\xi_U, \gamma) - \bar{W}_D(\pi) &= \pi\big[H(\gamma) - H(1)\big] + (1-\pi)\big[L(\gamma) - L(1)\big] + (1-\pi)\,\delta k X_1 \\ &\leq (1-\gamma)\big[(1-\pi)M_L - \pi w(1-\delta)(1-k)\big].\end{aligned}$$

For $\pi \geq \pi^\circ \equiv M_L / \big(M_L + w(1-\delta)(1-k)\big) \in [0,1)$, the right-hand side is non-positive for every γ, so that $\bar{W}_U \leq \bar{W}_D$ on $[\pi^\circ, 1]$. By Step 0(d) and continuity, $\bar{W}_D$ also exceeds $\bar{W}_\Lambda$ and $\bar{W}_R$ on some interval $(\pi'', 1]$ with $\pi'' < 1$. The stop-loss deadline is therefore optimal on

$[\hat{\pi}, 1]$ for any $\hat{\pi} \in (\max\{\pi^\circ, \pi''\}, 1)$, and $a_{\xi_D} < 1$.

Step 4: the attempt limit requires $\delta \leq \delta_\Lambda(k, \beta)$. Suppose the attempt limit is optimal at some $\pi \in (0,1)$. Then $\bar{W}_\Lambda(\pi) \geq \bar{W}_U(\pi) \geq \bar{W}(\xi_U, 0)$ and $\bar{W}_\Lambda(\pi) \geq \bar{W}_R(\pi)$. Using $H(0) = kw\big(1+\delta(1-k)\big)$, $L(0) = \delta X(0)$, $\beta\Phi(k,\beta) = (1-k)\tilde{\gamma}$ and Step 0(b), the first inequality reads $A(\pi) \equiv \bar{W}_\Lambda(\pi) - \bar{W}(\xi_U, 0) = \pi w(1-k)(\tilde{\gamma} - \delta k) - (1-\pi)\delta(1-k)|\kappa|\tilde{\gamma} \geq 0$, and the second reads $B(\pi) \geq 0$, with B as in Step 2. Since $\pi > 0$, the inequality $A(\pi) \geq 0$ gives $\tilde{\gamma} \geq \delta k$, that is, $\delta \leq \tilde{\gamma}/k$. Moreover, the terms in $|\kappa|$ cancel in $(1-\tilde{\gamma})A(\pi) + \tilde{\gamma}B(\pi) = \pi w\big[(1-\tilde{\gamma})(1-k)(\tilde{\gamma} - \delta k) - \tilde{\gamma}(\delta - \tilde{p}_H)\big]$, which is therefore non-negative. Substituting $\tilde{p}_H = k + (1-k)\tilde{\gamma}$, the terms $(1-k)\tilde{\gamma}^2$ cancel, and dividing by $\pi w > 0$ gives $\delta\big[\tilde{\gamma} + k(1-k)(1-\tilde{\gamma})\big] \leq \tilde{\gamma}k + (1-k)\tilde{\gamma} = \tilde{\gamma}$. Together with $\delta \leq \tilde{\gamma}/k$, this gives $\delta \leq \delta_\Lambda(k, \beta)$.

Step 5: residency requires $\delta \geq \delta_R(k, \beta)$. Suppose residency is optimal at some $\pi \in (0,1)$. Using Step 0(b), the inequalities $\bar{W}_R(\pi) \geq \bar{W}_\Lambda(\pi)$ and $\bar{W}_R(\pi) \geq \bar{W}_U(\pi) \geq \bar{W}(\xi_U, 0)$ read respectively $\pi w(\delta - \tilde{p}_H) \geq (1-\pi)\delta(1-k)|\kappa|(1-\tilde{\gamma})$ and $\pi w\big[\delta - k\big(1+\delta(1-k)\big)\big] \geq (1-\pi)\,\delta(1-k)|\kappa|$. Both right-hand sides are non-negative and $\pi > 0$, so $\delta \geq \tilde{p}_H = k + \beta\Phi(k,\beta)$ and $\delta\big(1-k(1-k)\big) \geq k$. Because $1-k(1-k) > 0$, the latter is equivalent to $\delta \geq k/\big(1-k(1-k)\big)$. Hence $\delta \geq \delta_R(k,\beta)$; if $\kappa < 0$, both right-hand sides are positive and the inequality is strict.

Step 6: the case $k = \frac{1-\beta}{1+w}$. Let $\kappa = 0$. By Step 0(b), $X(\gamma) = X(0) = \rho k w > 0$ for every γ, so that $X_\Gamma = X_1 = X(0)$ and the test is leniency-tolerant; by Step 3, the stop-loss deadline is not optimal at any prior in $(0,1)$. Let $E \equiv \bar{W}_\Lambda$ if $\delta \leq \tilde{p}_H$ and $E \equiv \bar{W}_R$ if $\delta \geq \tilde{p}_H$; the two definitions agree when $\delta = \tilde{p}_H$, and by Step 2(a), $E = \max\{\bar{W}_\Lambda, \bar{W}_R\}$ on $[0,1]$. Since $X_1 > 0$, Step 0(d) gives $\bar{W}_U(\pi) \geq \bar{W}(\xi_U, 1) > \bar{W}_D(\pi)$ for $\pi < 1$, so that $\mathcal{W} = \max\{E, \bar{W}_U\}$ on $[0,1)$.

Let $f \equiv E - \bar{W}_U$, which is continuous and concave by Step 1(a). At $\pi = 0$, $L(\gamma) = \delta X(0) - (1-k)\gamma\big(1+\delta X(0)\big) \leq L(0) = \delta X(0)$, so that $\bar{W}_U(0) = \delta X(0) = E(0)$ and $f(0) = 0$. At $\pi = 1$, $f(1) = E(1) - w < 0$ by Step 0(d). Let $\bar{\pi} \equiv \min\{\pi \in [0,1] : f \leq 0 \text{ on } [\pi, 1]\}$, which exists by continuity and satisfies $\bar{\pi} < 1$ because $f(1) < 0$. I claim that $f > 0$ on $(0, \bar{\pi})$. Suppose instead that $f(\check{\pi}) \leq 0$ for some $\check{\pi} \in (0, \bar{\pi})$. By the minimality of $\bar{\pi}$, there is $\pi' \in [\check{\pi}, 1]$ with $f(\pi') > 0$, and $\pi' \in (\check{\pi}, \bar{\pi})$ because $f(\check{\pi}) \leq 0$ and $f \leq 0$ on $[\bar{\pi}, 1]$. Concavity on $[0, \pi']$ then gives $f(\check{\pi}) \geq \frac{\check{\pi}}{\pi'} f(\pi') + \big(1 - \frac{\check{\pi}}{\pi'}\big) f(0) > 0$, a contradiction.

Hence, for $\pi \in (0,1)$, unrestricted testing is optimal if and only if $f(\pi) \leq 0$, that is, if and only if $\pi \geq \bar{\pi}$. For $\pi \leq \bar{\pi}$, $f(\pi) \geq 0$, so that E attains $\mathcal{W}(\pi)$: the attempt limit is optimal if $\delta \leq \tilde{p}_H$, and residency is optimal if $\delta \geq \tilde{p}_H$. This proves part (b).

Step 7: conclusion for part (a). Let $\kappa < 0$. Set $\pi_D \equiv a_{\xi_D}$. By Step 2(b), $\Delta_{\xi_D} = [\pi_D, 1]$, so the stop-loss deadline is optimal if and only if $\pi \geq \pi_D$; by Step 3, $\pi_D < 1$ if and only if $X_1 \leq 0$, that is, if and only if the test is leniency-intolerant, which is (iii).

If the attempt limit is optimal at some prior in $(0, 1)$, then $\delta \leq \delta_\Lambda(k, \beta)$ by Step 4; set $\pi_1 \equiv a_{\xi_\Lambda}$ and $\pi_2 \equiv b_{\xi_\Lambda}$, so that by Step 1(b) it is optimal if and only if $\pi_1 \leq \pi \leq \pi_2$, which is (i). Otherwise, set $\pi_1 = \pi_2 \equiv 0$. Likewise, if residency is optimal at some prior in $(0, 1)$, then $\delta \geq \delta_R(k, \beta)$ by Step 5; set $\pi_3 \equiv a_{\xi_R}$ and $\pi_4 \equiv b_{\xi_R}$, so that it is optimal if and only if $\pi_3 \leq \pi \leq \pi_4$, which is (ii). Otherwise, set $\pi_3 = \pi_4 \equiv \pi_2$.

These thresholds satisfy $0 \leq \pi_1 \leq \pi_2 \leq \pi_3 \leq \pi_4 \leq \pi_D \leq 1$. The inequality $\pi_2 \leq \pi_3$ follows from Step 2(a) when both settings are optimal at some prior, and holds by construction otherwise. The inequality $\pi_4 \leq \pi_D$ follows from Step 2(b): if residency is optimal at some prior, then $b_{\xi_R} \leq a_{\xi_D}$; otherwise $\pi_4 = \pi_2$, which is either $b_{\xi_\Lambda} \leq a_{\xi_D}$ or 0.

Finally, because the sets Δ_ξ cover $[0, 1]$, every prior in $(0, 1)$ outside the intervals in (i)–(iii) belongs to Δ_{ξ_U}, where $(\xi_U, \gamma_U^\star)$ is optimal, which is (iv). □

A.9 Proof of Lemma 3

For every design $(\gamma, k) \in [0, 1]^2$, define the net contribution of a deferred incompetent candidate,

$$\ell(\gamma, k) \equiv X(\gamma, k) - (1 - \delta)k = \rho(k)\, p_H(\gamma, k)\, w - \big(1 - \rho(k)\big)\, p_L(\gamma, k) - (1 - \delta)k,$$

where $\rho(k) = \min\{\frac{k}{1-\beta}, 1\}$. The design $(\gamma, k) = (0, 0)$ gives $p_H = p_L = 0$ and no cost, so $\ell(0, 0) = 0$. Hence $\max_{(\gamma,k)} \ell(\gamma, k) \geq 0$, and residency is worthwhile if and only if $\ell(\gamma, k) > 0$ for some design.

If. Suppose $w > (1-\delta)(1-\beta)$, and consider the design $(\gamma, k) = (1, 1-\beta)$. Then $\rho(1-\beta) = 1$, so the candidate reskills for sure, and $p_H(1, 1 - \beta) = 1$, so he then passes for sure. Hence $\ell(1, 1 - \beta) = w - (1 - \delta)(1 - \beta) > 0$ and residency is worthwhile.

Only if. Suppose $w \leq (1 - \delta)(1 - \beta)$, and fix any design (γ, k). Two bounds hold. First, since $p_H \leq 1$ and the term $\big(1 - \rho(k)\big)p_L(\gamma, k)$ is non-negative, $X(\gamma, k) \leq \rho(k)w$. Second, since $\rho(k) \leq \frac{k}{1-\beta}$, we have $(1 - \beta)\rho(k) \leq k$, and therefore $(1 - \delta)(1 - \beta)\rho(k) \leq (1 - \delta)k$ Combining the two bounds,

$$\ell(\gamma, k) \leq \rho(k)\big[w - (1 - \delta)(1 - \beta)\big] \leq 0.$$

Since the design was arbitrary, $\max_{(\gamma,k)} \ell(\gamma,k) = 0$, and residency is not worthwhile. □

A.10 Proof of Proposition 7

Throughout, let $\mu \equiv (1-\delta)(1-\beta)$, so that $w \leq \mu$, and let $\pi_0 \equiv \frac{1}{1+w}$. For a design (γ, k), let

$$h(\gamma,k) \equiv p_H(\gamma,k)\, w - (1-\delta)k \qquad \text{and} \qquad \ell(\gamma,k) \equiv X(\gamma,k) - (1-\delta)k$$

be the net contributions of a competent candidate and of a deferred incompetent candidate who sit the test once, and let $A(\pi) \equiv \pi w - (1-\pi)$ be the welfare from admitting every candidate without information, which is attained by $(\xi_D, 1, 0)$. Welfare $\bar{W}(\xi,\gamma,k) = (1-\delta)W(\xi,\gamma,k)$ under each menu is

$$\begin{aligned}
\bar{W}(\xi_R,\gamma,k) &= \delta\big[\pi h + (1-\pi)\ell\big],\\
\bar{W}(\xi_D,\gamma,k) &= \pi h - (1-\pi)\big[p_L + (1-\delta)k\big],\\
\bar{W}(\xi_U,\gamma,k) &= \pi h\big[1+\delta(1-p_H)\big] + (1-\pi)\big[\delta(1-p_L)\ell - p_L - (1-\delta)k\big],\\
\bar{W}(\xi_\Lambda,\gamma,k) &= \pi h + (1-\pi)\delta\ell, \qquad \text{for } \gamma \leq \tilde{\gamma}(k).
\end{aligned}$$

Each of these is affine in π and is defined on $[0,1]$; the endpoints $\pi = 0$ and $\pi = 1$ are used only as limits. Also, denote by $\bar{W}(\xi_\emptyset)$ the hiring freeze welfare $(1-\delta)W(\xi_\emptyset,\gamma,k)$ that produces 0 for any (γ,k).

Step 1: two bounds. By the proof of Lemma 3, every design satisfies

$$\ell(\gamma,k) \leq \rho(k)\,(w-\mu) \leq 0. \tag{A.3}$$

Moreover, $w \leq \mu < 1-\delta$, so for every $\pi \in [0,1]$

$$\pi w - (1-\delta) < 0. \tag{A.4}$$

Step 2: only three options matter. Let $R_1(\pi) \equiv \max_{k\in[0,1]} \bar{W}(\xi_R, 1, k)$ be the best welfare from residency with a fully lenient test, and let $M(\pi) \equiv \max\{0, A(\pi), R_1(\pi)\}$. I show that no menu and design yields more than $M(\pi)$. Since $M(\pi)$ is attained by $\xi_\emptyset$, by $(\xi_D, 1, 0)$, or by residency with $\gamma = 1$, this proves

$$\max_{\xi,\gamma,k} \bar{W}(\xi,\gamma,k) = M(\pi). \tag{A.5}$$

Residency. At $\gamma = 0$, $p_H = k$ and $p_L = 0$, so $\bar{W}(\xi_R, 0, k) = \delta k\big[\pi w + (1-\pi)\rho(k)w - (1-\delta)\big] \le \delta k\big[w - (1-\delta)\big] \le 0$. Since $\bar{W}(\xi_R, \gamma, k)$ is affine in γ, it follows that $\bar{W}(\xi_R, \gamma, k) \le \max\{\bar{W}(\xi_R, 0, k), \bar{W}(\xi_R, 1, k)\} \le \max\{0, R_1(\pi)\}$.

stop-loss deadline. Substituting $p_H = k + (1-k)\gamma$ and $p_L = (1-k)\gamma$,

$$\bar{W}(\xi_D, \gamma, k) = (1-k)\gamma\, A(\pi) + k\big[\pi w - (1-\delta)\big] \le (1-k)\max\{0, A(\pi)\} \le M(\pi),$$

where the first inequality uses (A.4).

Unrestricted testing. Let $x \equiv (1-k)\gamma$ and $g \equiv (1-k)(1-\gamma)$, so that $p_H = k + x$, $p_L = x$ and $1 - p_H = g$. Direct substitution gives

$$\bar{W}(\xi_U, \gamma, k) = x\, A(\pi) + k\big[\pi w - (1-\delta) + (1-\pi)\delta\,\ell\big] + g\,\bar{W}(\xi_R, \gamma, k).$$

The middle term is non-positive by (A.3) and (A.4), and $\bar{W}(\xi_R, \gamma, k) \le \max\{0, R_1(\pi)\}$ by the residency step. Hence $\bar{W}(\xi_U, \gamma, k) \le (x+g)M(\pi) \le M(\pi)$, because $x + g = 1 - k \le 1$ and $M(\pi) \ge 0$.

Attempt limit. I first show that every incentive-compatible design satisfies $p_H \le \frac{k}{1-\beta}$. If $k \ge 1 - \beta$, this holds because $p_H \le 1$. If $k < 1 - \beta$, then $\Phi(\beta, k) = \frac{k^2}{2(1-\beta)^2}$, and

$$p_H = k + (1-k)\gamma \le k + \frac{\beta k^2}{2(1-\beta)^2} \le k + \frac{\beta k}{1-\beta} = \frac{k}{1-\beta},$$

where the inequalities use $\gamma \le \tilde{\gamma}(k) \le \frac{\beta\Phi(\beta,k)}{1-k}$ and $k < 1 - \beta < 2(1-\beta)$. Therefore $h \le \frac{kw}{1-\beta} - (1-\delta)k = \frac{k}{1-\beta}(w - \mu) \le 0$, and together with (A.3) this gives $\bar{W}(\xi_\Lambda, \gamma, k) \le 0 \le M(\pi)$.

Together with $\bar{W}(\xi_\emptyset) = 0$, this proves (A.5).

Step 3: the function R_1. At $\gamma = 1$, $p_H = 1$ and $p_L = 1 - k$. For $k \ge 1 - \beta$, $\rho = 1$ and $\bar{W}(\xi_R, 1, k) = \delta\big[w - (1-\delta)k\big]$, which is decreasing in k; so it suffices to consider $k \in [0, 1-\beta]$. There $\rho = \frac{k}{1-\beta}$, and a direct computation gives

$$\bar{W}(\xi_R, 1, k) = \delta\Big[A(\pi) + a(\pi)\,k - \frac{1-\pi}{1-\beta}\,k^2\Big], \qquad a(\pi) \equiv \frac{(1-\pi)(2+w-\beta)}{1-\beta} - (1-\delta). \quad \text{(A.6)}$$

This expression is strictly concave in k for $\pi < 1$, with unconstrained maximizer

$$\frac{(1-\beta)\,a(\pi)}{2(1-\pi)} = \tfrac{1}{2}\Big[2 + w - \beta - \frac{\mu}{1-\pi}\Big] = k_R(\pi).$$

Define $G(\pi) \equiv \max_{k\in[0,1-\beta]} \big[a(\pi)k - \frac{1-\pi}{1-\beta}k^2\big] \geq 0$, so that $R_1(\pi) = \delta\big[A(\pi) + G(\pi)\big]$. Note that $G(\pi) > 0$ if and only if $a(\pi) > 0$, which is the case if and only if $k_R(\pi) > 0$. Four properties of R_1 follow:

(a) R_1 is continuous and convex on $[0,1]$, since it is the maximum over a compact set of continuous functions that are affine in π.

(b) $R_1(1) = \delta w < w = A(1)$.

(c) $R_1(0) = \delta \max_k \ell(1,k)$. If $w < \mu$, then $R_1(0) < 0$: by (A.3), $\ell(1,k) < 0$ whenever $k > 0$, and $\ell(1,0) = -1$.

(d) If $k_R(\pi) \geq 1-\beta$, the expression in (A.6) is increasing on $[0, 1-\beta]$, so $R_1(\pi) = \bar{W}(\xi_R, 1, 1-\beta) = \delta(w-\mu) \leq 0$.

Step 4: the threshold $\hat{w}$. At π_0, $A(\pi_0) = 0$ and $1 - \pi_0 = \frac{w}{1+w}$, so $R_1(\pi_0) = \delta G(\pi_0)$. Hence $R_1(\pi_0) > 0$ if and only if $a(\pi_0) > 0$, that is, if and only if $w(2 + w - \beta) > \mu(1 + w)$. This is equivalent to $q(w) \equiv w^2 + \big(1 + \delta(1-\beta)\big)w - \mu > 0$. On $w \geq 0$, q is strictly increasing, with $q(0) = -\mu < 0$ and $q(\mu) = \mu^2 + \delta(1-\beta)\mu > 0$. Its unique positive root is $\hat{w}$, so $0 < \hat{w} < \mu$, and $R_1(\pi_0) > 0$ if and only if $w > \hat{w}$. When $w \leq \hat{w}$, $a(\pi_0) \leq 0$, so $G(\pi_0) = 0$ and $R_1(\pi_0) = 0$.

Step 5: the case $w \leq \hat{w}$. On $[0, \pi_0]$, $A \leq 0$, so $M = \max\{0, R_1\}$. The function R_1 is convex, with $R_1(0) \leq 0$ by (A.3) and $R_1(\pi_0) = 0$. Hence $R_1 \leq 0$ on $[0, \pi_0]$, and $M = 0$ is attained by $\xi_\emptyset$. On $[\pi_0, 1]$, $A \geq 0$, and $R_1 - A$ is convex, with $(R_1 - A)(\pi_0) = 0$ and $(R_1 - A)(1) < 0$ by (b). Hence $R_1 \leq A$ on $[\pi_0, 1]$, and $M = A$ is attained by $(\xi_D, 1, 0)$. Setting $\underline{\pi}_R = \bar{\pi}_R = \pi_0$ proves part (i) of the proposition in this case, with no prior at which residency must be optimal.

Step 6: the case $w > \hat{w}$. Now $R_1(\pi_0) > 0$.

On $[0, \pi_0]$, the set $\{\pi \in [0, \pi_0] : R_1(\pi) \leq 0\}$ is closed and convex by (a). It contains $\pi = 0$, because $R_1(0) \leq 0$, and excludes π_0. It is therefore an interval $[0, \underline{\pi}_R]$ with $\underline{\pi}_R < \pi_0$, and $R_1(\underline{\pi}_R) = 0$ by continuity. I claim $\underline{\pi}_R > 0$. If $w < \mu$, this follows from (c). If $w = \mu$, (d) gives $R_1 = 0$ on $[0, \pi_{cap}]$, where $\pi_{cap} \equiv 1 - \frac{\mu}{w+\beta} > 0$ is the prior at which $k_R(\pi_{cap}) = 1 - \beta$; so $\underline{\pi}_R \geq \pi_{cap} > 0$.

On $[\pi_0, 1]$, the set $\{\pi \in [\pi_0, 1] : R_1(\pi) \leq A(\pi)\}$ is closed and convex, because $R_1 - A$ is convex. It contains $\pi = 1$ by (b) and excludes π_0, where $R_1 > 0 = A$. It is then an interval $[\bar{\pi}_R, 1]$ with $\pi_0 < \bar{\pi}_R < 1$, where $\bar{\pi}_R < 1$ because $R_1 - A < 0$ in a neighbourhood of 1.

Combining these facts with (A.5): for $\pi \leq \underline{\pi}_R$, $R_1 \leq 0$ and $A < 0$, so $M = 0$ and $\xi_\emptyset$ is optimal. For $\pi \in [\underline{\pi}_R, \pi_0]$, $R_1 \geq 0 \geq A$. For $\pi \in [\pi_0, \bar{\pi}_R]$, $R_1 \geq A \geq 0$. Hence $M = R_1$ on $[\underline{\pi}_R, \bar{\pi}_R]$, and residency with $\gamma = 1$ is optimal. For $\pi \geq \bar{\pi}_R$, $A \geq R_1$ and $A > 0$, so $M = A$ and $(\xi_D, 1, 0)$ is optimal. This proves part (i) of the proposition in this case, with $\underline{\pi}_R < \pi_0 < \bar{\pi}_R$, once Step 7 identifies residency's optimal accuracy. Together with Step 5, it also proves part (ii): the thresholds coincide when $w \leq \hat{w}$ and differ when $w > \hat{w}$.

Step 7: residency's accuracy. Suppose $w > \hat{w}$, and fix $\pi \in [\underline{\pi}_R, \bar{\pi}_R]$. I show that $0 < k_R(\pi) \leq 1 - \beta$, so that $k_R(\pi)$ maximizes (A.6) and $\big(\xi_R, 1, k_R(\pi)\big)$ attains $R_1(\pi)$.

Strictly positive accuracy. Suppose $k_R(\pi) \leq 0$. Then $G(\pi) = 0$ and $R_1(\pi) = \delta A(\pi)$. If $\pi \leq \pi_0$, then $R_1(\pi) \geq 0$ forces $A(\pi) \geq 0$, hence $\pi = \pi_0$, which contradicts $R_1(\pi_0) > 0 = \delta A(\pi_0)$. If $\pi > \pi_0$, then $A(\pi) > 0$ and $R_1(\pi) = \delta A(\pi) < A(\pi)$, which contradicts $R_1(\pi) \geq A(\pi)$. Hence $k_R(\pi) > 0$.

Accuracy at most $1 - \beta$. By Step 6, $R_1 > 0$ on $(\underline{\pi}_R, \bar{\pi}_R]$: on $(\underline{\pi}_R, \pi_0]$ by the definition of $\underline{\pi}_R$, and on $[\pi_0, \bar{\pi}_R]$ because $R_1 \geq A \geq 0$ there, with $R_1(\pi_0) > 0$ and $A > 0$ for $\pi > \pi_0$. Suppose $k_R(\pi) > 1 - \beta$. Then (d) gives $R_1(\pi) = \delta(w - \mu) \leq 0$, so $\pi = \underline{\pi}_R$. Since k_R is continuous in π, priors slightly above $\underline{\pi}_R$ also satisfy $k_R > 1 - \beta$, and (d) gives $R_1 \leq 0$ at those priors. This contradicts $R_1 > 0$ on $(\underline{\pi}_R, \pi_0]$. Hence $k_R(\pi) \leq 1 - \beta$. □

A.11 Proof of Proposition 8

Throughout, let $\mu \equiv (1-\delta)(1-\beta)$, so that $w > \mu$, and use h, ℓ, A and the welfare expressions $\bar{W}$ from the proof of Proposition 7. Let $v \equiv w - (1-\delta)\bar{k}$. By the definition of $\bar{k}$, the attempt limit is incentive compatible at full leniency, $\tilde{\gamma}(\bar{k}) = 1$; equivalently, $\beta\Phi(\beta, \bar{k}) = 1 - \bar{k}$. Since $\bar{k} > 1 - \beta$, the design $(1, \bar{k})$ gives $\rho = 1$ and $p_H = 1$, so $h = \ell = v$ and

$$V_\Lambda(\pi) \equiv \bar{W}(\xi_\Lambda, 1, \bar{k}) = \big(\pi + (1 - \pi)\delta\big)v.$$

Let $R^*(\pi) \equiv \max_{k \in [0,1]} \bar{W}(\xi_R, 1, k)$ be the best welfare from residency with a fully lenient test, and let $M(\pi) \equiv \max\{A(\pi), R^*(\pi), V_\Lambda(\pi)\}$. When the dependence on w matters, I write $R^*(\pi; w)$, $V_\Lambda(\pi; w)$ and so on.

Step 1: three bounds. First, by the proof of Lemma 3, every design satisfies $\ell(\gamma, k) \leq \rho(k)(w - \mu) \leq w - \mu < w$. Second, every design satisfies $\ell(\gamma, k) \leq h(\gamma, k)$, because $\ell = \rho p_H w - (1 - \rho)p_L - (1 - \delta)k \leq p_H w - (1 - \delta)k = h$. Third, since $\big(\pi + (1 - \pi)\delta\big)\bar{k} \leq 1$, for

every $\pi \in [0,1]$ and every design,

$$\max\Big\{\pi w-(1-\delta),\ \pi w-(1-\delta)+(1-\pi)\delta\,\ell(\gamma,k)\Big\} \le \big(\pi+(1-\pi)\delta\big)w-(1-\delta) \le V_\Lambda(\pi). \quad \text{(A.7)}$$

Step 2: only three options matter. The design $(1, 1-\beta)$ gives $\rho = 1$ and $p_H = 1$, hence $h = \ell = w - \mu$, so for every π

$$R^*(\pi) \ge \bar{W}(\xi_R, 1, 1-\beta) = \delta(w-\mu) > 0. \quad \text{(A.8)}$$

In particular $M(\pi) > 0 = \bar{W}(\xi_\emptyset)$, so the hiring freeze is never optimal. I show that no menu and design yields more than $M(\pi)$. Since $M(\pi)$ is attained by $(\xi_D, 1, 0)$, by residency with $\gamma = 1$, or by $(\xi_\Lambda, 1, \bar{k})$, this proves

$$\max_{\xi,\gamma,k} \bar{W}(\xi,\gamma,k) = M(\pi). \quad \text{(A.9)}$$

Residency. At $\gamma = 0$, $\bar{W}(\xi_R, 0, k) = \delta k\big[\pi w+(1-\pi)\rho(k)w-(1-\delta)\big] \le \delta k\big[w-(1-\delta)\big]$. If $k < 1-\beta$, this is at most $\delta\max\{0,(1-\beta)w-\mu\} \le \delta(w-\mu) \le R^*(\pi)$. If $k \ge 1-\beta$, it is at most $\delta\big[w-(1-\delta)k\big] = \bar{W}(\xi_R,1,k) \le R^*(\pi)$. Since $\bar{W}(\xi_R,\gamma,k)$ is affine in γ, $\bar{W}(\xi_R,\gamma,k) \le R^*(\pi)$ for every design.

stop-loss deadline. As in the proof of Proposition 7, $\bar{W}(\xi_D,\gamma,k) = (1-k)\gamma\,A(\pi)+k\big[\pi w-(1-\delta)\big]$. By (A.7), this is at most $(1-k)\gamma\,M(\pi) + k\,M(\pi) \le M(\pi)$, since $M(\pi) > 0$ and $(1-k)\gamma + k \le 1$.

Unrestricted testing. With $x \equiv (1-k)\gamma$ and $g \equiv (1-k)(1-\gamma)$,

$$\bar{W}(\xi_U,\gamma,k) = x\,A(\pi)+k\big[\pi w-(1-\delta)+(1-\pi)\delta\,\ell\big]+g\,\bar{W}(\xi_R,\gamma,k) \le (x+k+g)M(\pi) = M(\pi),$$

by (A.7) and the residency step.

Attempt limit. By the second bound of Step 1, $\bar{W}(\xi_\Lambda,\gamma,k) = \pi h + (1-\pi)\delta\ell \le \big(\pi+(1-\pi)\delta\big)h$. I show that every incentive-compatible design satisfies $h \le \max\{0,v\}$, which gives $\bar{W}(\xi_\Lambda,\gamma,k) \le \max\{0, V_\Lambda(\pi)\} \le M(\pi)$. If $k \ge \bar{k}$, then $h \le w-(1-\delta)k \le v$. If $k < \bar{k}$, then $\gamma \le \tilde{\gamma}(k) \le \frac{\beta\Phi(\beta,k)}{1-k}$, so $p_H \le f(k) \equiv k + \beta\Phi(\beta,k)$. The function $\Phi(\beta,\cdot)$ is convex, because its integrand $\max\{\frac{k}{1-\beta} - c, 0\}$ is convex in k; hence f is convex, with $f(0) = 0$ and $f(\bar{k}) = \bar{k} + (1-\bar{k}) = 1$. Convexity gives $f(k) \le \frac{k}{\bar{k}}$ on $[0,\bar{k}]$, so $h \le \frac{k}{\bar{k}}w - (1-\delta)k = \frac{k}{\bar{k}}\,v \le \max\{0,v\}$.

Together, these prove (A.9).

Step 3: residency. For $k \ge 1-\beta$, $\bar{W}(\xi_R,1,k) = \delta\big[w-(1-\delta)k\big]$ is decreasing in k. For $k \le 1-\beta$, it is the strictly concave quadratic (A.6), with unconstrained maximizer $k_R(\pi)$.

Hence the maximizer of $\bar{W}(\xi_R, 1, \cdot)$ over $[0,1]$ is unique and equals $\min\{\max\{k_R(\pi), 0\}, 1-\beta\}$. Four properties follow:

(a) R^* is continuous and convex on $[0,1]$, and jointly continuous in (π, w).

(b) $R^*(1) = \delta w < w = A(1)$, and $R^*(0) = \delta \max_k \ell(1,k) = \delta(w-\mu)$, by Step 1 and (A.8).

(c) For each k, $\bar{W}(\xi_R, 1, k)$ is increasing in w with slope $\delta\big[\pi + (1-\pi)\rho(k)\big] \leq \delta$. Hence $R^*(\pi; w') - R^*(\pi; w) \leq \delta(w'-w)$ for $w < w'$.

(d) At every prior with $R^*(\pi) \geq A(\pi)$, $k_R(\pi) > 0$, so residency's optimal design is $\big(1, k_R^*(\pi)\big)$. Indeed, if $k_R(\pi) \leq 0$, the maximizer is $k = 0$ and $R^*(\pi) = \delta A(\pi)$. By (A.8) this requires $A(\pi) > 0$, but then $R^*(\pi) = \delta A(\pi) < A(\pi)$.

Step 4: the attempt limit against admitting everyone. The function $V_\Lambda - A$ is affine in π. At $\pi = 0$ it equals $1 + \delta v \geq 1 - \delta(1-\delta)\bar{k} > 0$, and at $\pi = 1$ it equals $-(1-\delta)\bar{k} < 0$. Hence $V_\Lambda > A$ on $[0, \pi_\Lambda^+)$ and $V_\Lambda < A$ on $(\pi_\Lambda^+, 1]$, where

$$\pi_\Lambda^+(w) \equiv \frac{1+\delta v}{1 + w - (1-\delta)v} \in (0,1),$$

so π_Λ^+ is continuous and strictly increasing in w. Similarly, $R^* - A$ is convex, positive at $\pi = 0$ by (b), and negative at $\pi = 1$. Hence there is a unique $r(w) \in (0,1)$ such that $R^* > A$ on $[0, r)$ and $R^* < A$ on $(r, 1]$.

Step 5: the threshold w^*. Let $\Delta(\pi; w) \equiv R^*(\pi; w) - V_\Lambda(\pi; w)$, which is convex in π, and define

$$\Psi(w) \equiv \min_{\pi \in [0, \pi_\Lambda^+(w)]} \Delta(\pi; w) = \min_{l \in [0,1]} \Delta\big(l\, \pi_\Lambda^+(w); w\big),$$

which is continuous in w by (a) and Step 4. Four facts about Ψ follow.

(a) At $\pi = 0$, by (b), $\Delta(0; w) = \delta(w-\mu) - \delta v = \delta(1-\delta)\big(\bar{k} - (1-\beta)\big) \equiv D_0 > 0$, which does not depend on w.

(b) For $w < w'$, (c) and $V_\Lambda(\pi; w') - V_\Lambda(\pi; w) = \big(\pi + (1-\pi)\delta\big)(w'-w)$ give

$$\Delta(\pi; w') \leq \Delta(\pi; w) - (1-\delta)\pi\,(w'-w).$$

Since also $[0, \pi_\Lambda^+(w)] \subseteq [0, \pi_\Lambda^+(w')]$, any minimizer π_w of $\Delta(\cdot; w)$ satisfies $\Psi(w') \leq \Delta(\pi_w; w') \leq \Psi(w) - (1-\delta)\pi_w(w'-w)$. So Ψ is non-increasing. If $\Psi(w) < D_0$, then $\pi_w > 0$ by (a), and $\Psi(w') < \Psi(w)$.

(c) If $w \leq (1-\delta)\bar{k}$, then $v \leq 0$, so $V_\Lambda \leq 0 < R^*$ by (A.8), and $\Psi(w) > 0$.

(d) Let $\hat{\pi} \equiv \pi_\Lambda^+\big((1-\delta)\bar{k}\big) > 0$. For $w \geq (1-\delta)\bar{k}$, $\hat{\pi} \in [0, \pi_\Lambda^+(w)]$, and (b) gives $\Psi(w) \leq \Delta\big(\hat{\pi}; (1-\delta)\bar{k}\big) - (1-\delta)\hat{\pi}\big(w - (1-\delta)\bar{k}\big)$, which is negative for w large enough.

By continuity, (c) and (d), Ψ has a root, and every root exceeds $(1-\delta)\bar{k}$. The root is unique: if $\Psi(w_1) = \Psi(w_2) = 0$ with $w_1 < w_2$, then $\Psi(w_1) = 0 < D_0$, and (b) gives $\Psi(w_2) < 0$, a contradiction. Let w^* be this root. By (b), $\Psi(w) > 0$ for $w < w^*$ and $\Psi(w) < 0$ for $w > w^*$.

Step 6: the case $w \leq w^*$. Now $\Psi(w) \geq 0$, so $R^* \geq V_\Lambda$ on $[0, \pi_\Lambda^+]$, while $V_\Lambda < A$ on $(\pi_\Lambda^+, 1]$. Set $\pi_D(\zeta) \equiv r$.

Priors $\pi \leq r$. Here $R^* \geq A$. Moreover $V_\Lambda \leq R^*$: if $\pi \leq \pi_\Lambda^+$ this holds directly, and if $\pi > \pi_\Lambda^+$ then $V_\Lambda < A \leq R^*$. Hence $M = R^*$.

Priors $\pi \geq r$. Here $A \geq R^*$. Moreover $V_\Lambda \leq A$: if $\pi \geq \pi_\Lambda^+$ this holds directly, and if $\pi < \pi_\Lambda^+$ then $V_\Lambda \leq R^* \leq A$. Hence $M = A$.

Since $R^* > A$ on $[0, r)$, admitting every candidate is optimal if and only if $\pi \geq \pi_D(\zeta)$. By (A.9) and (d), residency $\big(\xi_R, 1, k_R^*(\pi)\big)$ is optimal for every $\pi \leq \pi_D(\zeta)$.

Step 7: the case $w > w^*$. Now $\Psi(w) < 0$. The set $S \equiv \{\pi \in [0, \pi_\Lambda^+] : \Delta(\pi) \leq 0\}$ is closed and convex, because Δ is convex and continuous. It excludes $\pi = 0$ by Step 5(a), and it contains a prior with $\Delta < 0$, hence a neighbourhood of that prior relative to $[0, \pi_\Lambda^+]$. Therefore $S = [\underline{\pi}_\Lambda, \bar{\pi}_\Lambda]$ with $0 < \underline{\pi}_\Lambda < \bar{\pi}_\Lambda \leq \pi_\Lambda^+$. On $[0, \pi_\Lambda^+]$, $V_\Lambda \geq A$, so $M = V_\Lambda$ on S and $M = R^*$ on $[0, \pi_\Lambda^+] \setminus S$. Two subcases arise.

Subcase $\Delta(\pi_\Lambda^+) \leq 0$, so that $\bar{\pi}_\Lambda = \pi_\Lambda^+$. Set $\pi_D(\zeta) \equiv \pi_\Lambda^+$. On $[\pi_\Lambda^+, 1]$, $V_\Lambda \leq A$, and $R^* - A$ is convex with $(R^* - A)(\pi_\Lambda^+) = \Delta(\pi_\Lambda^+) \leq 0$ and $(R^* - A)(1) < 0$, so $R^* \leq A$. Hence $M = A$ on $[\pi_D, 1]$, while $A < V_\Lambda \leq M$ on $[0, \pi_D)$. The attempt limit is optimal on $[\underline{\pi}_\Lambda, \pi_D]$, and residency on $[0, \underline{\pi}_\Lambda]$.

Subcase $\Delta(\pi_\Lambda^+) > 0$, so that $\bar{\pi}_\Lambda < \pi_\Lambda^+$. Then $R^*(\pi_\Lambda^+) > V_\Lambda(\pi_\Lambda^+) = A(\pi_\Lambda^+)$, so $r > \pi_\Lambda^+$. Set $\pi_D(\zeta) \equiv r$. On $[\pi_\Lambda^+, r]$, $R^* \geq A \geq V_\Lambda$, so $M = R^*$. On $[r, 1]$, $A \geq R^*$ and $A \geq V_\Lambda$, because $r > \pi_\Lambda^+$, so $M = A$. Since $R^* > A$ on $[0, r)$ and $V_\Lambda > A$ on $[0, \pi_\Lambda^+)$, admitting every candidate is optimal if and only if $\pi \geq \pi_D$. The attempt limit is optimal on $[\underline{\pi}_\Lambda, \bar{\pi}_\Lambda]$, and residency on $[0, \underline{\pi}_\Lambda] \cup [\bar{\pi}_\Lambda, \pi_D]$.

In both subcases, $0 < \underline{\pi}_\Lambda < \bar{\pi}_\Lambda \leq \pi_D(\zeta)$. Admitting every candidate is optimal if and only if $\pi \geq \pi_D(\zeta)$, and the attempt limit $(\xi_\Lambda, 1, \bar{k})$ is optimal on $[\underline{\pi}_\Lambda, \bar{\pi}_\Lambda]$. At every other $\pi \leq \pi_D(\zeta)$, $M = R^* \geq A$, so residency $\big(\xi_R, 1, k_R^*(\pi)\big)$ is optimal by (A.9) and (d). □

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